สยามชำนาญกิจ
SIAM CHAMNAN KIT

# Generative AI for Software Development 2026

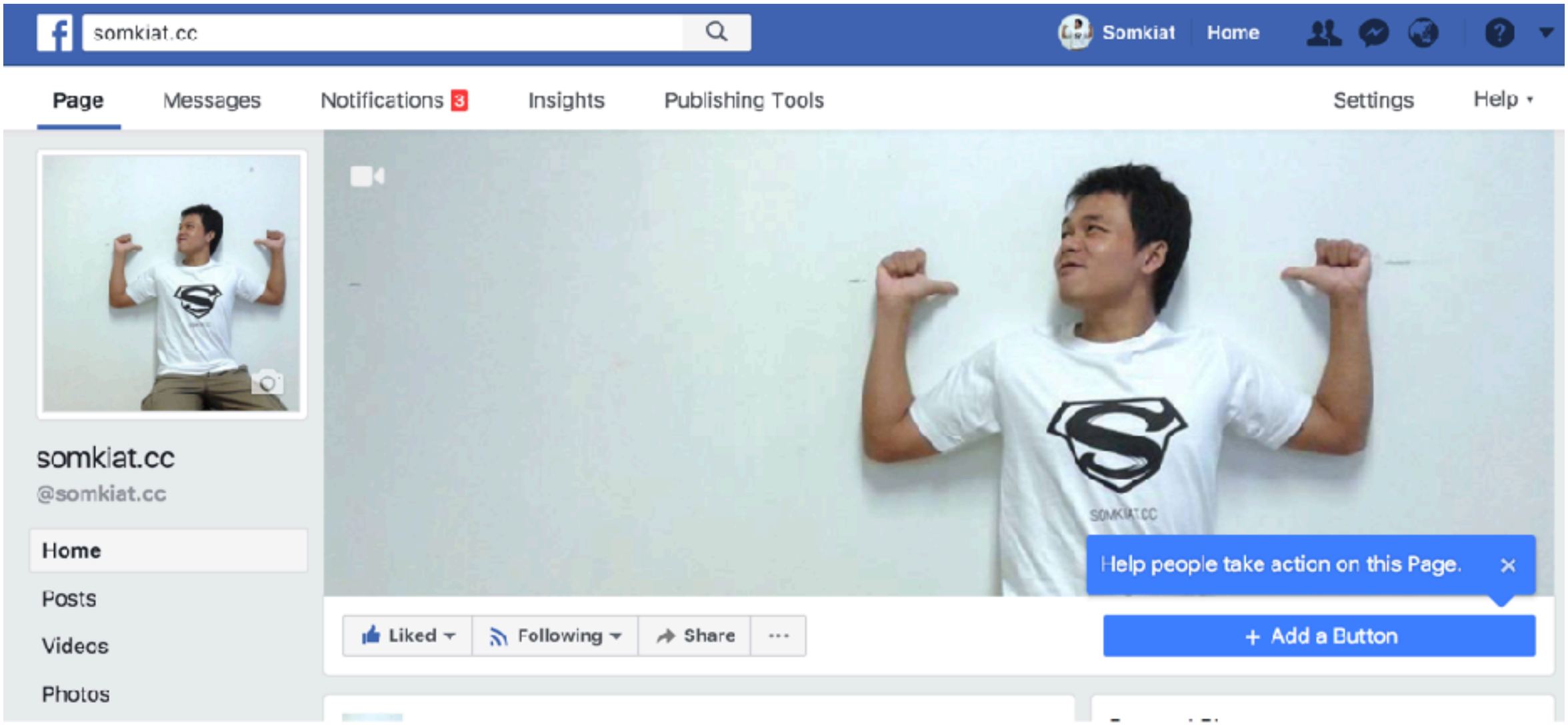
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# https://github.com/up1/workshop-ai-with-technical-team


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# Goals

Integrate Generative AI in Software Development Requirement, Coding and Testing Workshop with AI Agent





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# Learning Path

AI/ML/LLM

LLM Model

Prompt Engineer

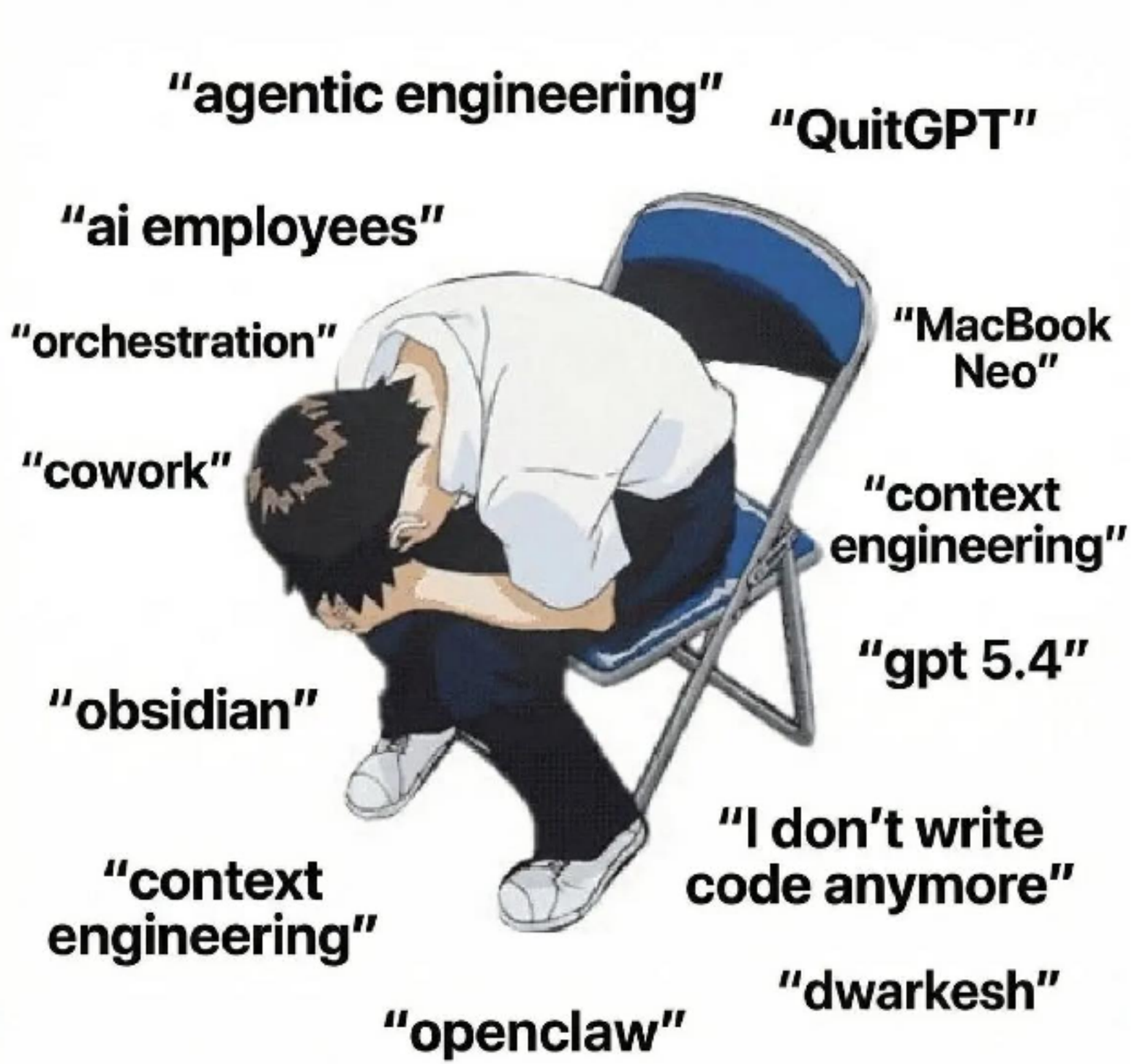
Context Engineering

AI Agent

AI in Software development

Use cases and workshops


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# How to use AI ?

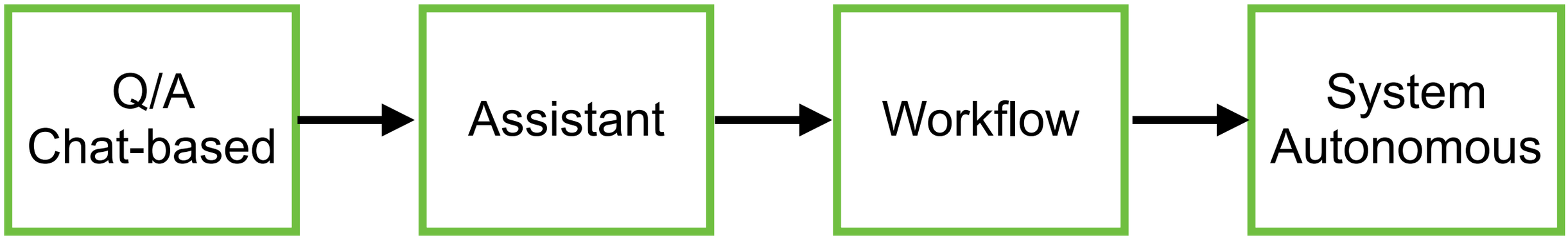
AI as
a **tool**

AI as
an **assistant**

AI as
an **agent**


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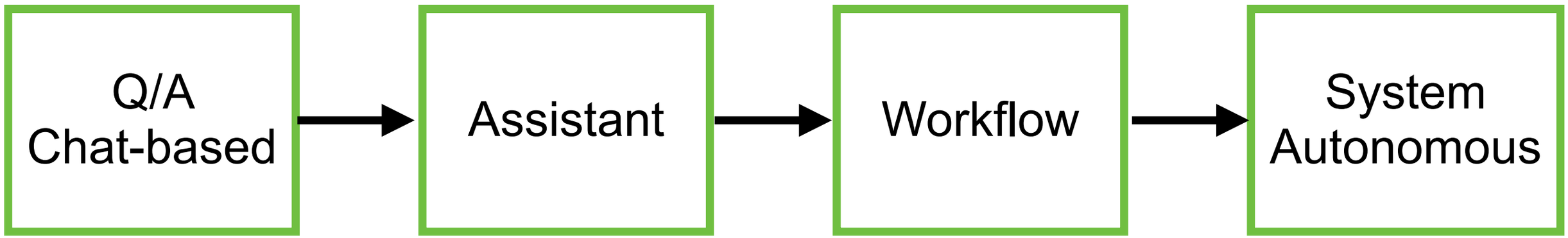
# Steps to use AI ?





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# Steps to use AI ?




Prompt engineering

Manage context and knowledge for AI (Context engineering)
Human in the Loop (HITL)


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# Software Development

Requirement | Design | Develop | Testing | Deploy

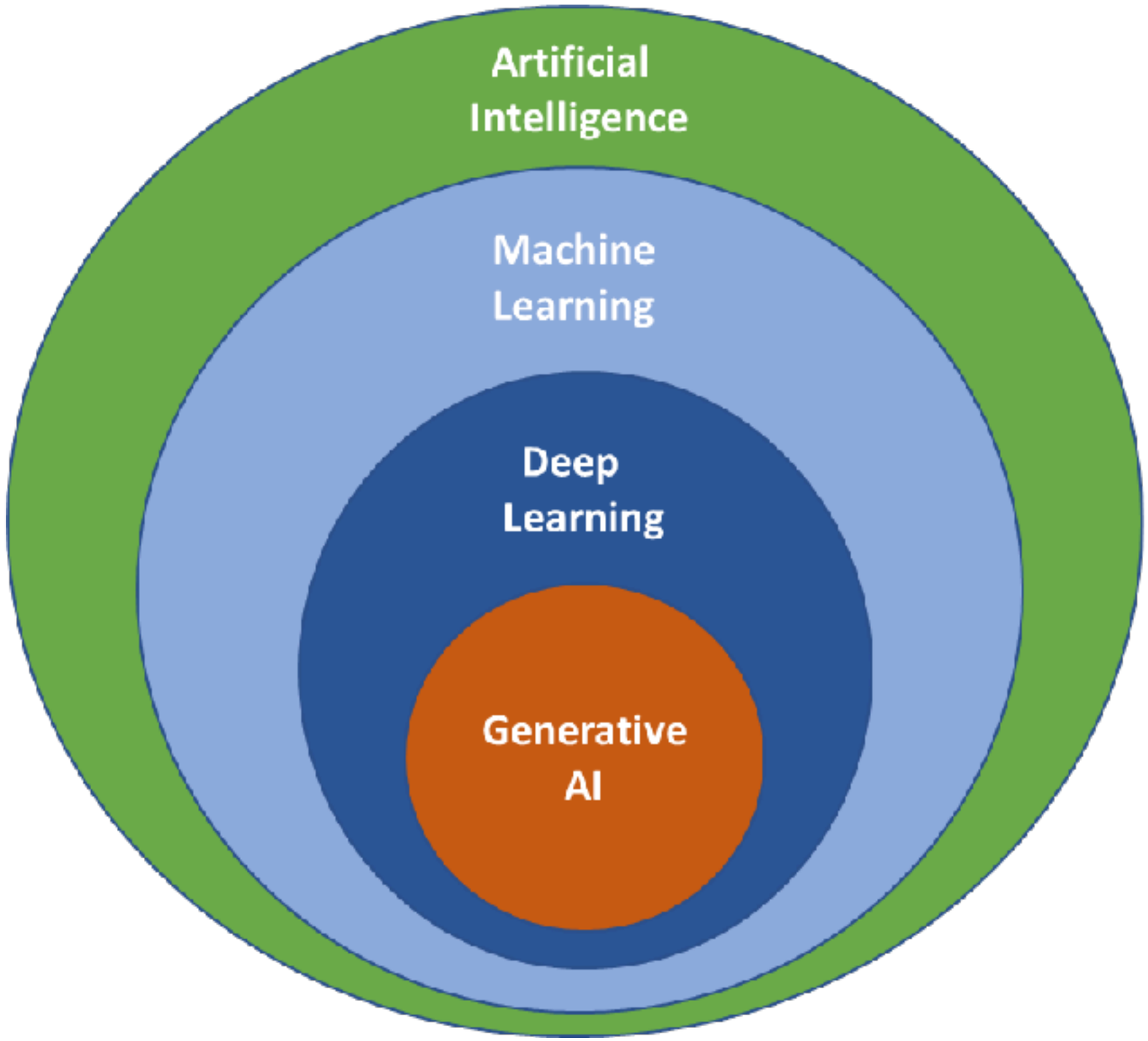
Generative AI

Improve Productivity … (Replace human !!)


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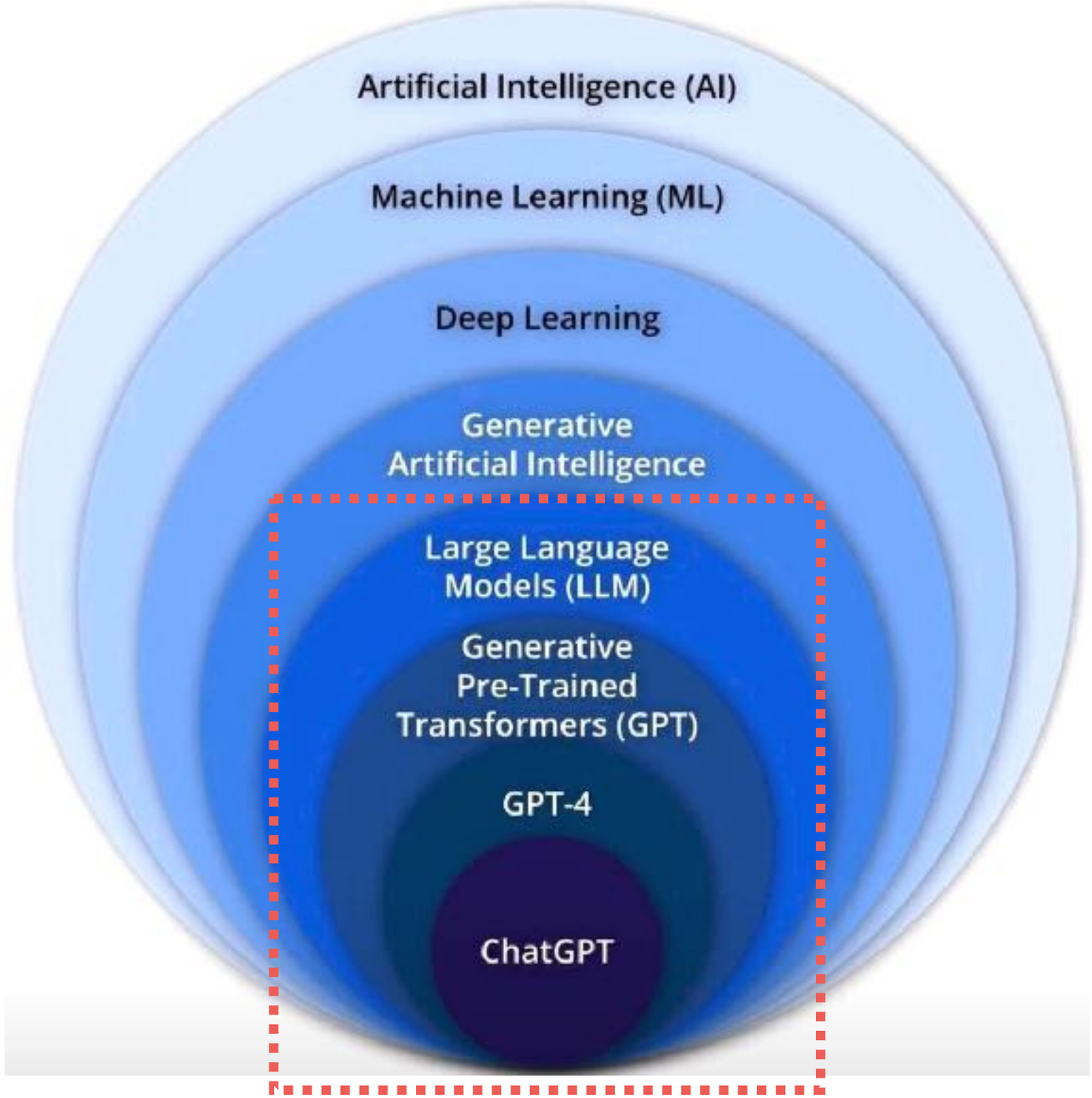
# Start with AI !!


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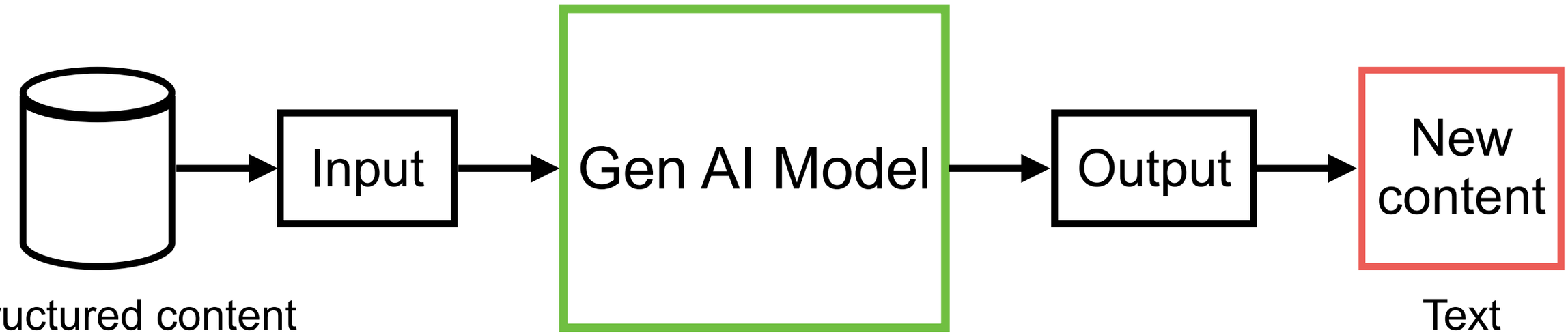
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# Generative AI




Unstructured content

Learn patterns from Unstructured content

Text
Image
Video
3D
Task

https://grow.google/ai-essentials/


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# Large Language Model (LLM)

Type of AI model
Process, understand
Generate human readable data

LLM

Training  data !!


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# LLMs in industry

| Model name | Company |
|---|---|
| Bidirectional Encoder Representation from Transformers (BERT) | Google AI |
| Generative Pre-trained transformer-5 (GPT-5) | OpenAI |
| Pathways Language Model-E (PaLM-E) | Google AI |
| BLOOM | NVIDIA AI |
| Llama 4 | Facebook |
| Claude’s Sonnet/Opus 5.5 | Anthropic |




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# LLM Development timeline

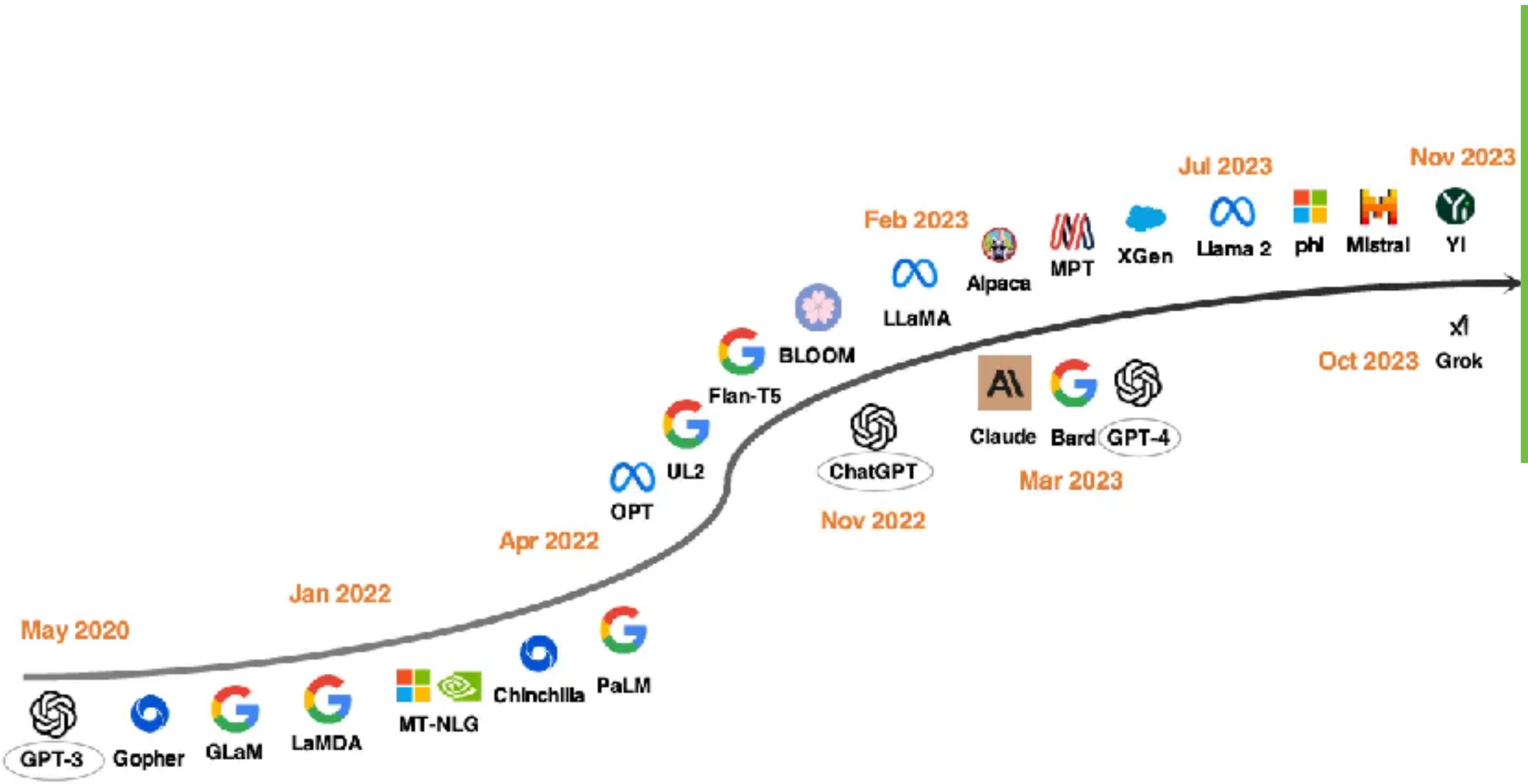



Figure 3: LLM development timeline. The models below the arrow are closed-source while those above the arrow are open-source.

https://arxiv.org/abs/2311.16989




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# Let’s go !!




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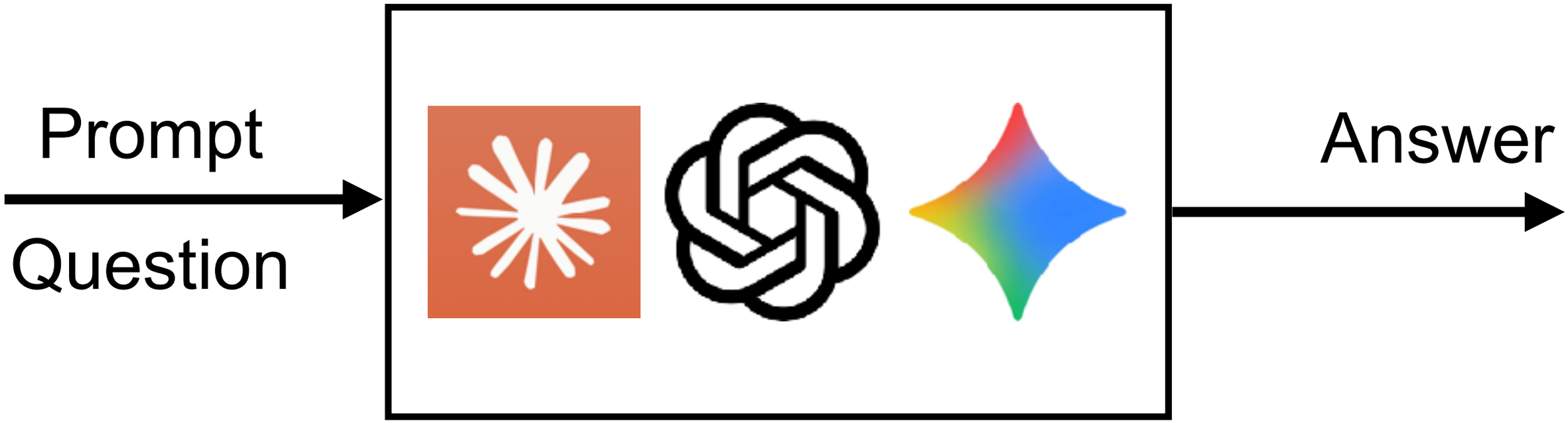
# Hello with Generative AI





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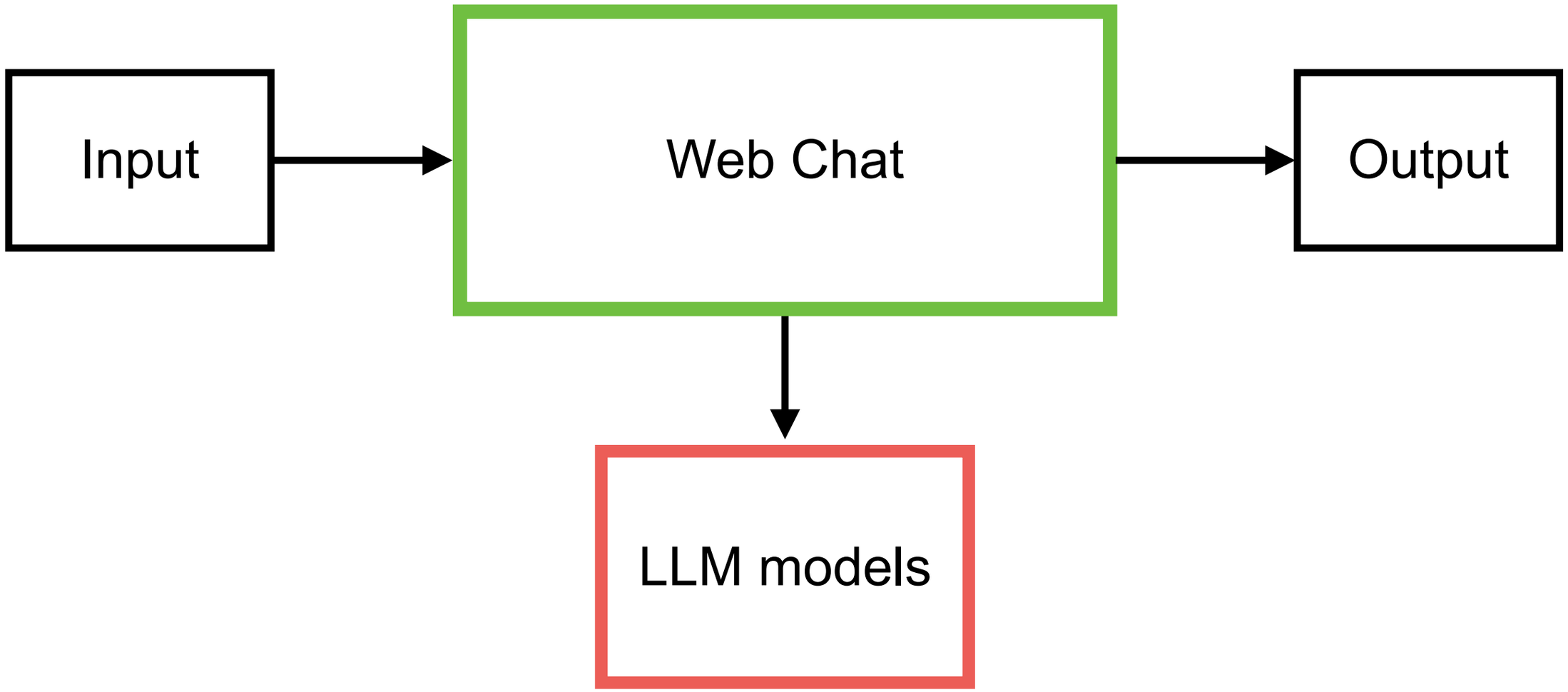
# Chat-based





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# Chat-based




https://github.com/copilot


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# LLM model card

**GPT-6 Astra** `gpt-6-astra`

Our most capable model for the most demanding work.

Compare | Try in Playground

| REASONING | SPEED | PRICE | INPUT | OUTPUT |
|---|---|---|---|---|
| Highest | Fast | $10 · $50 Input · Output | Text, Image | Text |

GPT-6 Astra is our most capable model for the most demanding work. Use it for complex reasoning, coding, computer use, research, and document creation. `reasoning.effort` supports `low`, `medium`, `high`, `xhigh`, and `max`.

Get started with GPT-6 Astra using the model guide.

- 1,050,000 context window
- 128,000 max output tokens
- Apr 30, 2026 knowledge cutoff
- Reasoning token support

**Pricing**

Pricing is based on the number of tokens used, or other metrics based on the model type. For tool-specific models, like search and computer use, there's a fee per tool call. See details in the pricing page.

Text tokens — Per 1M tokens

| Input | Cached input | Cache writes | Output |
|---|---|---|---|
| $10.00 | $1.00 | $12.50 | $50.00 |

https://developers.openai.com/api/docs/models/gpt-6-astra


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# LLM model card

| | Claude Fable 5.1 | Claude Opus 5.5 | Claude Sonnet 5.5 | Claude Haiku 5.5 |
|---|---|---|---|---|
| | For demanding reasoning and long-horizon agentic work | For long-running agentic coding and knowledge work | The best combination of speed and intelligence | For high-volume, latency-sensitive tasks such as classification, extraction, and routing |
| Comparative latency | Slower | Moderate | Fast | Fastest |
| Pricing | $10 / input MTok<br>$50 / output MTok | $4 / input MTok<br>$20 / output MTok | $2 / input MTok<br>$10 / output MTok | From $0.10 / input MTok<br>From $0.50 / output MTok |
| Claude API ID | `claude-fable-5-1` | `claude-opus-5-5` | `claude-sonnet-5-5` | `claude-haiku-5-5` |
| **Capabilities** | | | | |
| Thinking | Adaptive (always on) | Adaptive (always on) | Adaptive | Adaptive |
| Default effort | `high` | `medium` | `high` | `medium` |
| Context window | 1M tokens | 1M tokens | 1M tokens | 1M tokens |
| Max output | 128K tokens | 128K tokens | 128K tokens | 128K tokens |
| Reliable knowledge cutoff | Jun 2026 | Jun 2026 | Jun 2026 | Jun 2026 |
| > Additional details | | | | |

https://platform.claude.com/docs/en/models/overview




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# Token Cost ?

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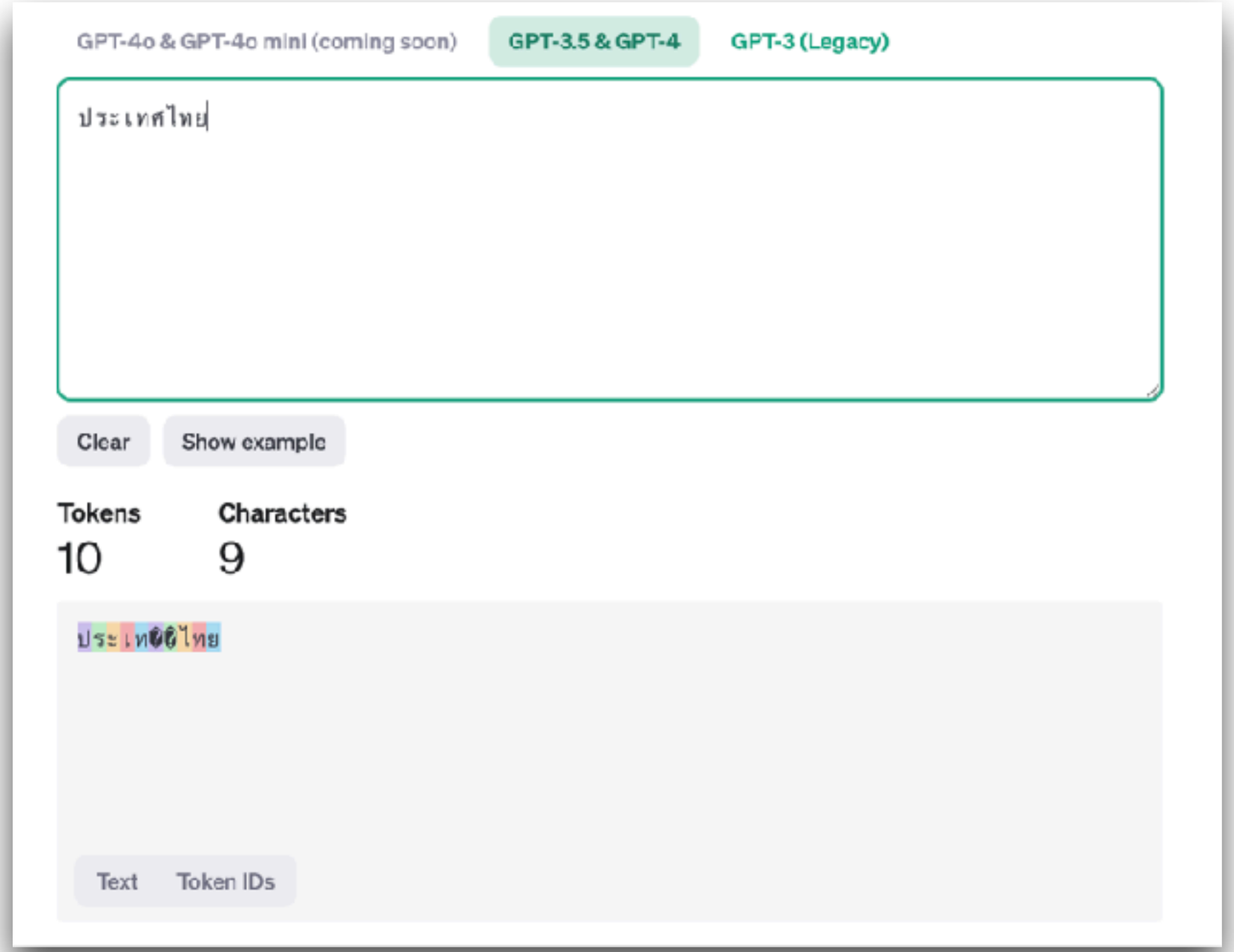
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https://developers.openai.com/api/docs/models/gpt-6-astra


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# OpenAI Tokenizer




https://platform.openai.com/tokenizer


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# Context window ?

**GPT-6 Astra** `gpt-6-astra`

Our most capable model for the most demanding work.

Compare | Try in Playground

| REASONING | SPEED | PRICE | INPUT | OUTPUT |
|---|---|---|---|---|
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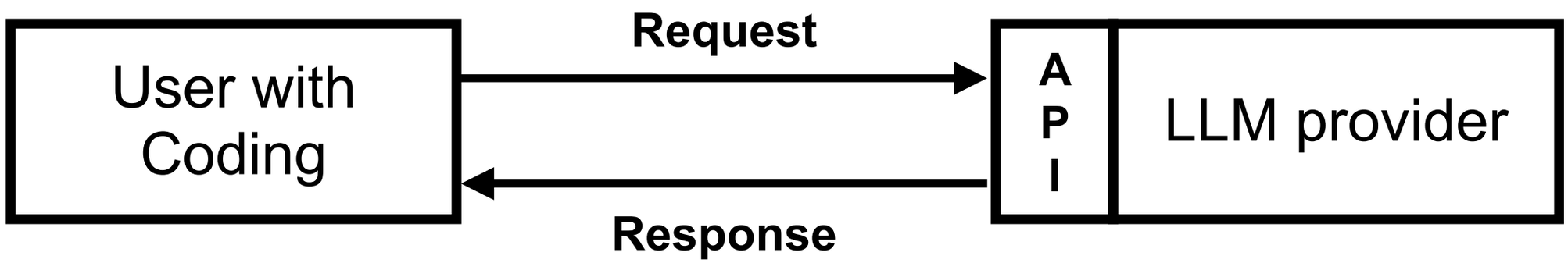
https://developers.openai.com/api/docs/models/gpt-6-astra


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# Context window?

Amount of text a model can process and remember at once time

Measure with **token**





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# Context window size of models

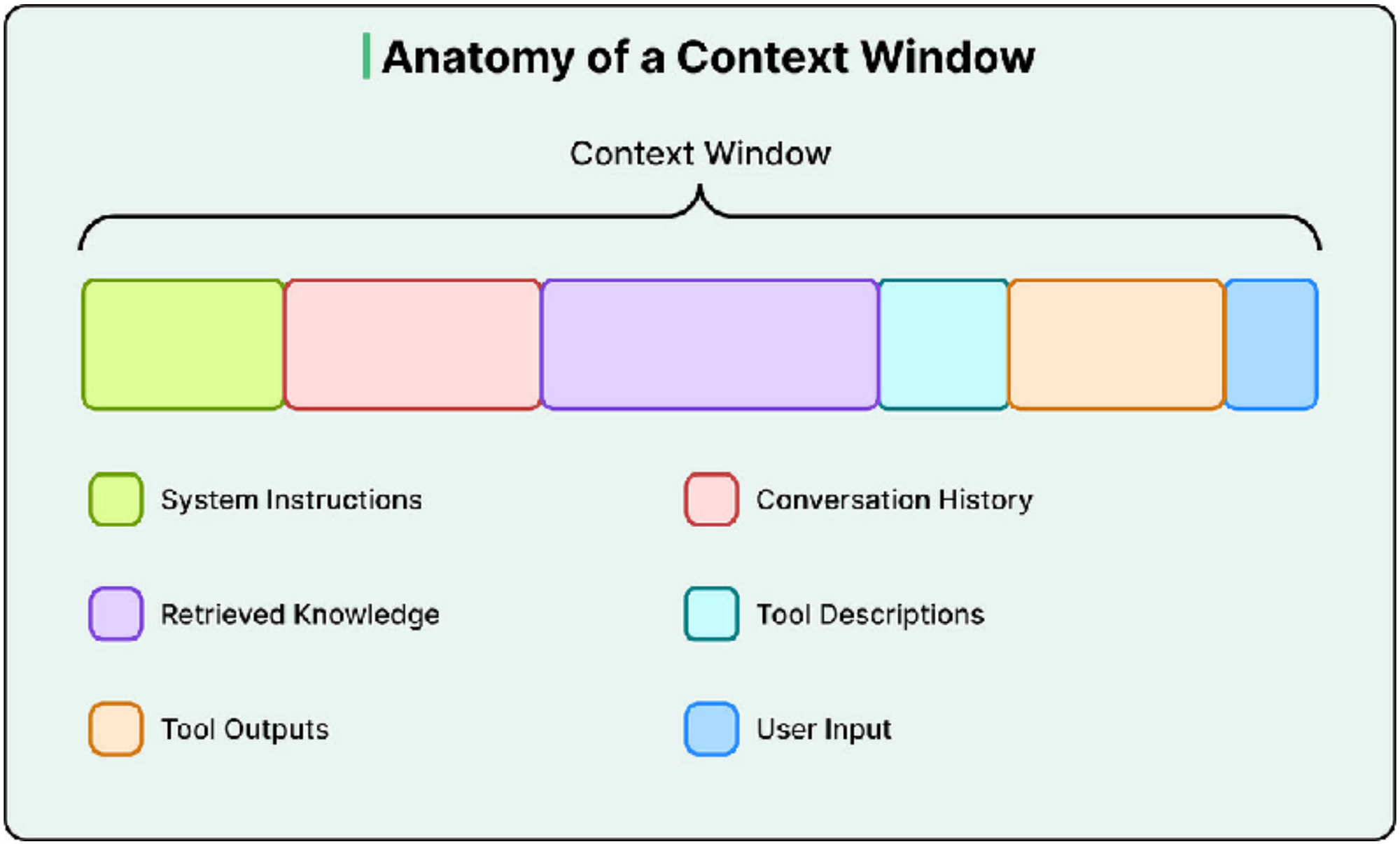
Different models may have different window size

| Model name | Context window size |
|---|---|
| GPT 5 | 1M |
| Gemini 3.5 | 1M |
| Llama 4 | 1M |
| Claude Opus 5.5 | 1M |
| Claude Haiku 4.5 | 200,000 |
| DeepSeek R1 | 1M |




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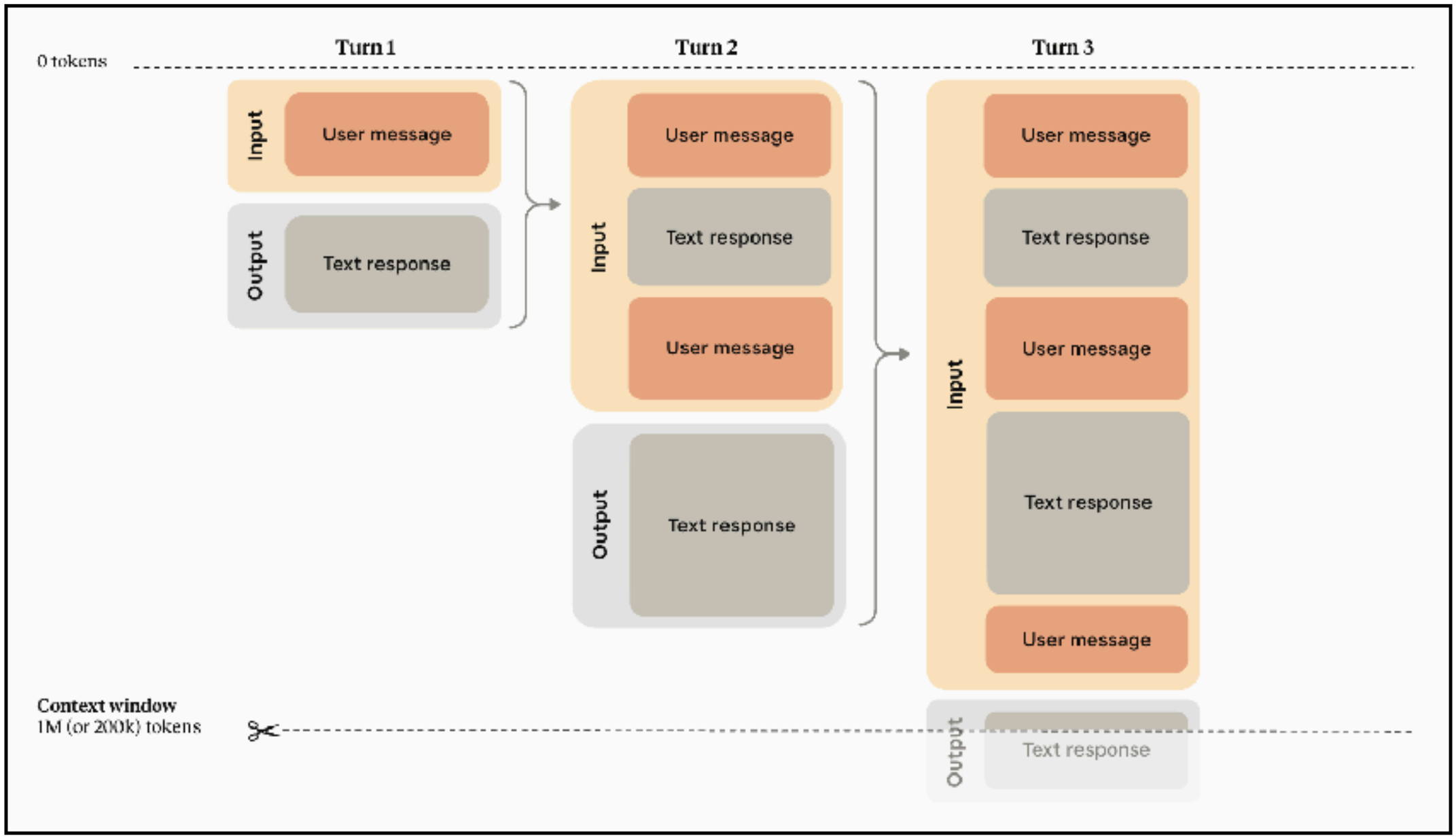
# Anatomy of Context Window





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# Context window ?




https://platform.claude.com/docs/en/build-with-claude/context-windows

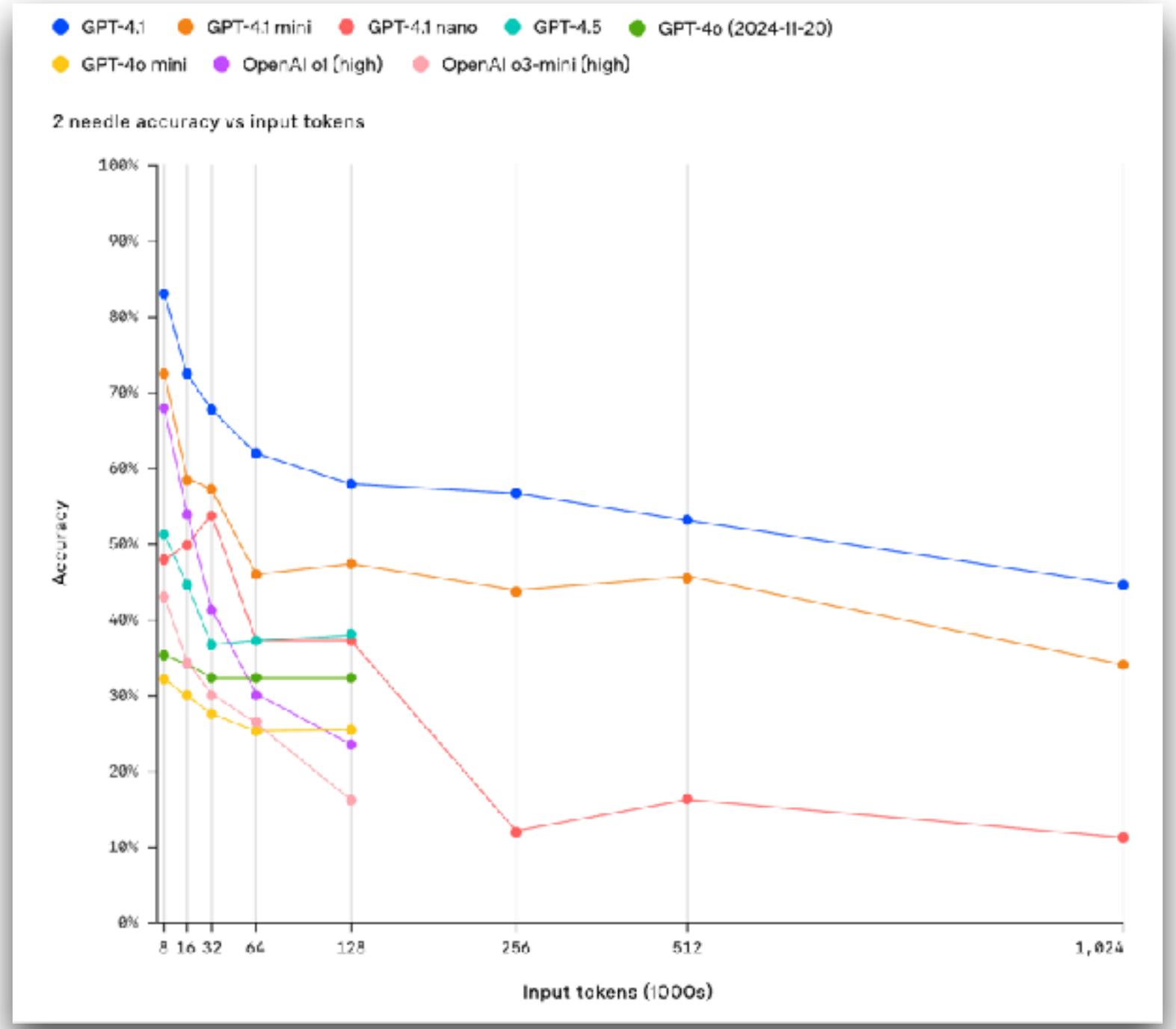

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# Long context == Better result ?




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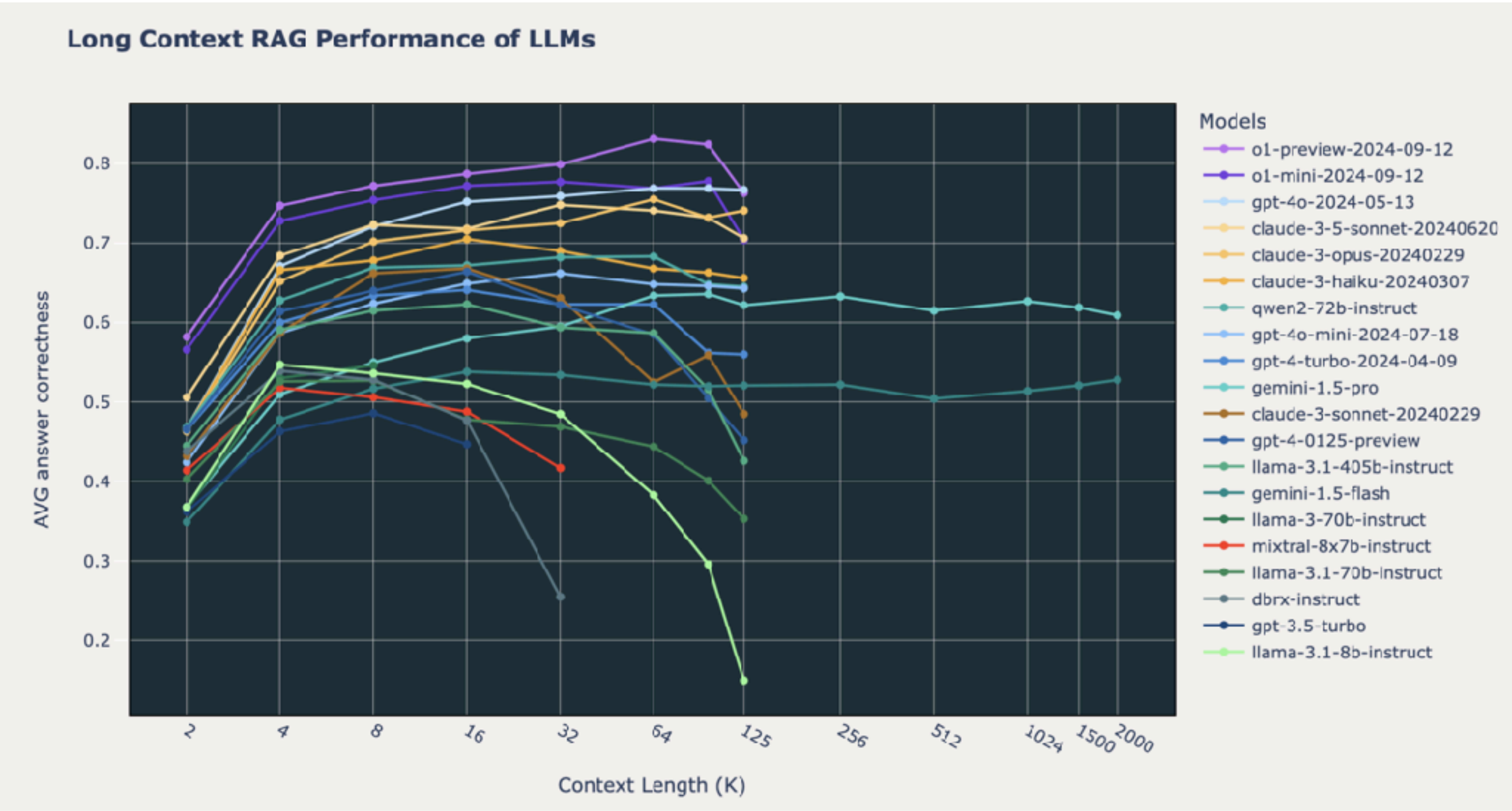
# Long context LLM is better !!




https://openai.com/index/gpt-4-1/


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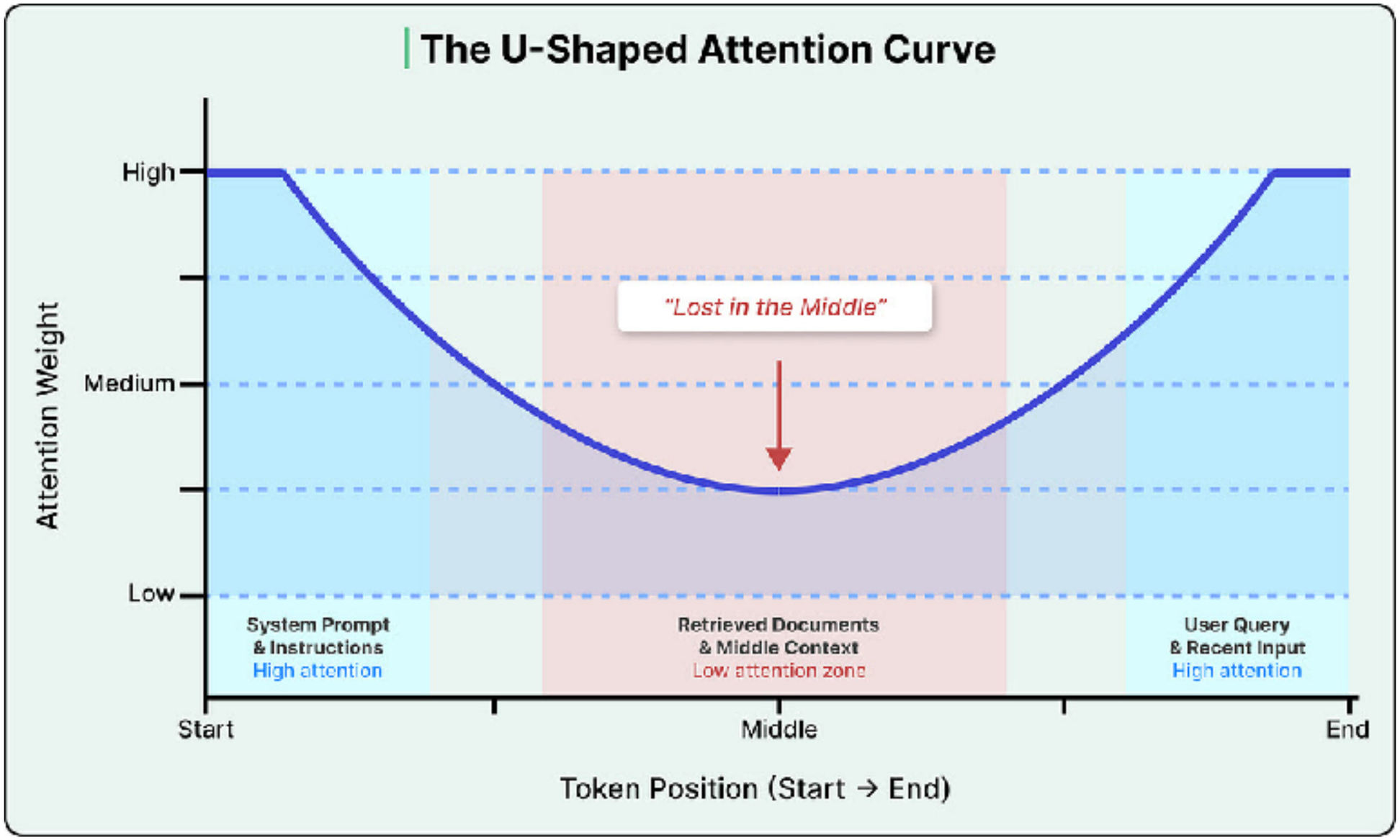
# Long context LLM is better !!




https://arxiv.org/abs/2411.03538


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# Loss in the middle of LLM

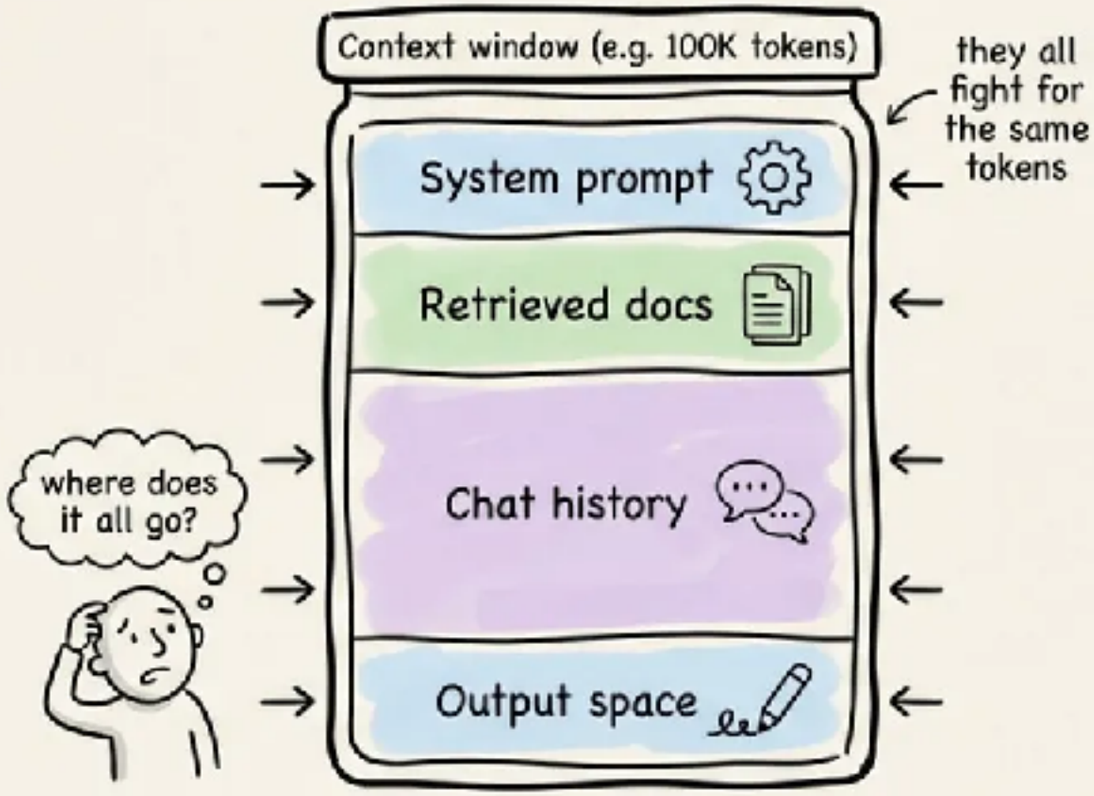



https://arxiv.org/abs/2307.03172

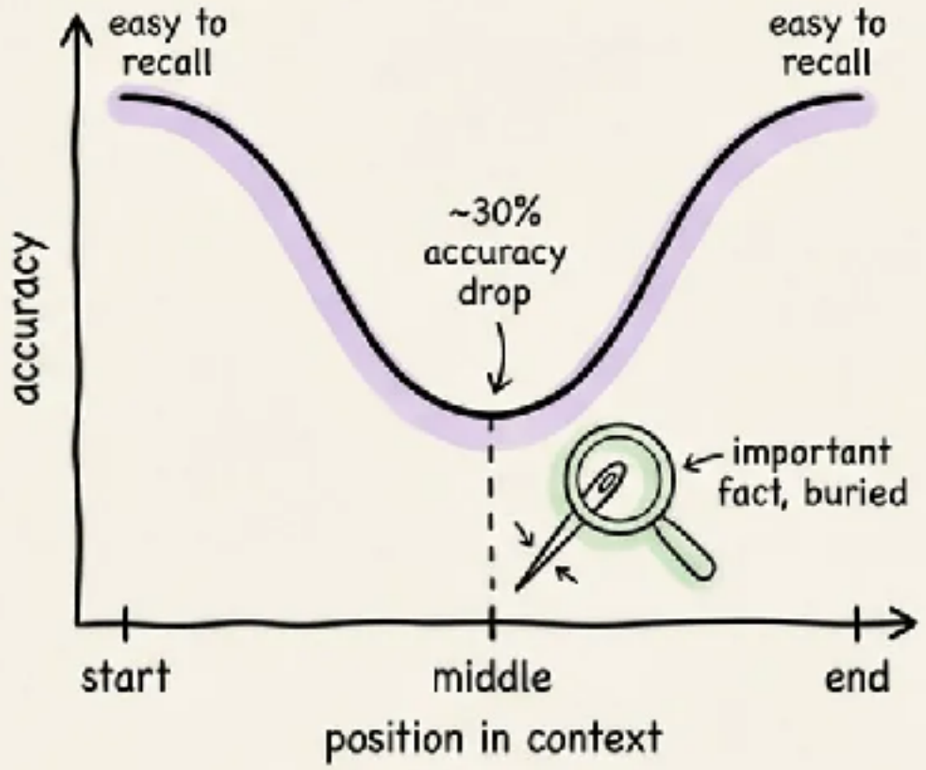

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# More tokens isn't more memory.

## Context = a shared budget



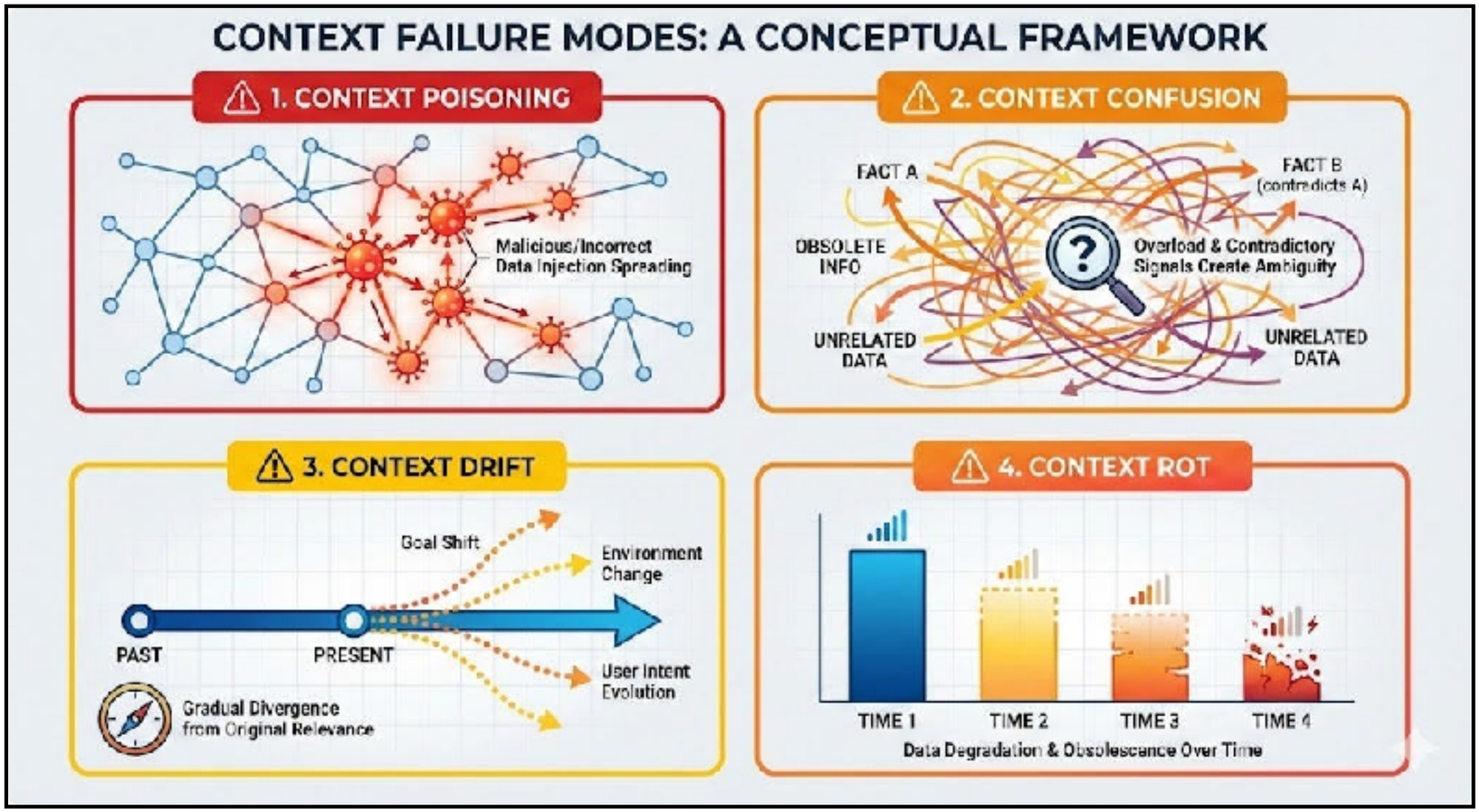

## Lost in the middle




**Bigger windows don't beat structured memory.**

https://blog.dailydoseofds.com/p/evolution-of-agent-landscape-from


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# Knowledge cutoff ?

**GPT-6 Astra** `gpt-6-astra`

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Compare | Try in Playground

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|---|---|---|---|---|
| Highest | Fast | $10 · $50<br>Input · Output | Text, Image | Text |

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- Apr 30, 2026 knowledge cutoff
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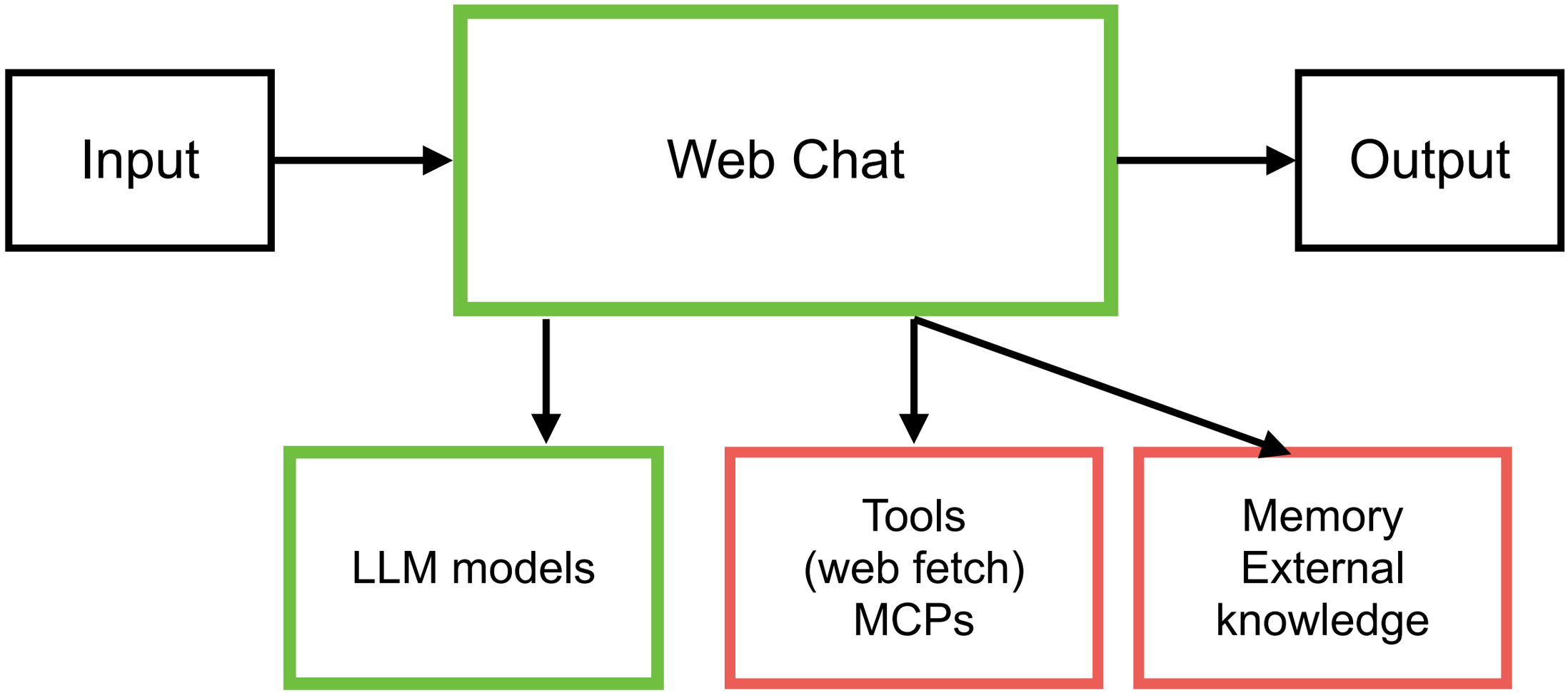
Text tokens — Per 1M tokens

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https://developers.openai.com/api/docs/models/gpt-6-astra


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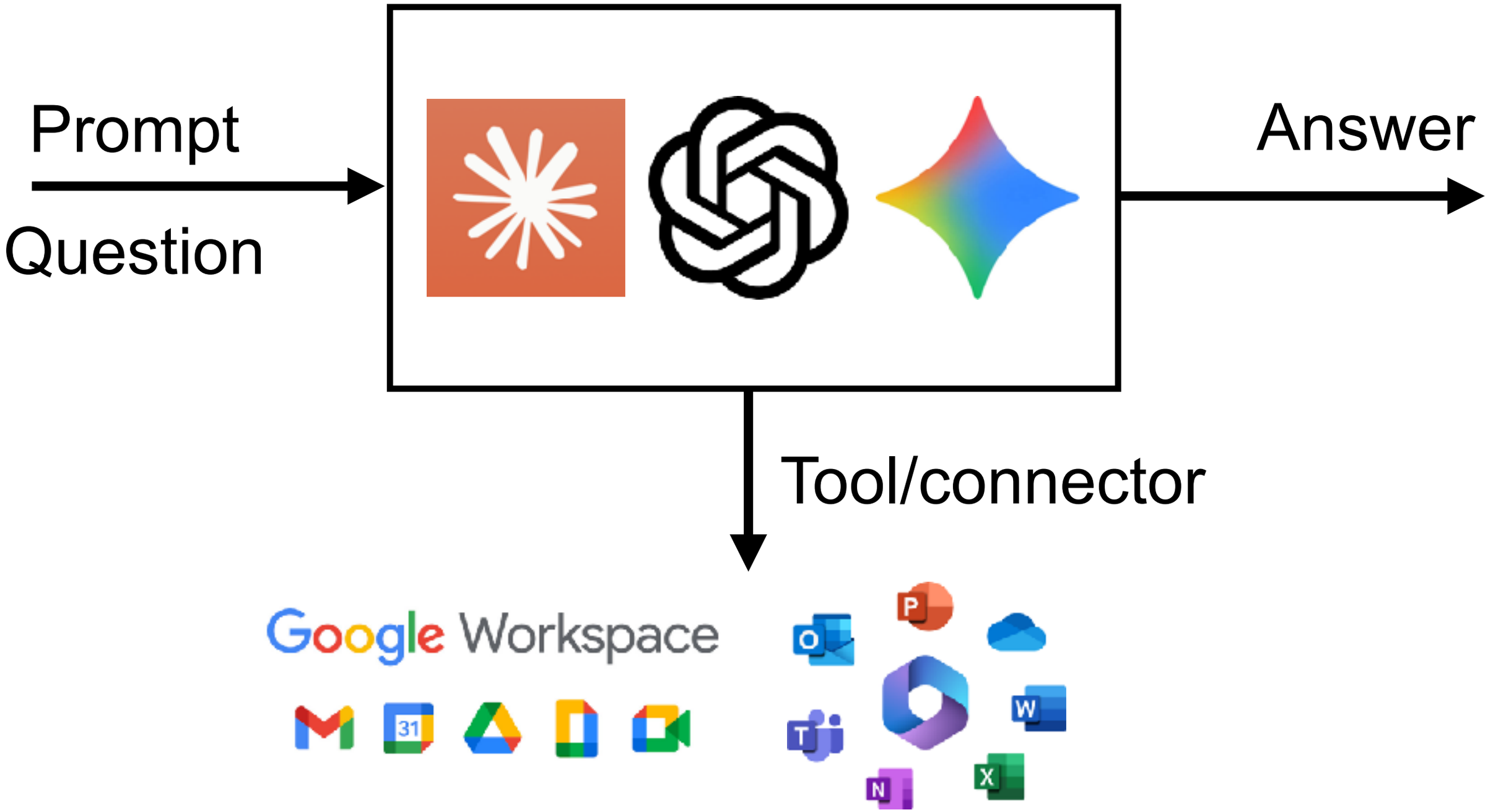
# Chat-based




https://github.com/copilot


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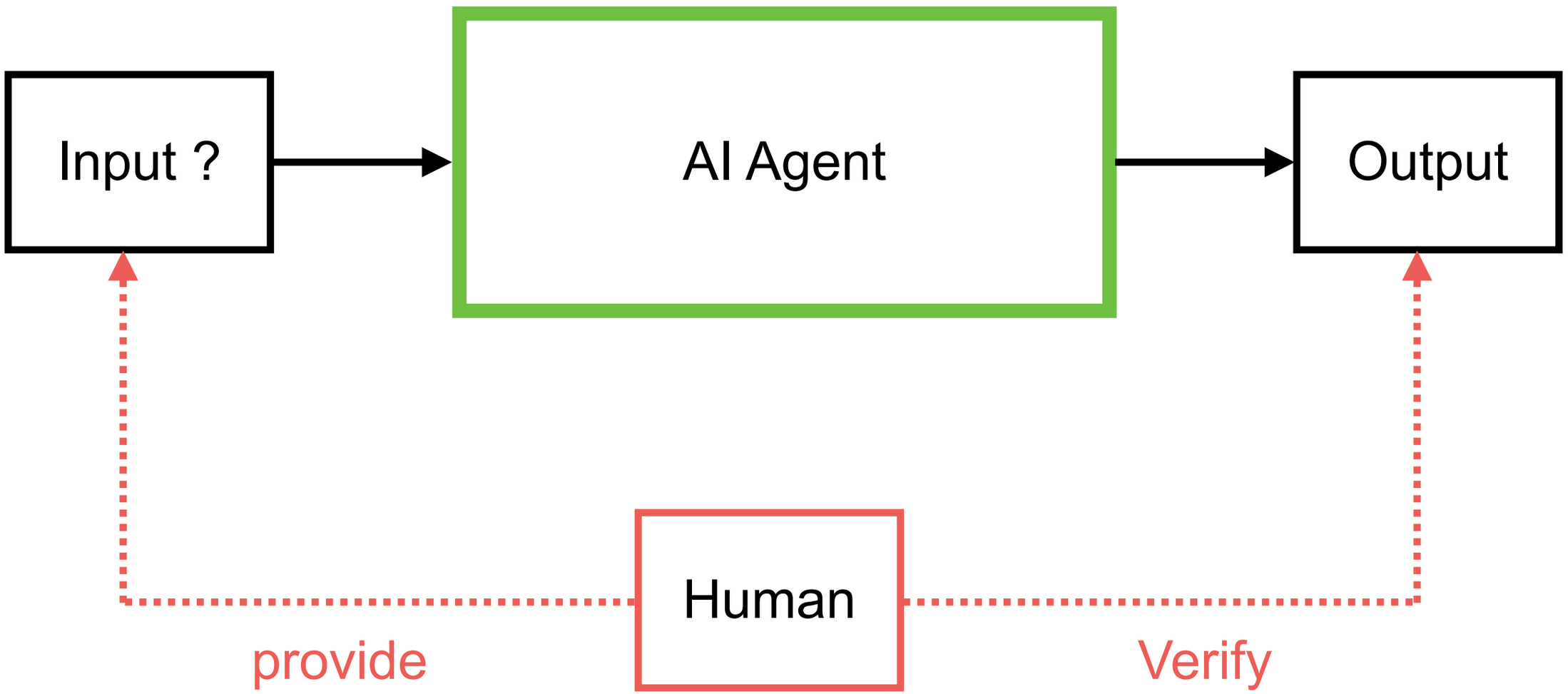
# Use tool to connect other app !!





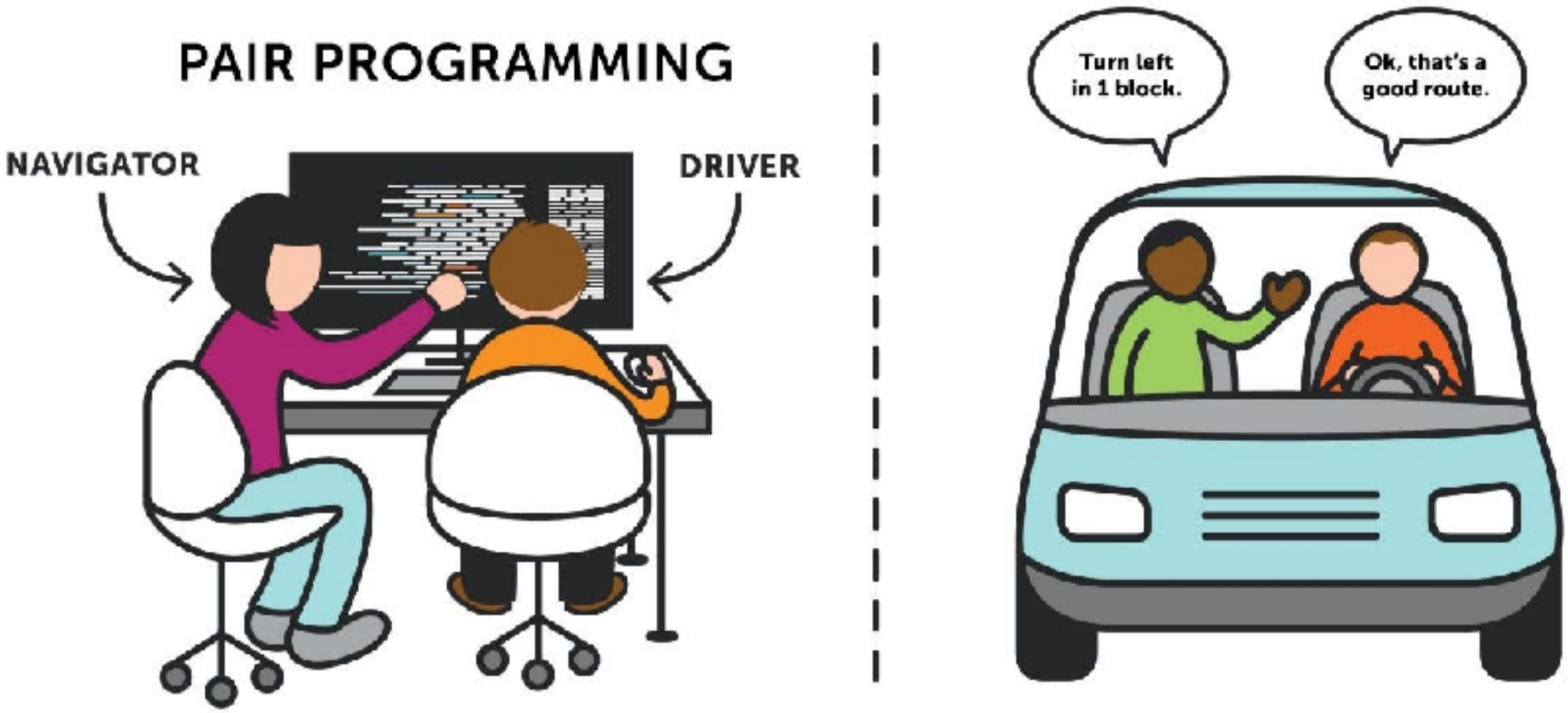
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# Hello with Generative AI





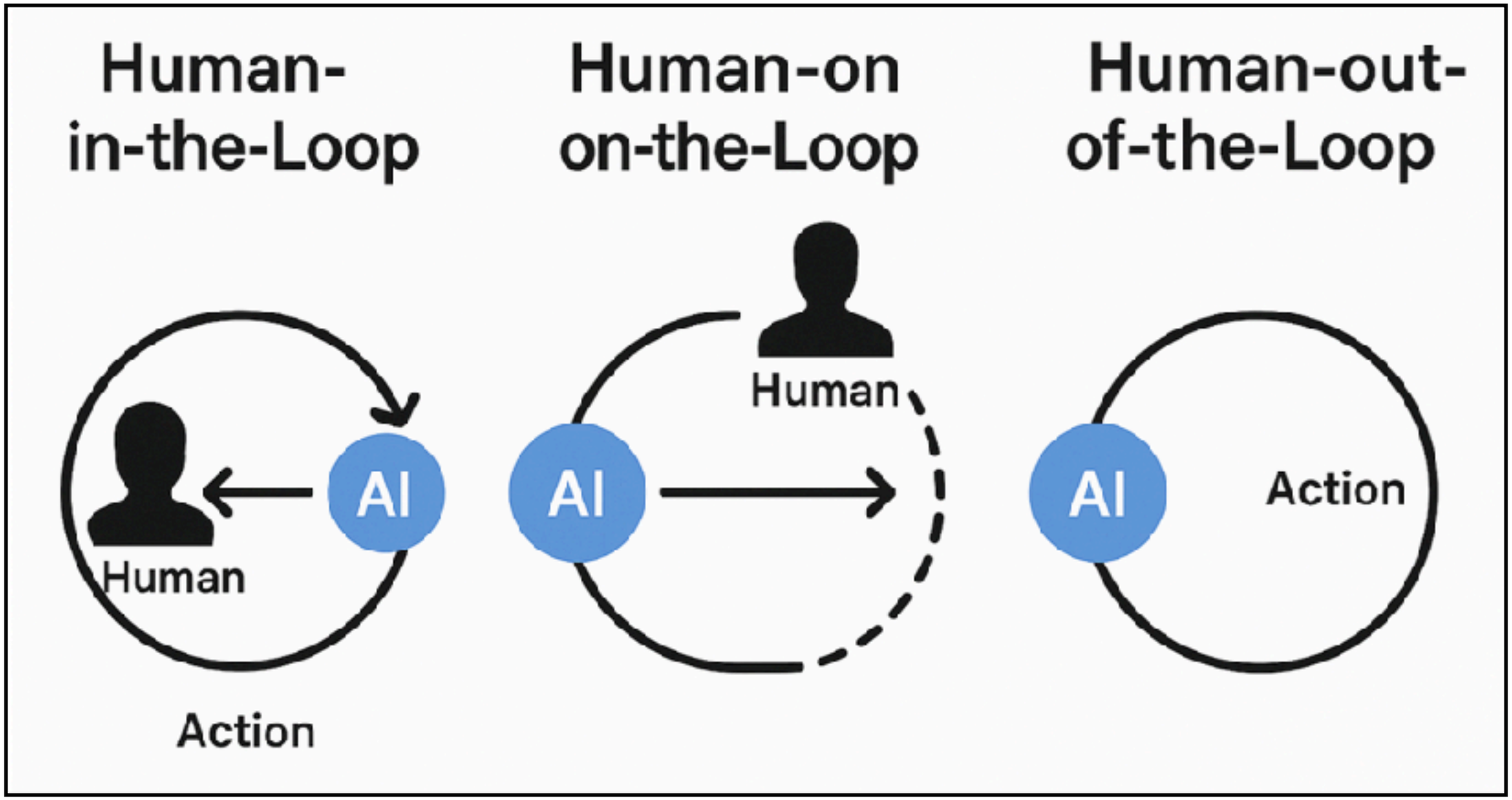
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Reviews, Tests and Understands
with experience


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# AI Agent with/without Human



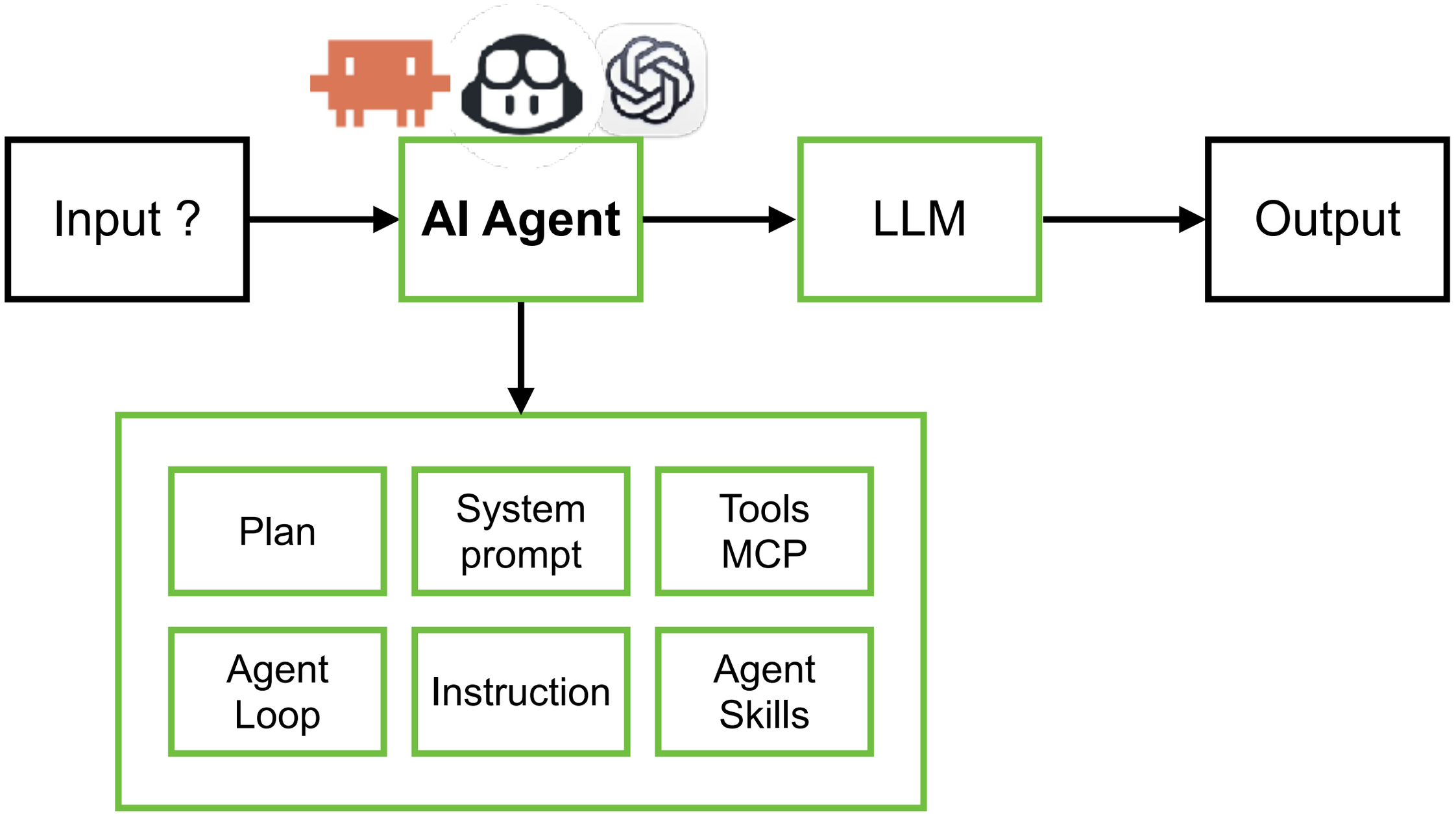


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# AI draft
# Human craft


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# Working with AI Agent





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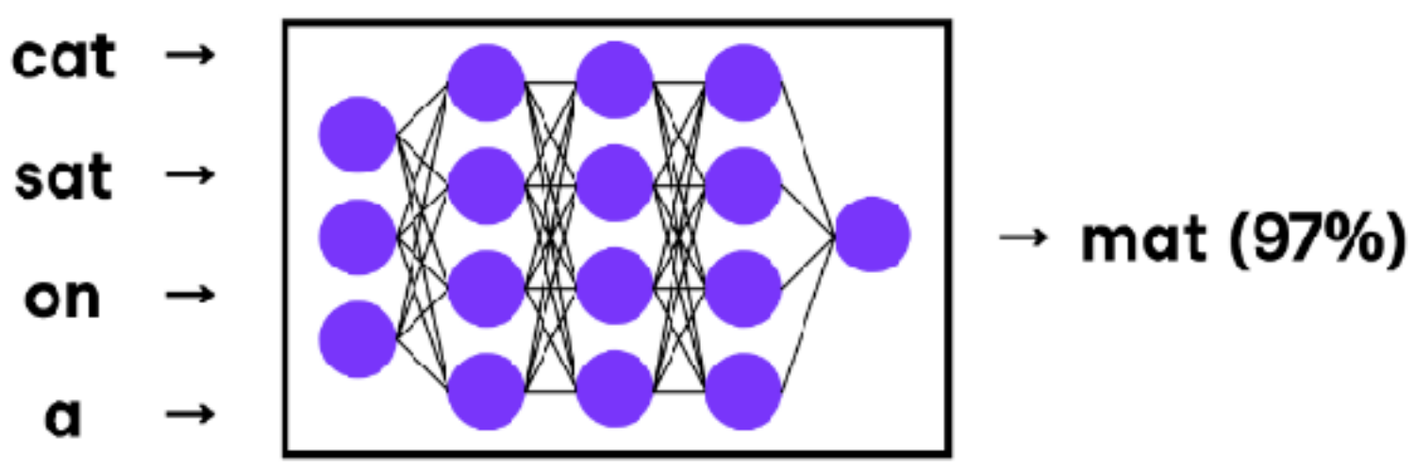
# สีเหลือง


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# Large Language Model (LLM)

Neural network

Predicts the next word in a sequence

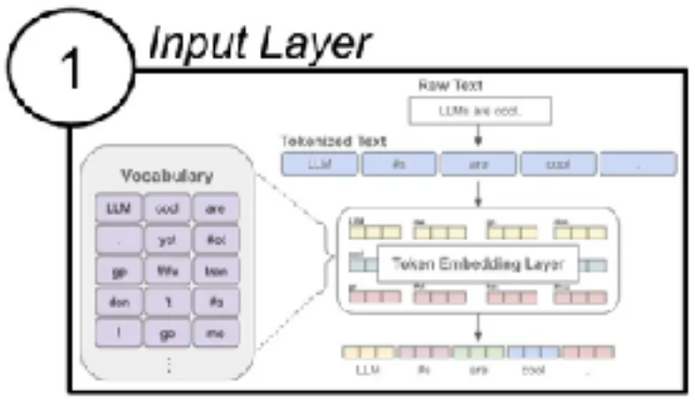


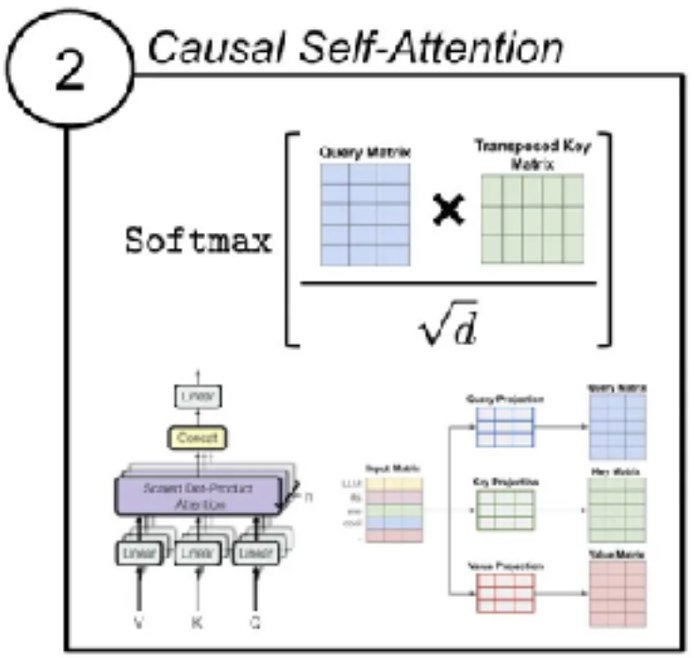


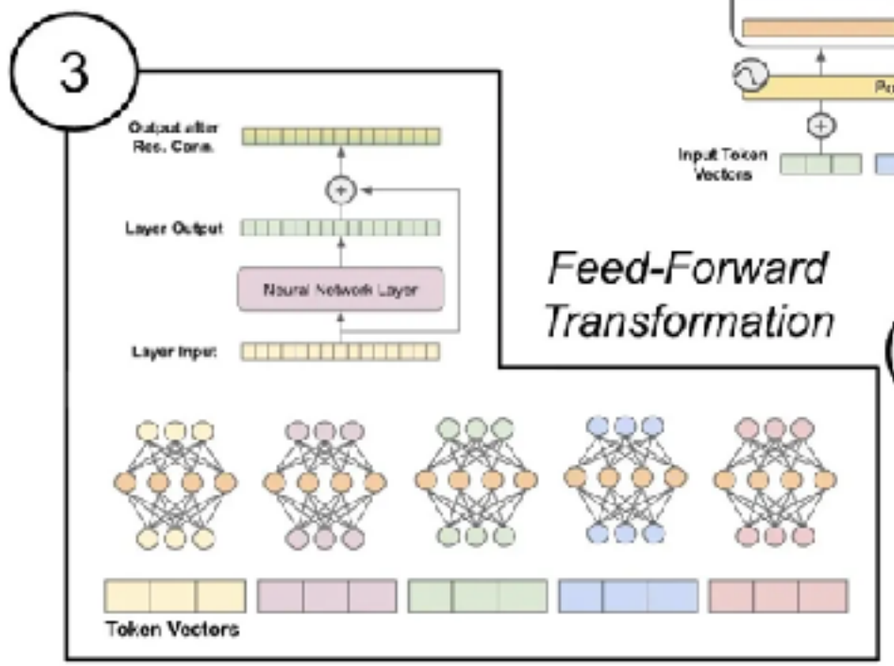
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# LLM components !!

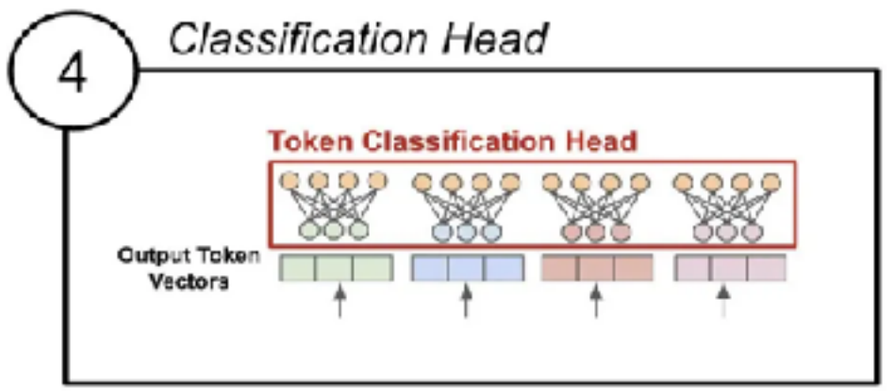
## Components of the Decoder-only Transformer

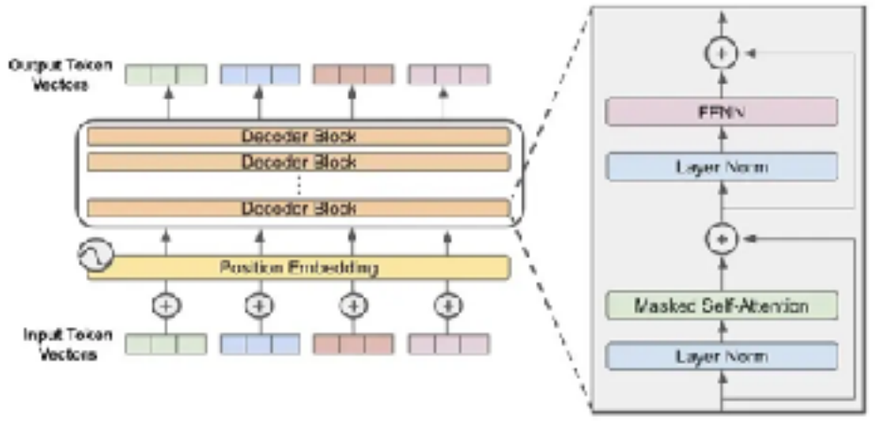



Feed-Forward Transformation



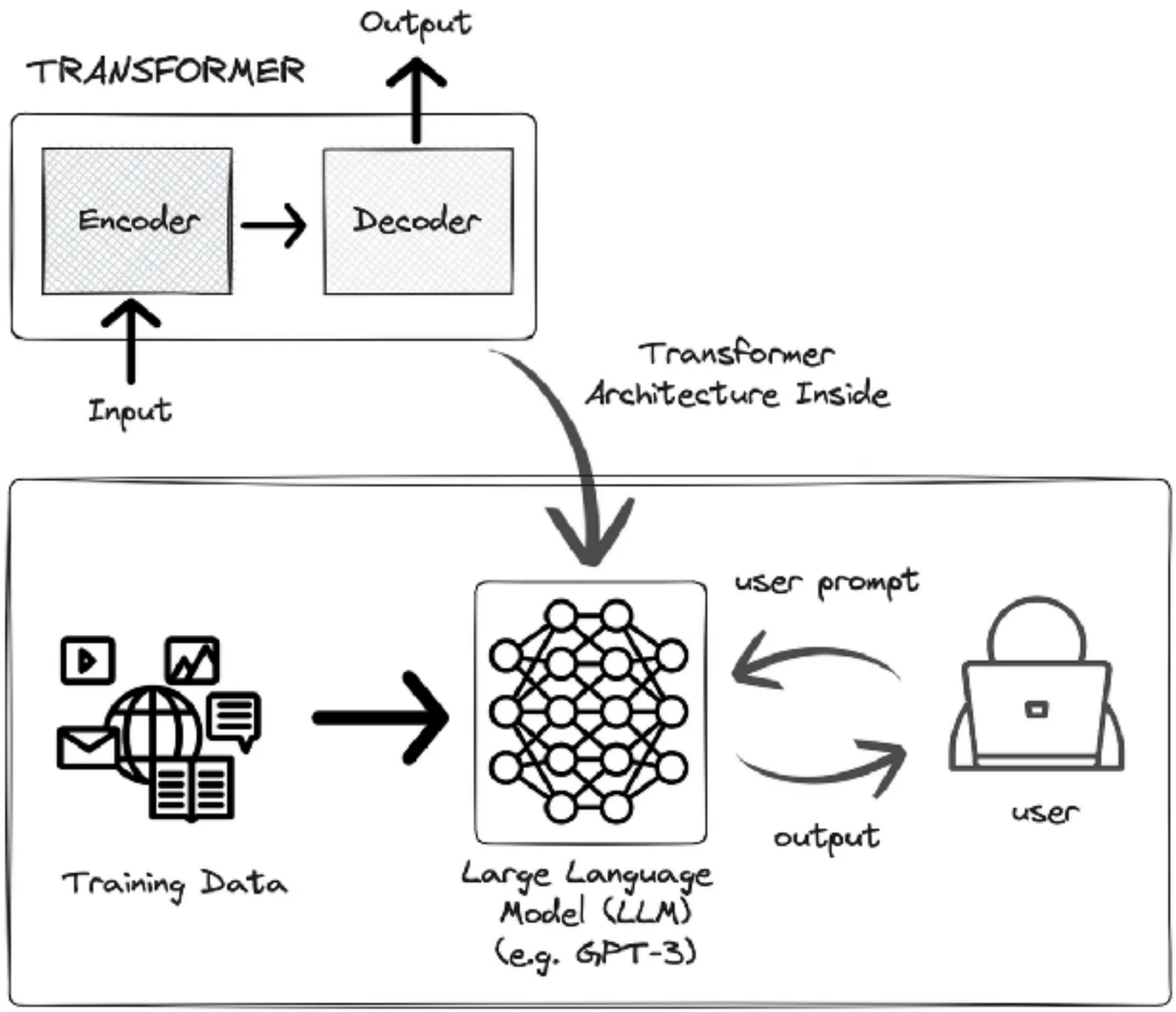




https://stackoverflow.blog/2024/08/22/llms-evolve-quickly-their-underlying-architecture-not-so-much


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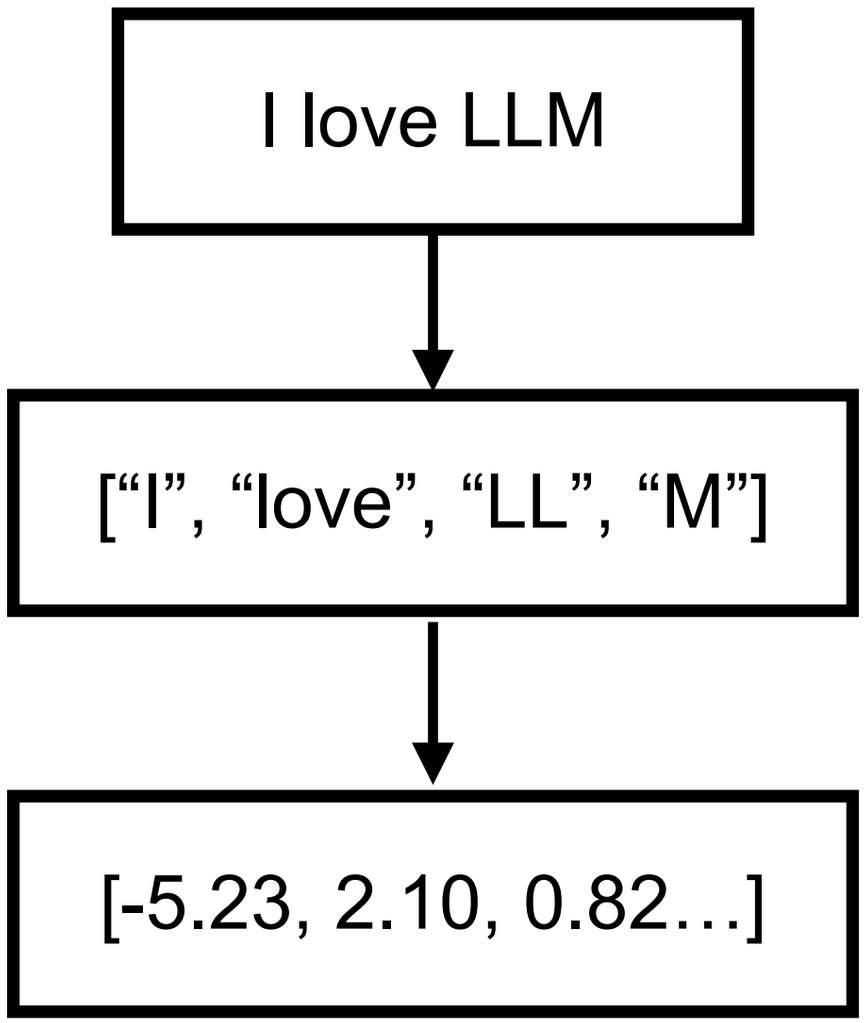
# Transformer inside





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# Transformer process



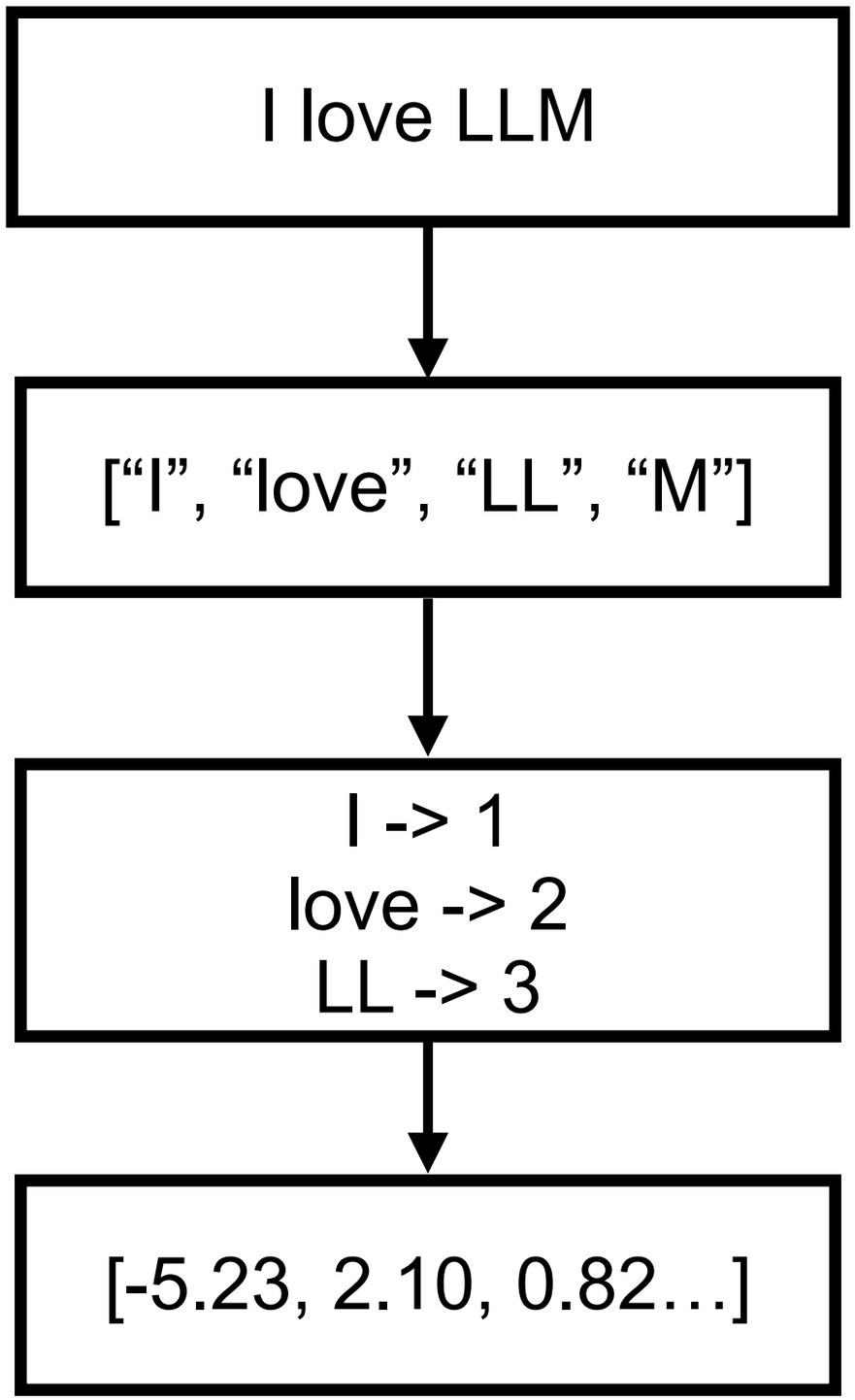

Sentence

Tokens

Embedding


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# Transformer process




Sentence

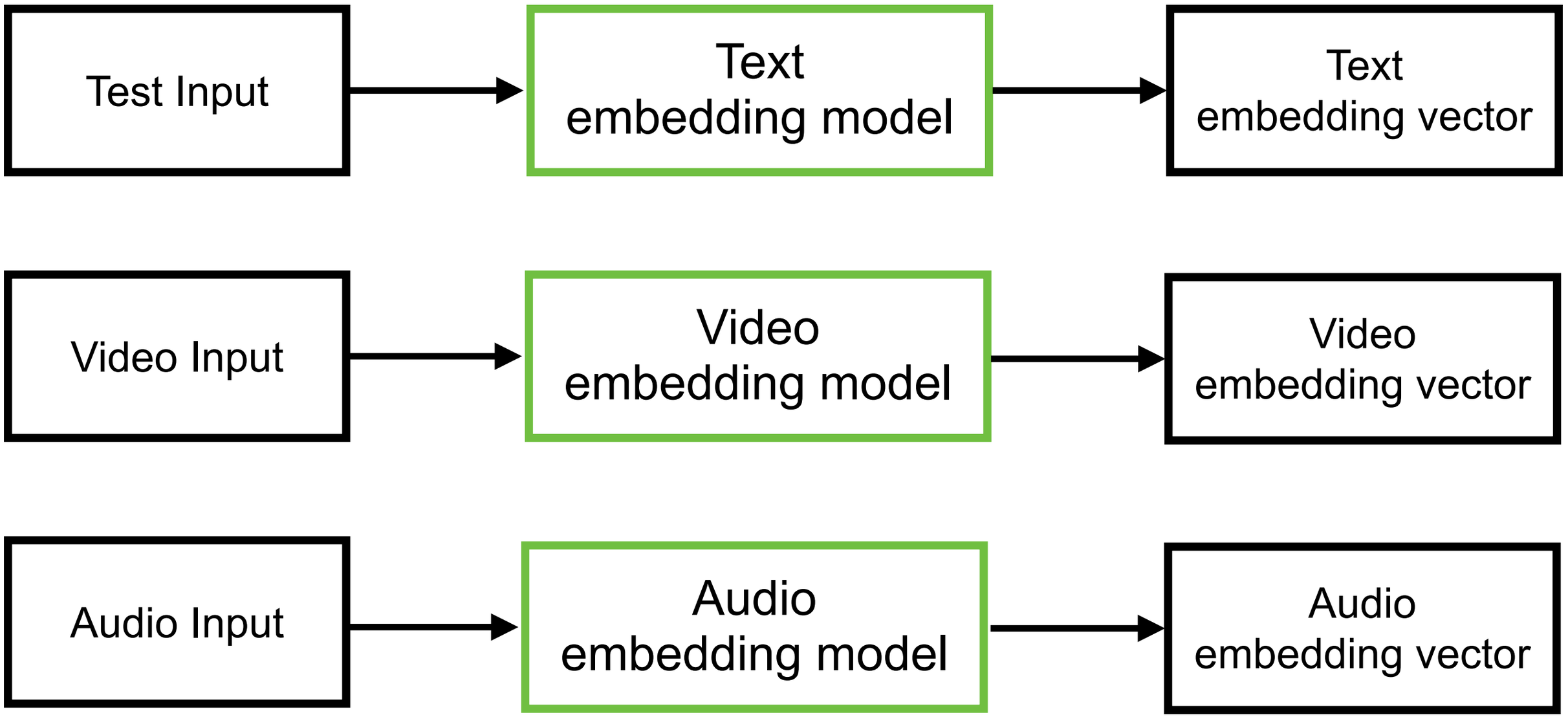
Tokens

Token index

Embedding


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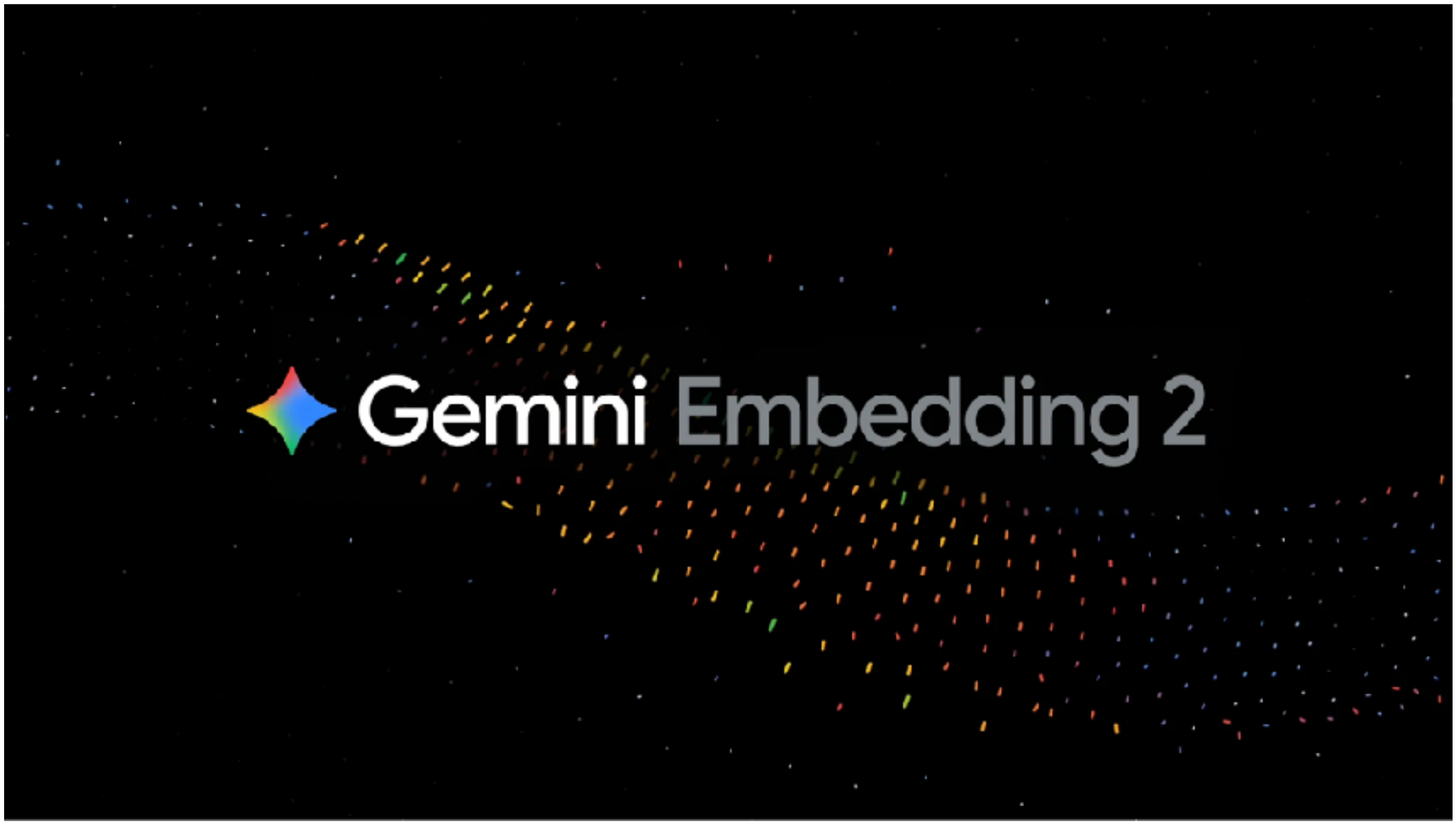
# Embedding





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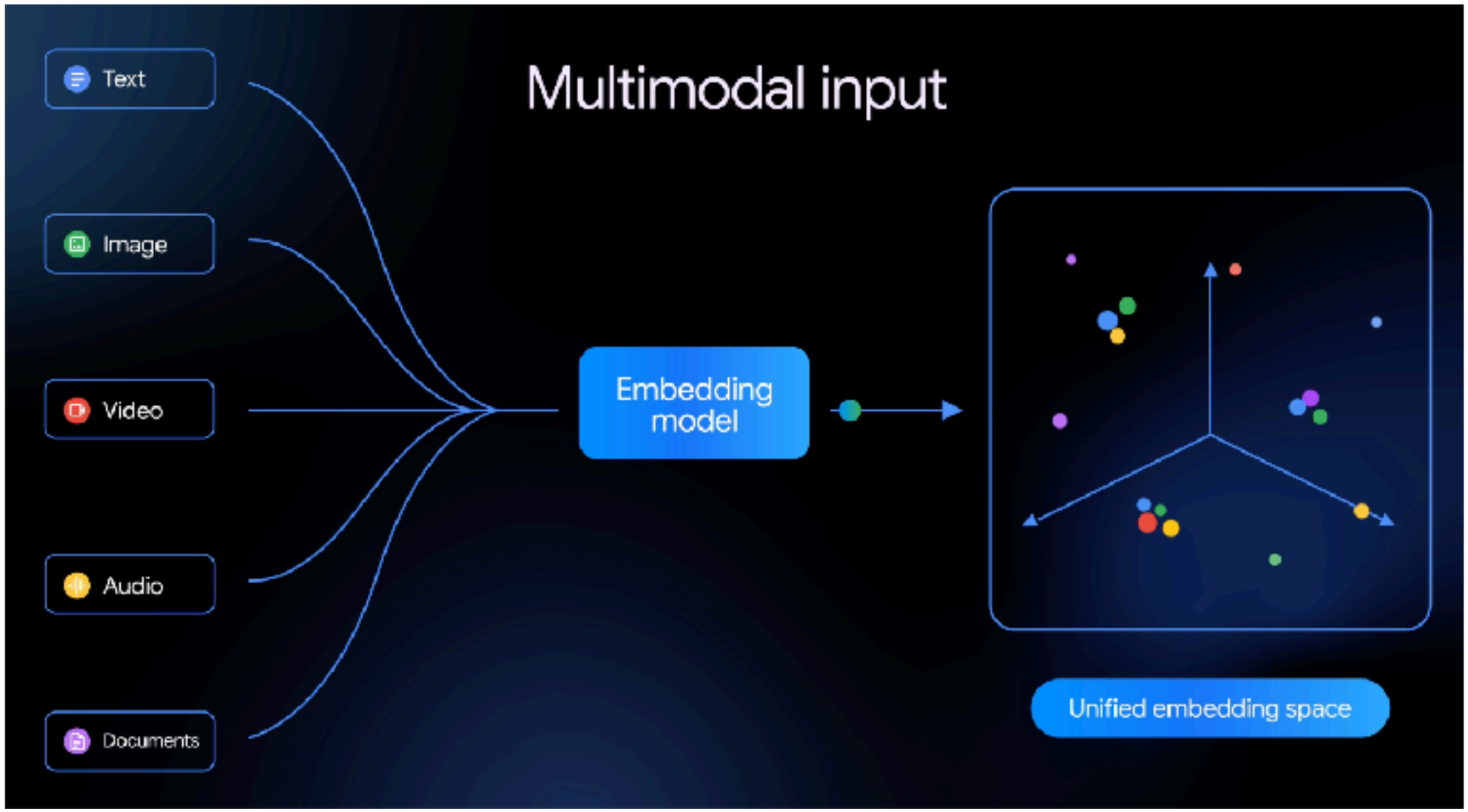
# Multimodal Embedding Model




https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/


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# Multimodal Embedding Model

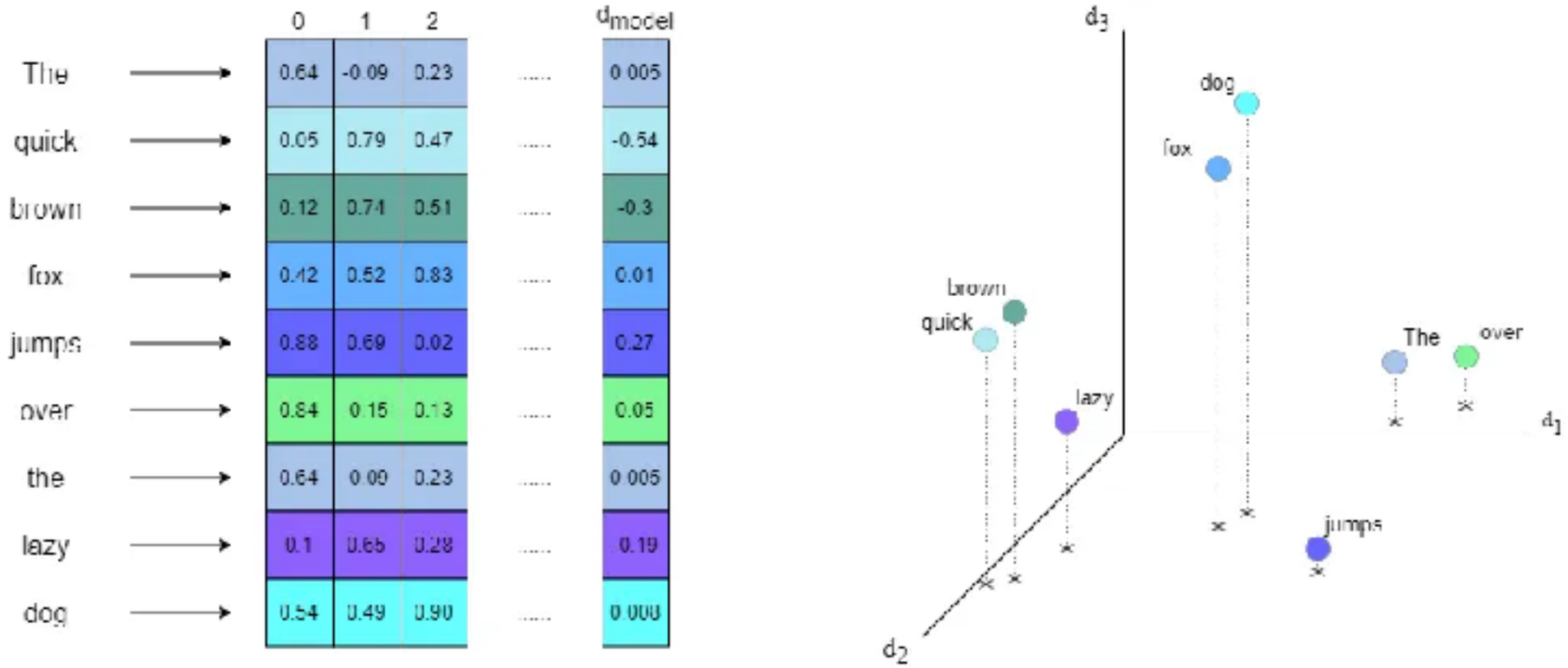



https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/


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# Embedding ?

Map items of **unstructured data** to high-dimensional real vectors




https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca


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# Embedding Leaderboard

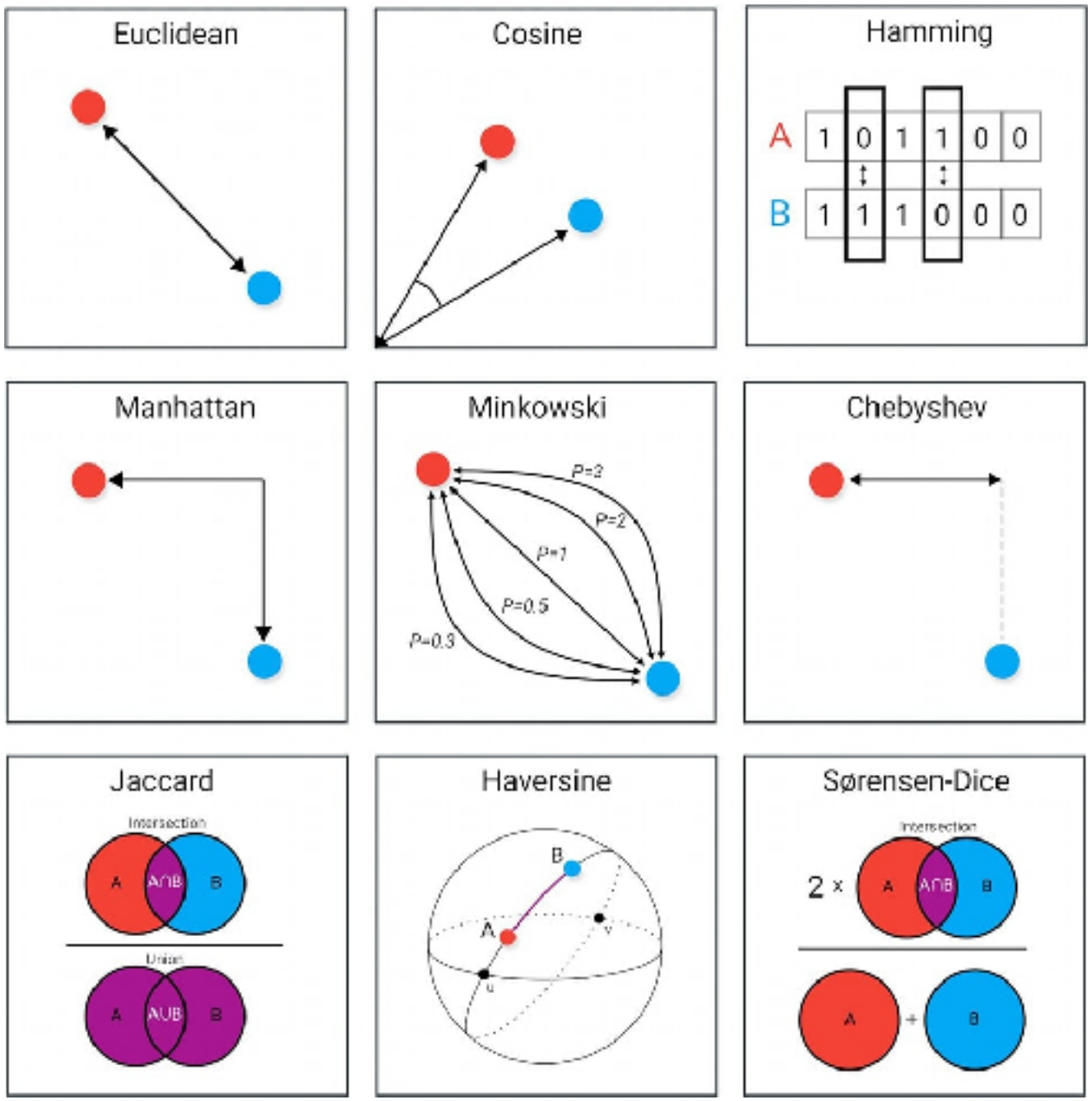
| Rank (Bor… | Model | Embedding D… | Max Tokens | Mean (T… | Mean (TaskT… | Bitext … | Classification | Clustering | Instructi |
|---|---|---|---|---|---|---|---|---|---|
| 1 | KaLM-Embedding-Gemma3-12B-2511 | 3840 | 32768 | **72.32** | **62.51** | **83.76** | **77.88** | 55.77 | 5.49 |
| 2 | llama-embed-nemotron-8b | 4096 | 32768 | 69.46 | 61.09 | 81.72 | 73.21 | 54.35 | 10.82 |
| 3 | Qwen3-Embedding-8B | 4096 | 32768 | 70.58 | 61.69 | 80.89 | 74.00 | 57.65 | 10.06 |
| 4 | gemini-embedding-001 | 3072 | 2048 | 68.37 | 59.59 | 79.28 | 71.82 | 54.59 | 5.18 |
| 5 | Qwen3-Embedding-4B | 2560 | 32768 | 69.45 | 60.86 | 79.36 | 72.33 | 57.15 | 11.56 |
| 6 | Octen-Embedding-8B | 4096 | 32768 | 67.84 | 60.28 | 80.35 | 66.68 | 55.68 | 8.98 |
| 7 | F2LLM-v2-14B | 5120 | 40960 | 68.74 | 59.45 | 77.15 | 73.60 | **60.91** | 0.62 |
| 8 | F2LLM-v2-8B | 4096 | 40960 | 68.09 | 58.99 | 75.96 | 71.93 | 60.62 | 0.85 |
| 9 | Seed1.6-embedding-1215 | 2048 | 32768 | 70.26 | 61.34 | 78.68 | 76.75 | 56.78 | -0.82 |
| 10 | F2LLM-v2-4B | 2560 | 40960 | 67.06 | 58.25 | 74.49 | 70.73 | 59.53 | 2.25 |
| 11 | jina-embeddings-v5-text-small | 1024 | 32768 | 67.00 | 58.90 | 69.71 | 71.32 | 53.41 | 1.35 |

https://huggingface.co/spaces/mteb/leaderboard




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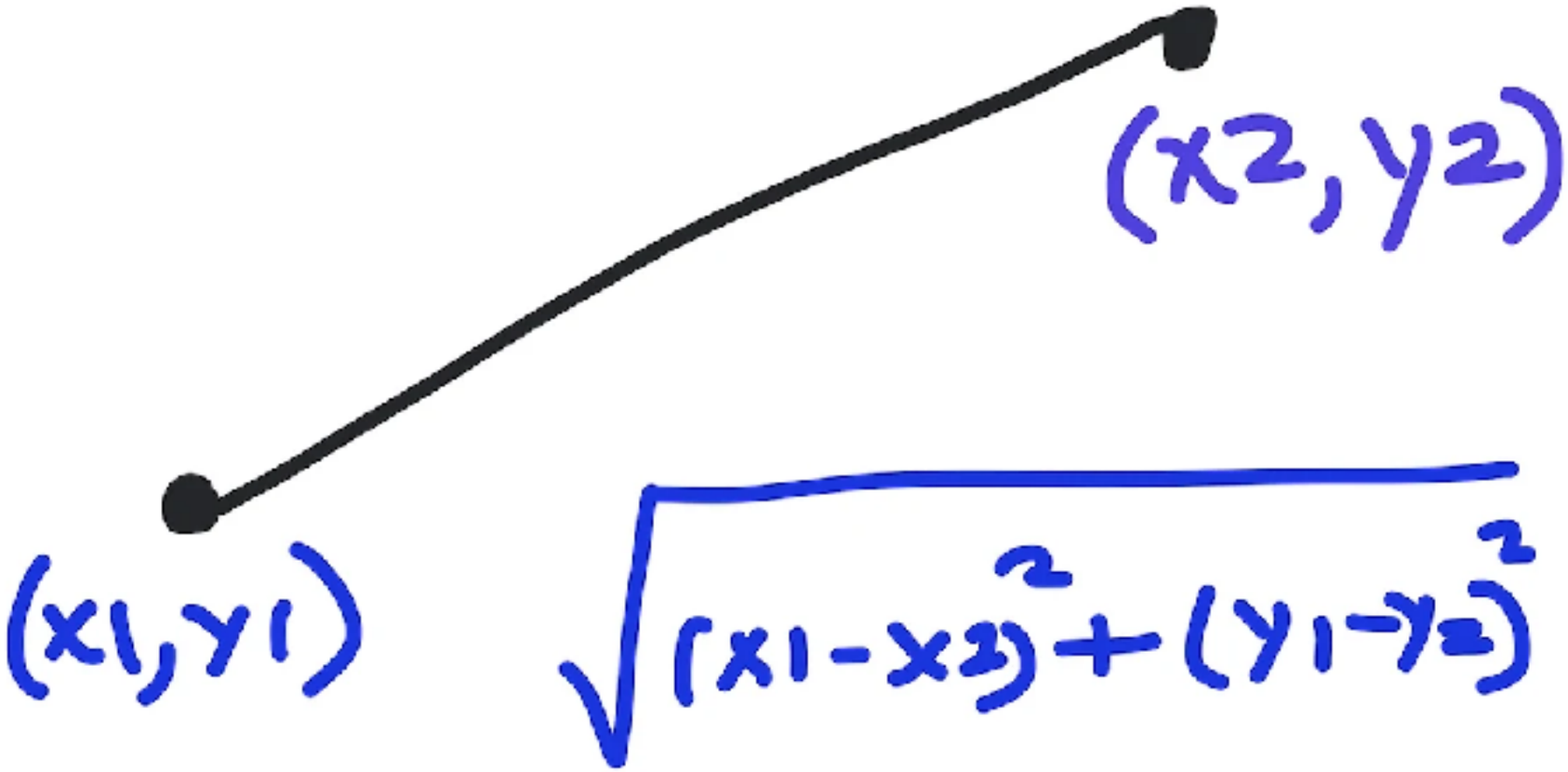
# Distance measure in Data Science




https://www.maartengrootendorst.com/blog/distances/


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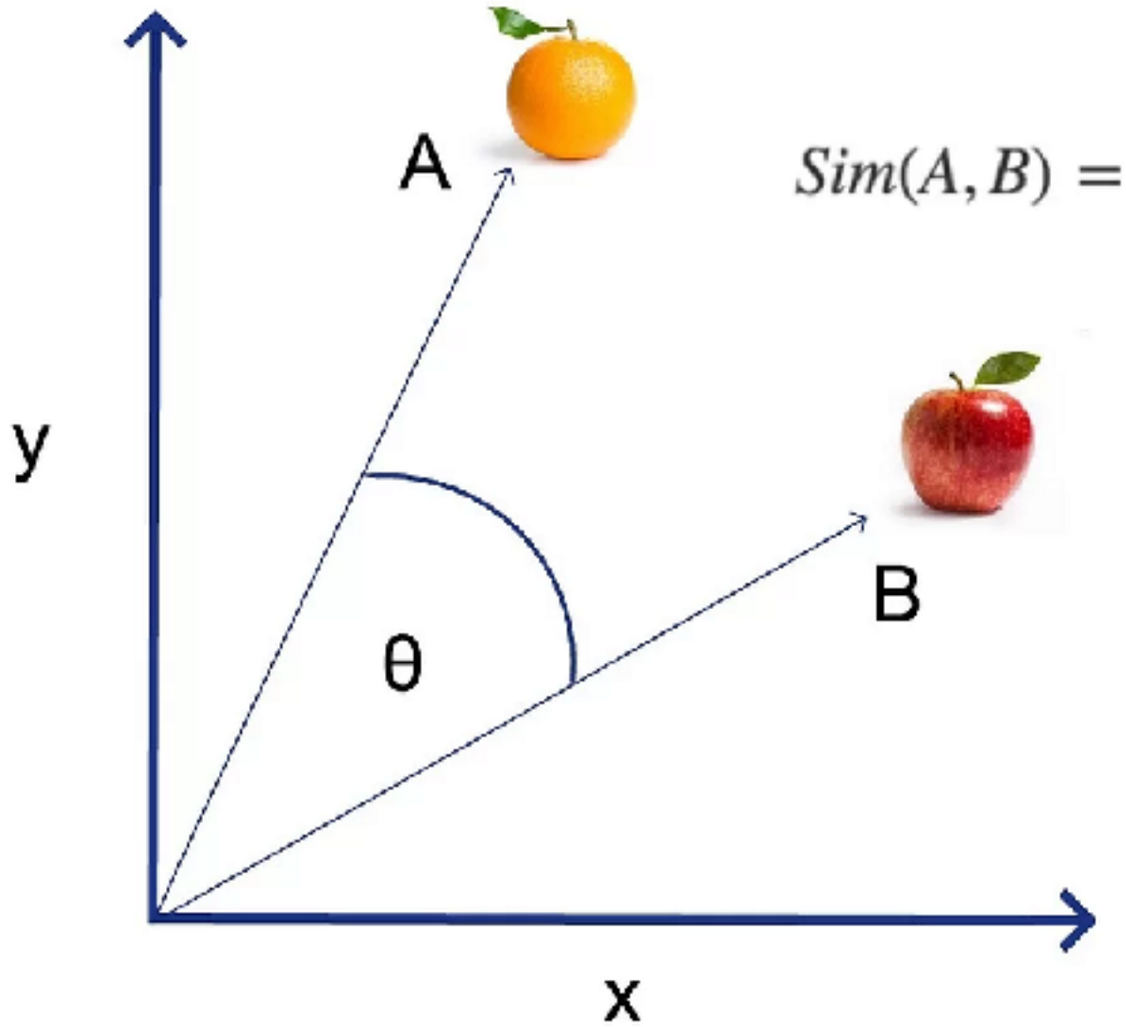
# Euclidean Distance





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# Cosine Similarity

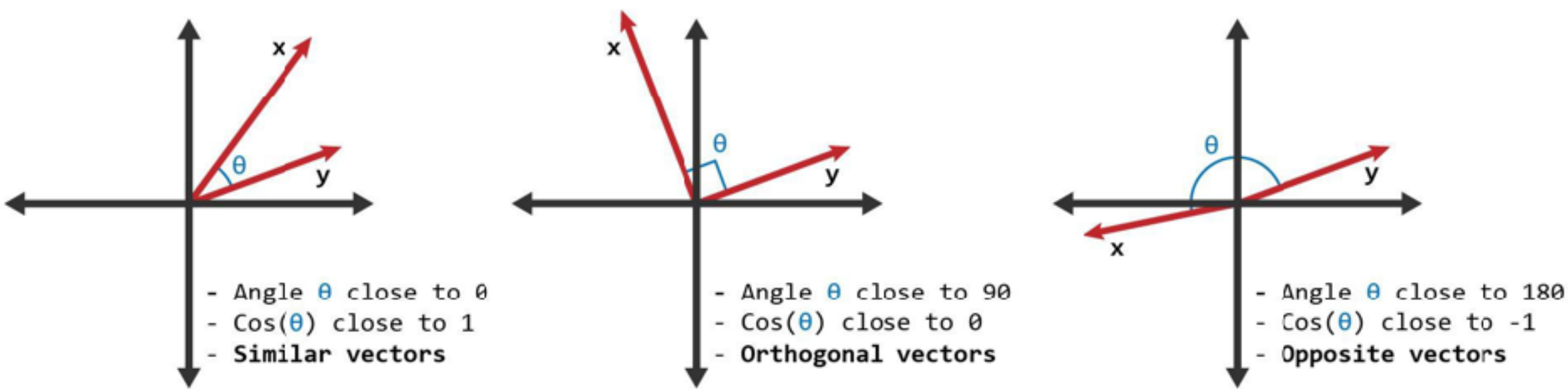



$$Sim(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

https://python.plainenglish.io/mastering-cosine-metrics-a-data-scientists-essential-toolkit-3ce95eda91f9


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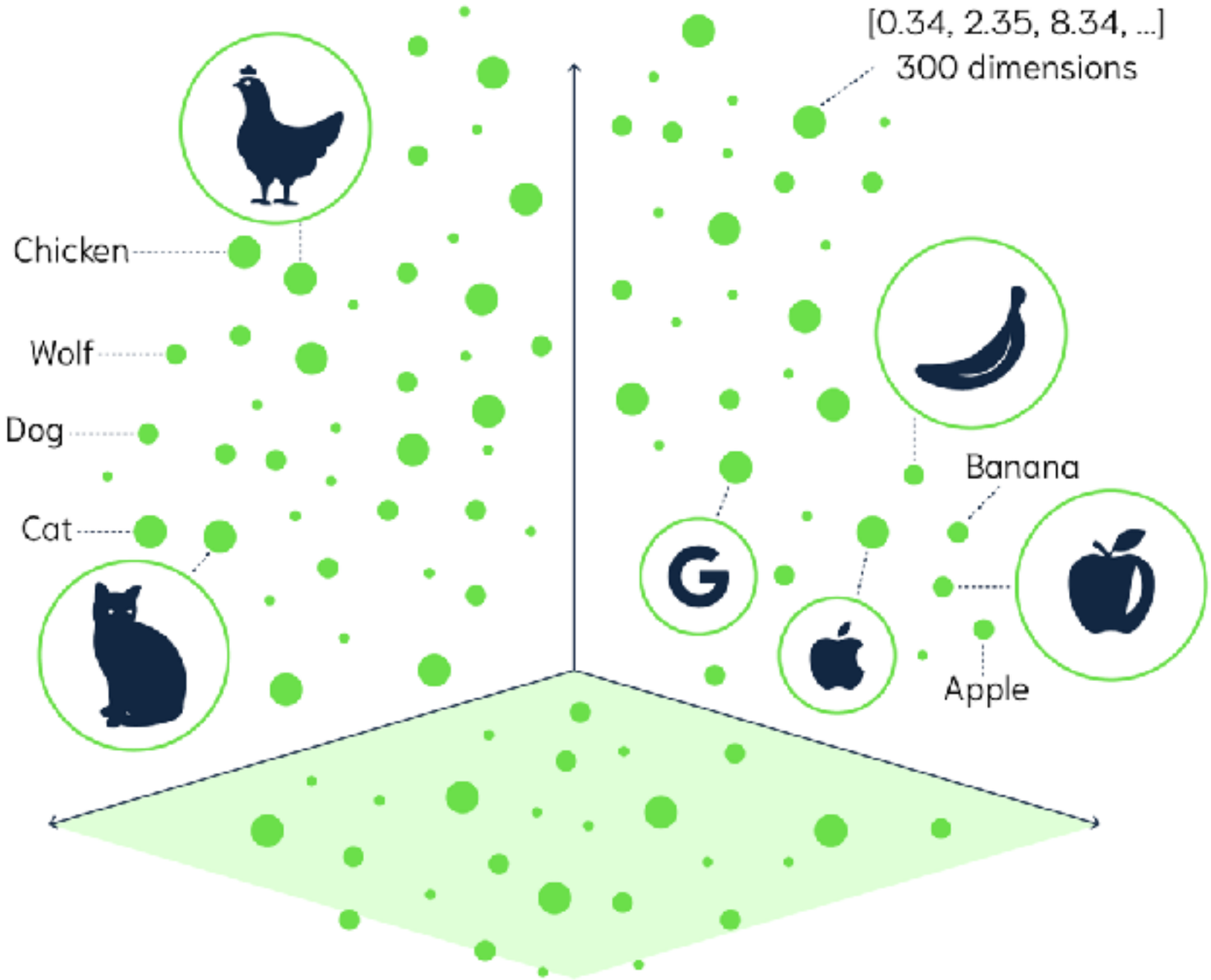
# Cosign Similarity




https://www.learndatasci.com/glossary/cosine-similarity/


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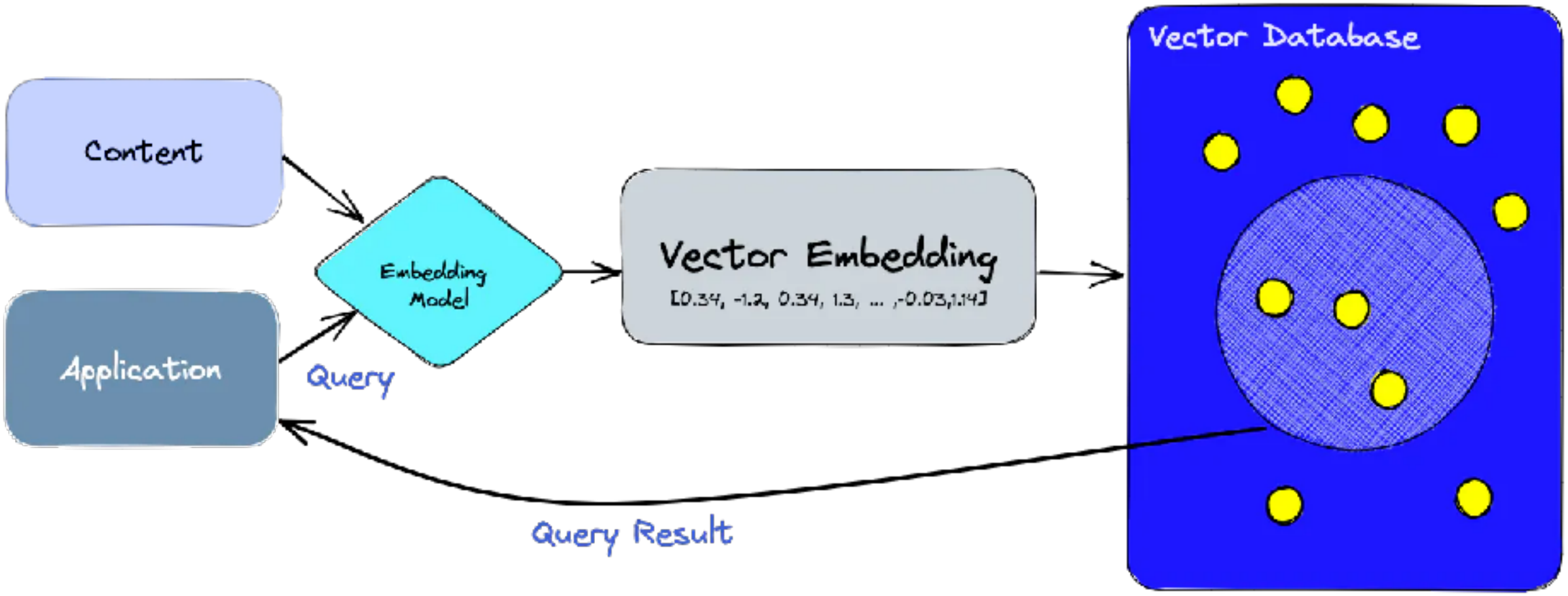
# Visual of Vector space





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# Store data in Vector Database

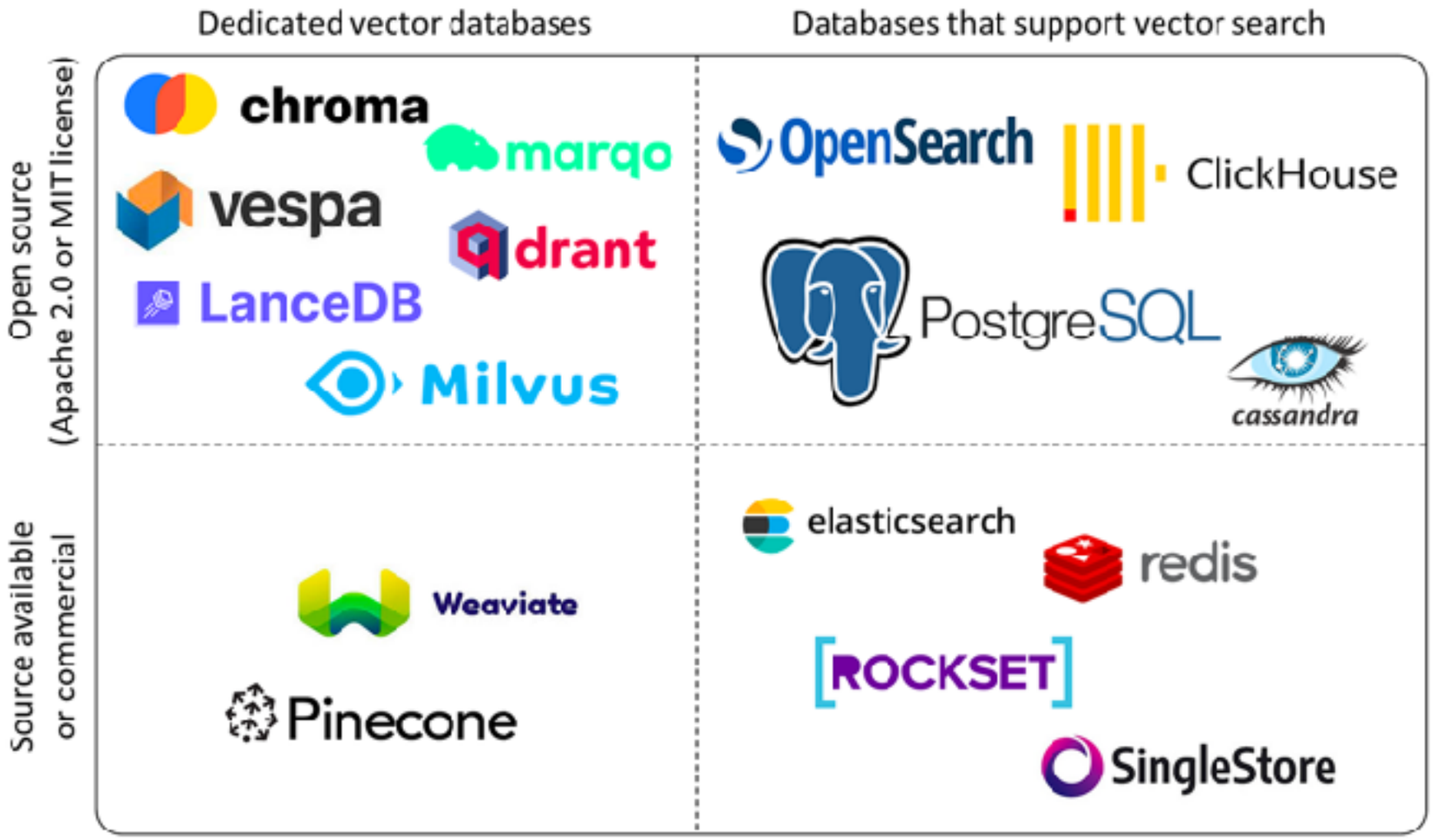




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# Vector Database ?

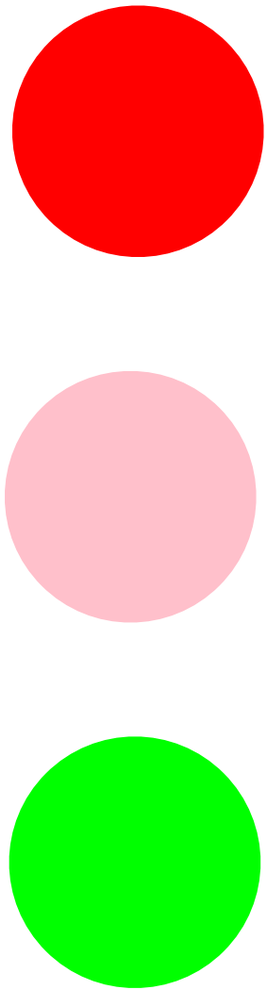
Map items of unstructured data to high-dimensional real **vectors**




https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca


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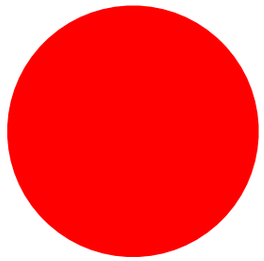
# RGB ?



https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb


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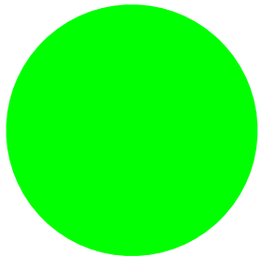
# RGB ?



[255, 0, 0]



[255, 192, 203]



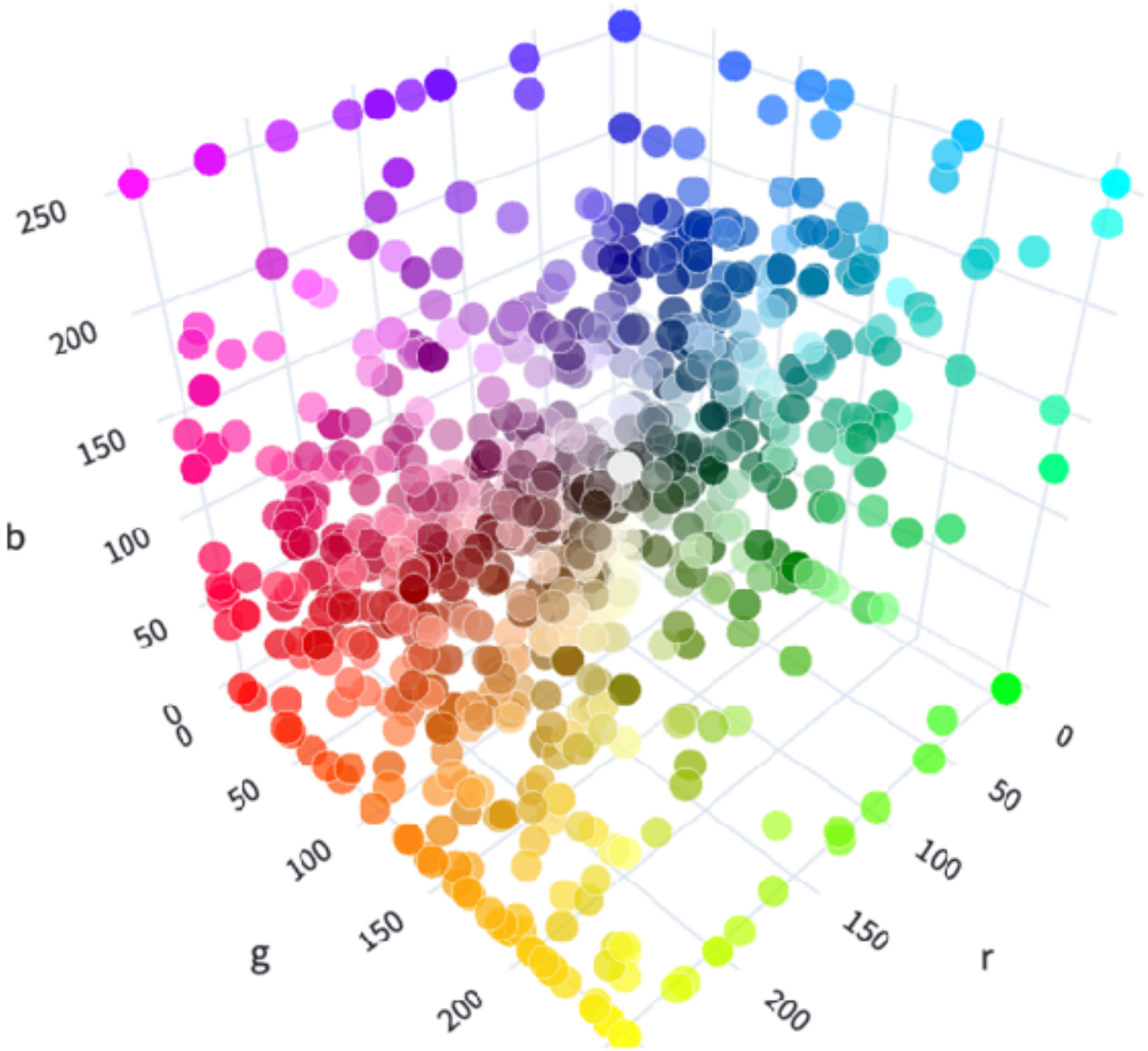
[0, 255, 0]

https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb



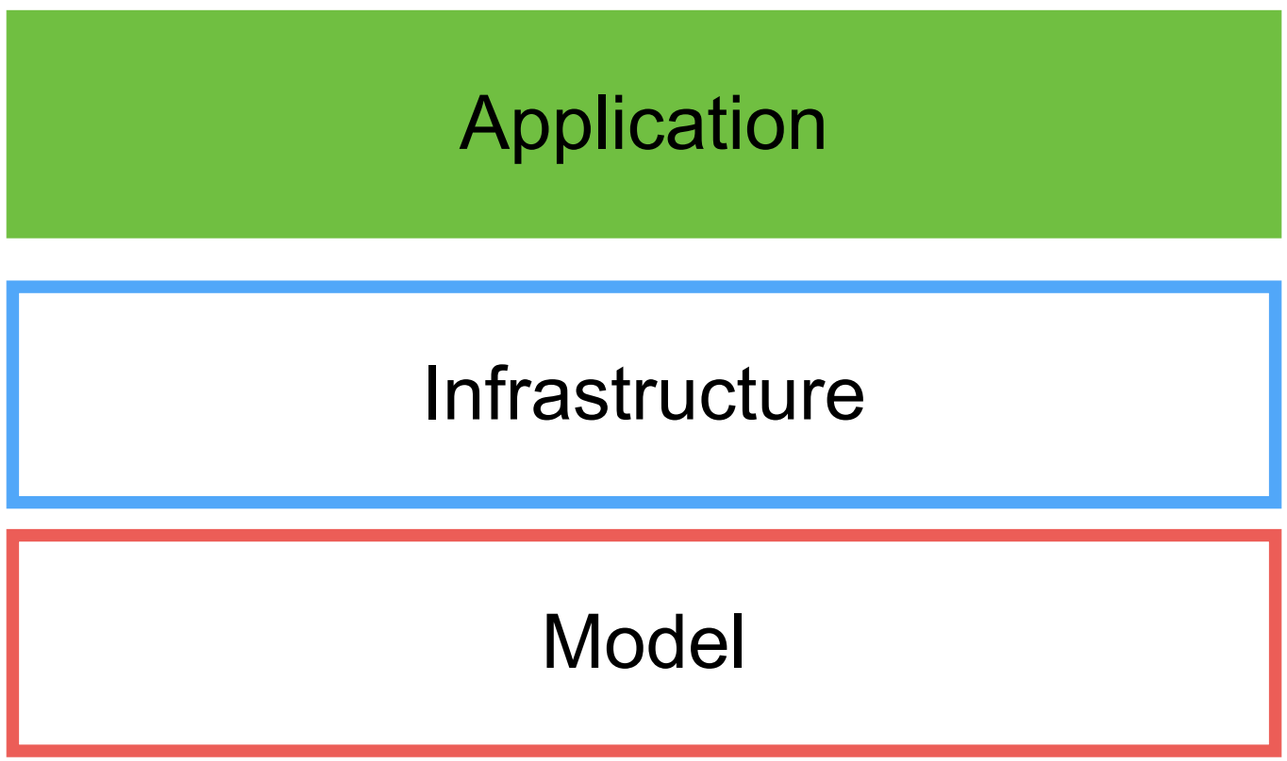

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# Visualize RGB



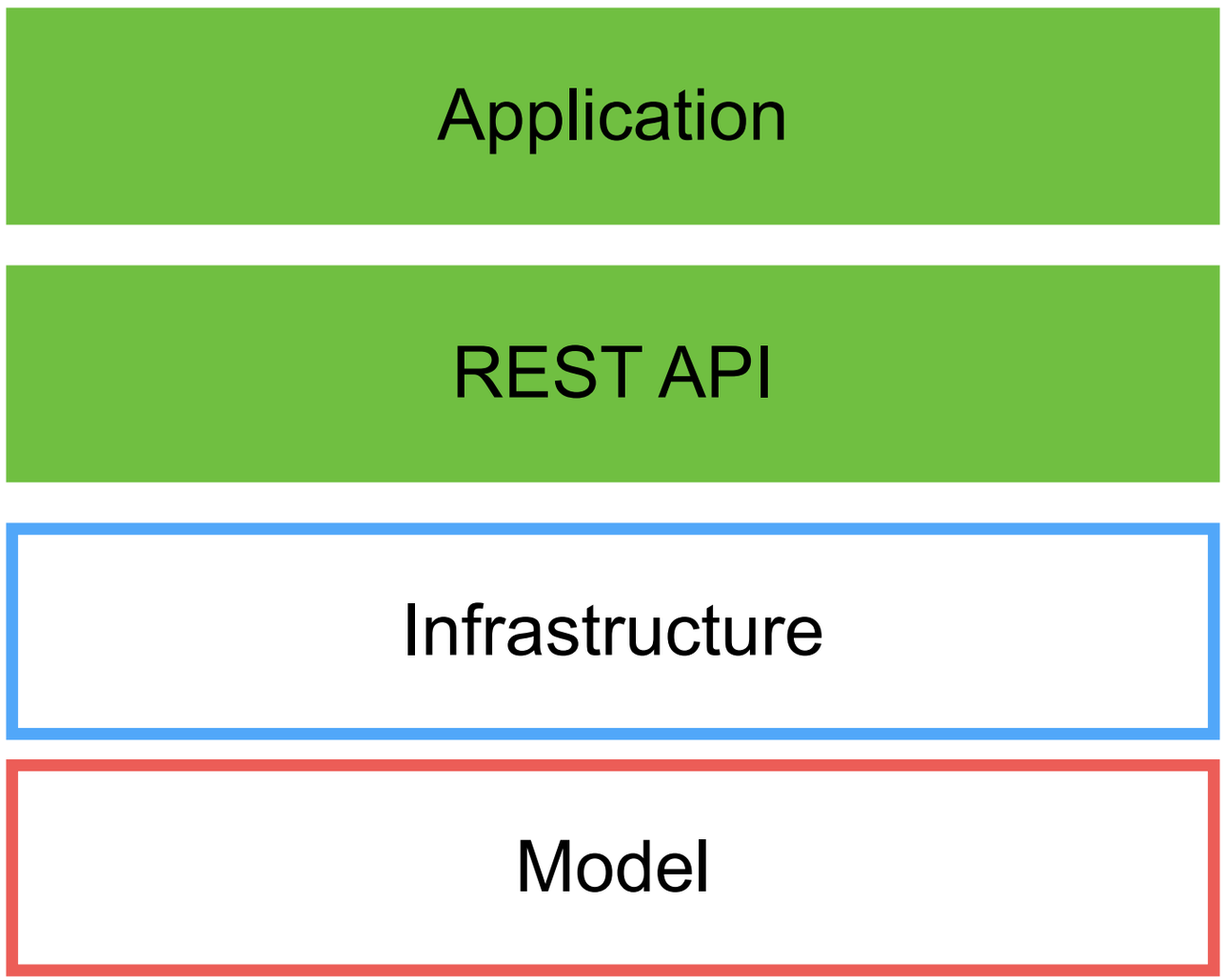

https://huggingface.co/spaces/jphwang/colorful_vectors


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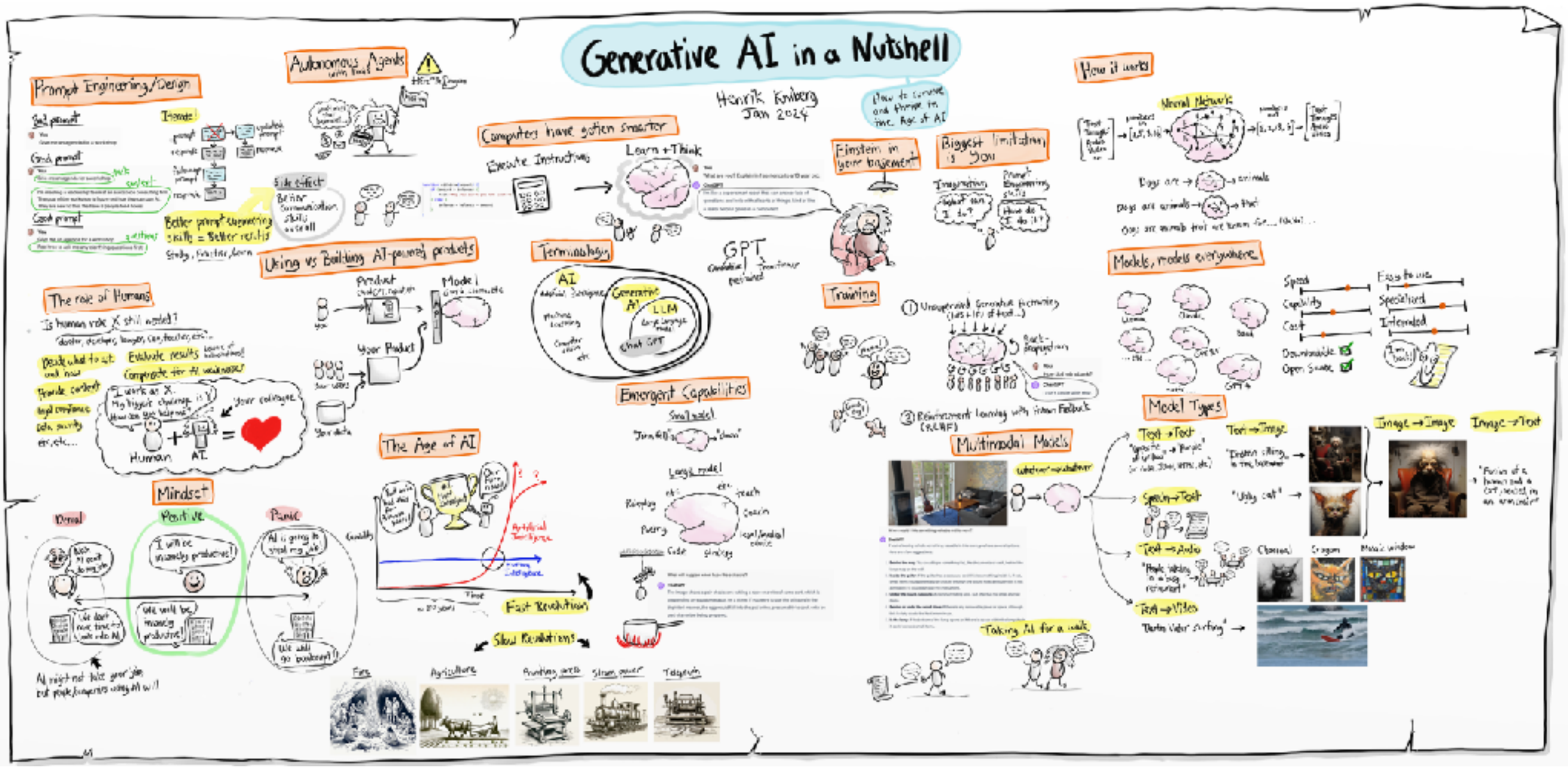
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# Generative AI in Nutshell




https://www.youtube.com/watch?v=2IK3DFHRFfw

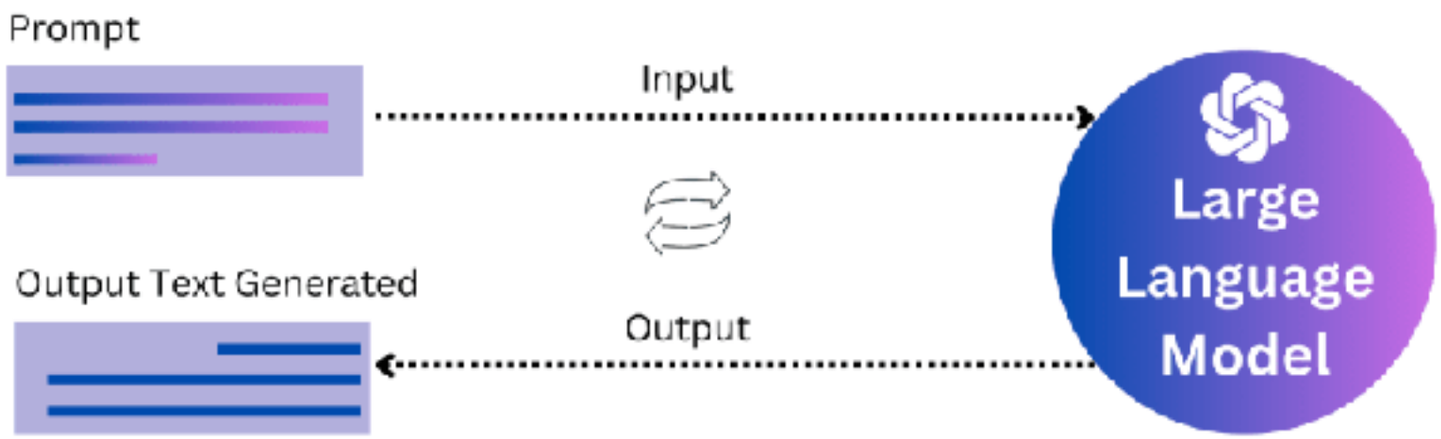

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# Prompt Engineering


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# Prompt composition

Prompt engineering
Compose prompts from user inputs and context




https://platform.openai.com/docs/guides/prompt-engineering


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# Better Prompt

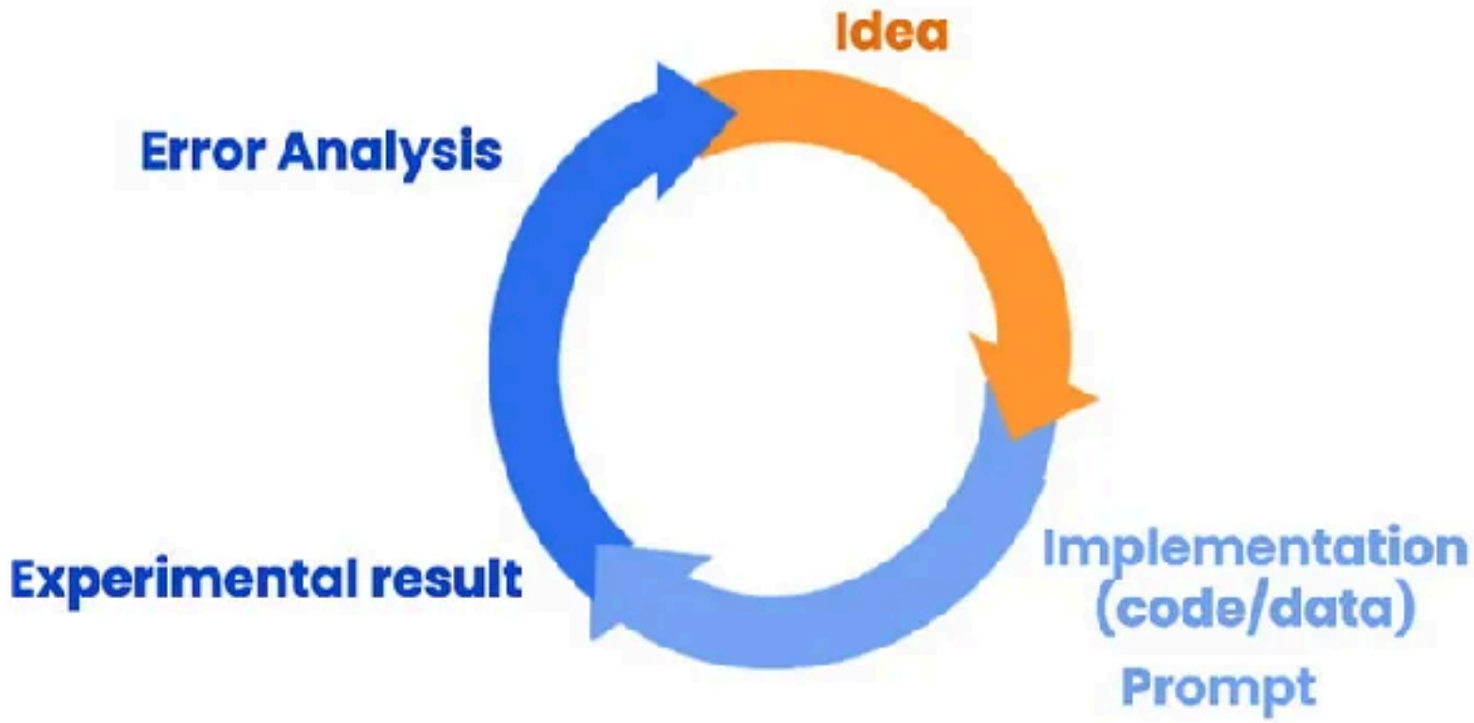
Write clear instructions
Provide reference text
Split complex tasks into simpler subtasks
Give the model time to think
Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering




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# Iterative Prompt Development



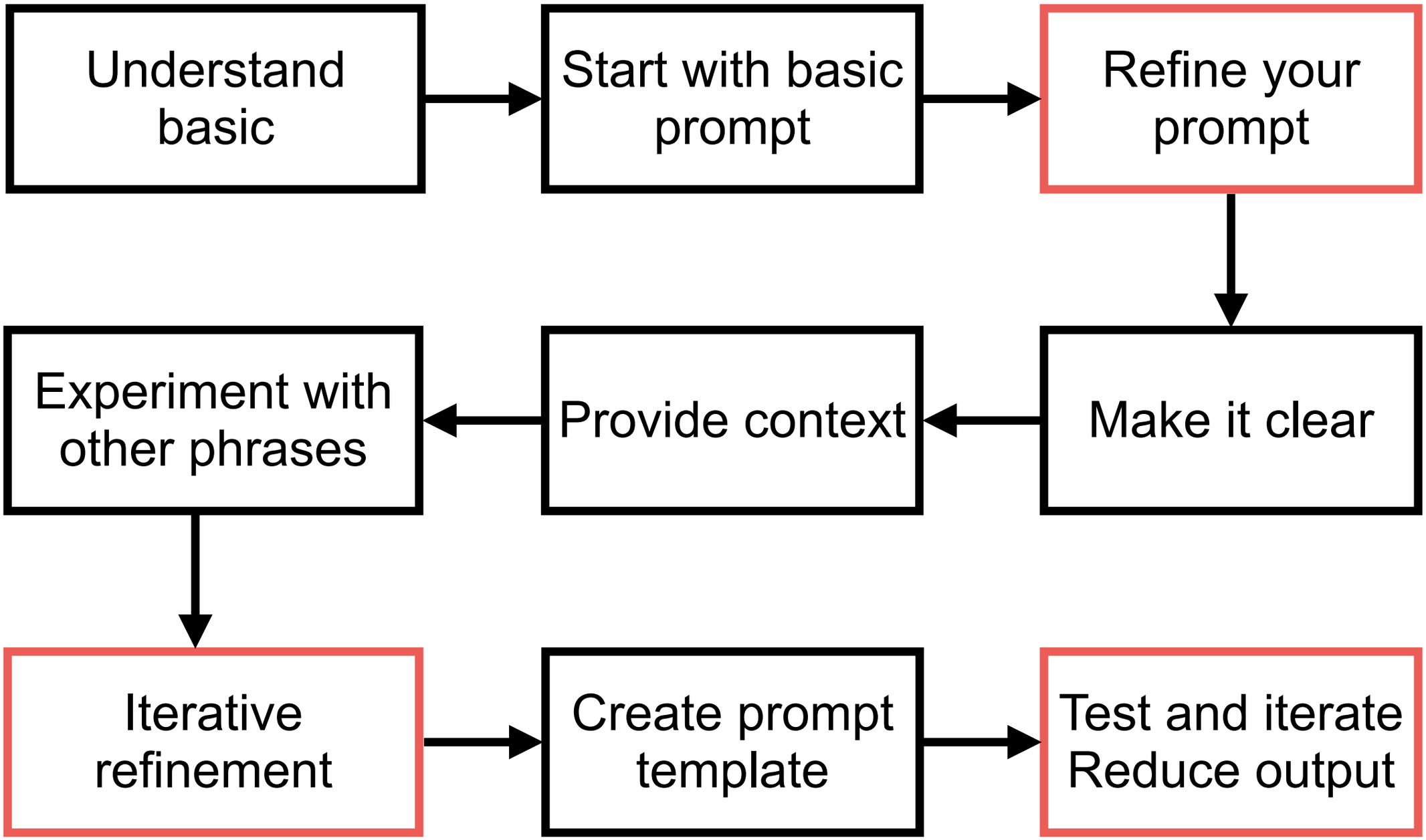

Iterative Process

- Try something
- Analyze where the result does not give what you want
- Clarify instructions, give more time to think
- Refine prompts with a batch of examples

https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/


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# Basic of Prompt Engineer





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# Better Prompt

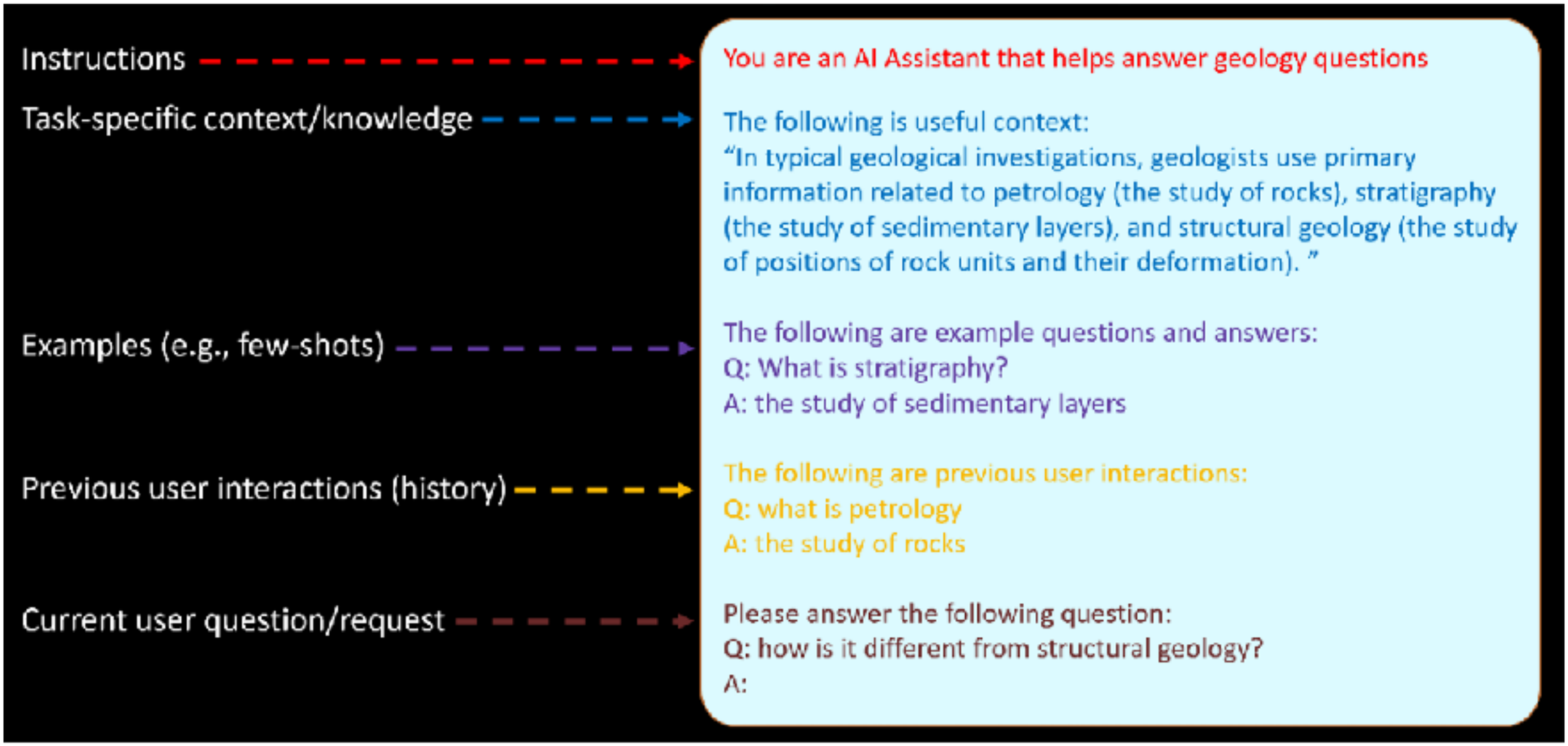
Write clear instructions
Provide reference text
Split complex tasks into simpler subtasks
Give the model time to think
Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering/six-strategies-for-getting-better-results




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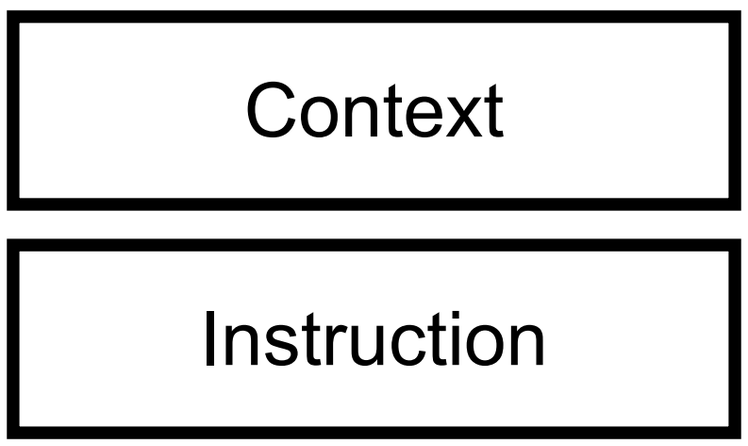
# Prompt Structure




https://learn.microsoft.com/en-us/ai/playbook/technology-guidance/generative-ai/working-with-llms/prompt-engineering


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# Structure of Prompt




https://platform.openai.com/docs/guides/prompt-engineering


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# Prompting Guide

Prompt Engineering

## Prompt Engineering Guide

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning. Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Prompt engineering is not just about designing and developing prompts. It encompasses a wide range of skills and techniques that are useful for interacting and developing with LLMs. It's an important skill to interface, build with, and understand capabilities of LLMs. You can use prompt engineering to improve safety of LLMs and build new capabilities like augmenting LLMs with domain knowledge and external tools.

https://www.promptingguide.ai/




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# Structure of Prompt !!

**APE**
Action, Purpose, Expectation

**RACE**
Role, Action, Context, Expectation

**TAG**
Task, Action, Goal

**COAST**
Context, Objective, Action, Scenario, Task

**RISE**
Role, Input, Step, Expectation

**RCCF**
Role, Context, Constraint, Format

https://twitter.com/pradeepeth/status/1673271866696544257




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# Prompt Techniques

Zero-shot

Chain-of Thought (CoT)

Few-shot

Meta or structure

https://www.promptingguide.ai/techniques


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# Chain of Thought Prompting (CoT)

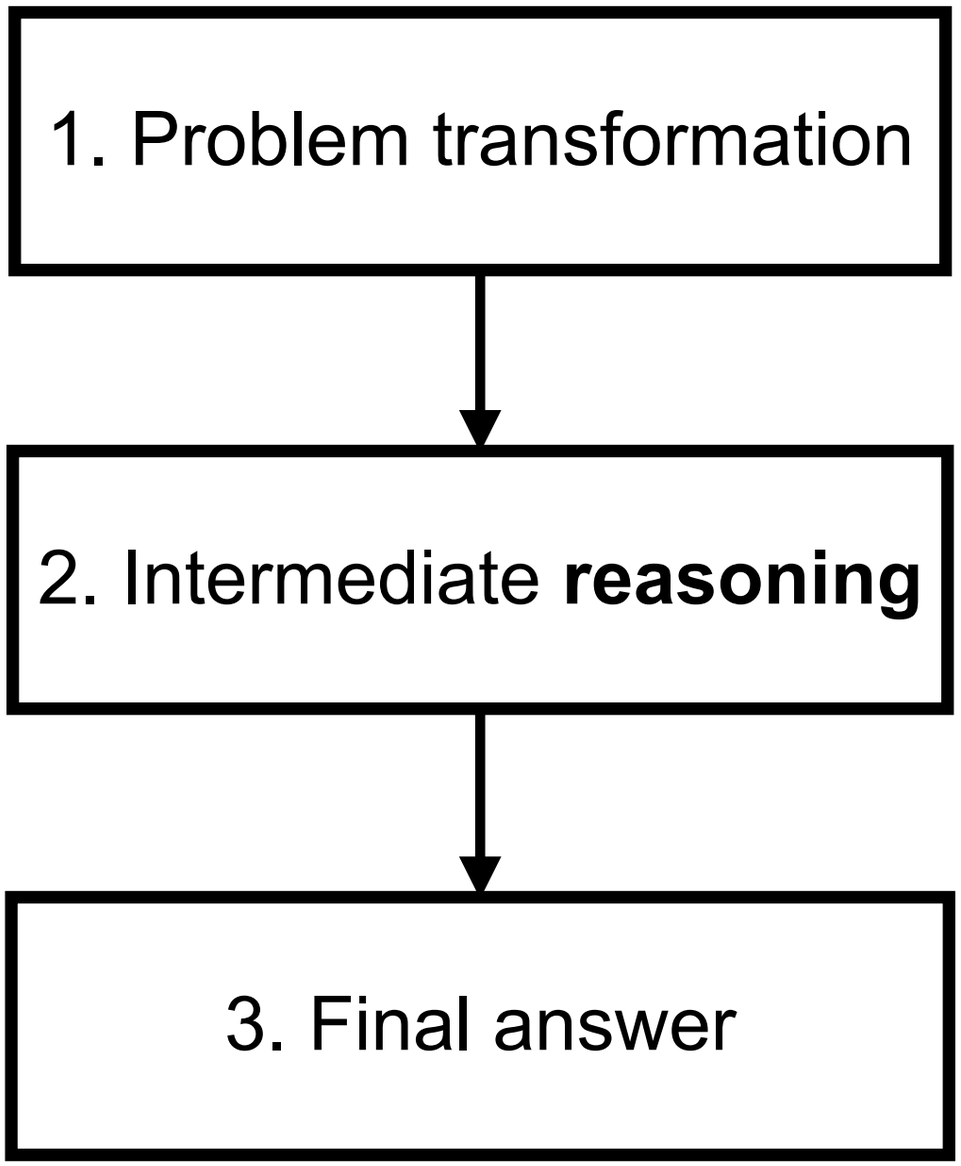
Technique used to improve the reasoning ability of LLM

Try to break down a complex problem into smaller, More manageable steps, lead to final answer

Reasoning model !!


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# Chain of Thought Prompting (CoT)




eg. step-by-step


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# Chain of Thought Prompting (CoT)

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

---

*(Output) The answer is 8.* ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

---

*(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4.* ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

---

*(Output) 8* ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

---

*(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls.* ✓

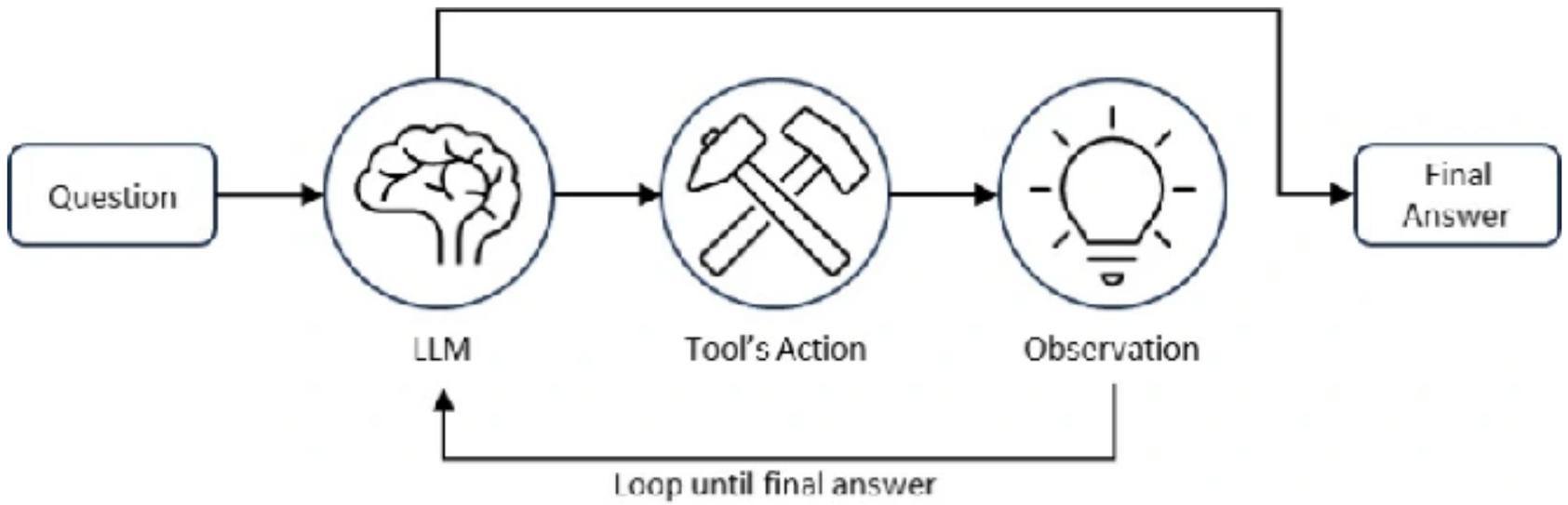
https://www.promptingguide.ai/techniques/cot




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# ReAct Prompt

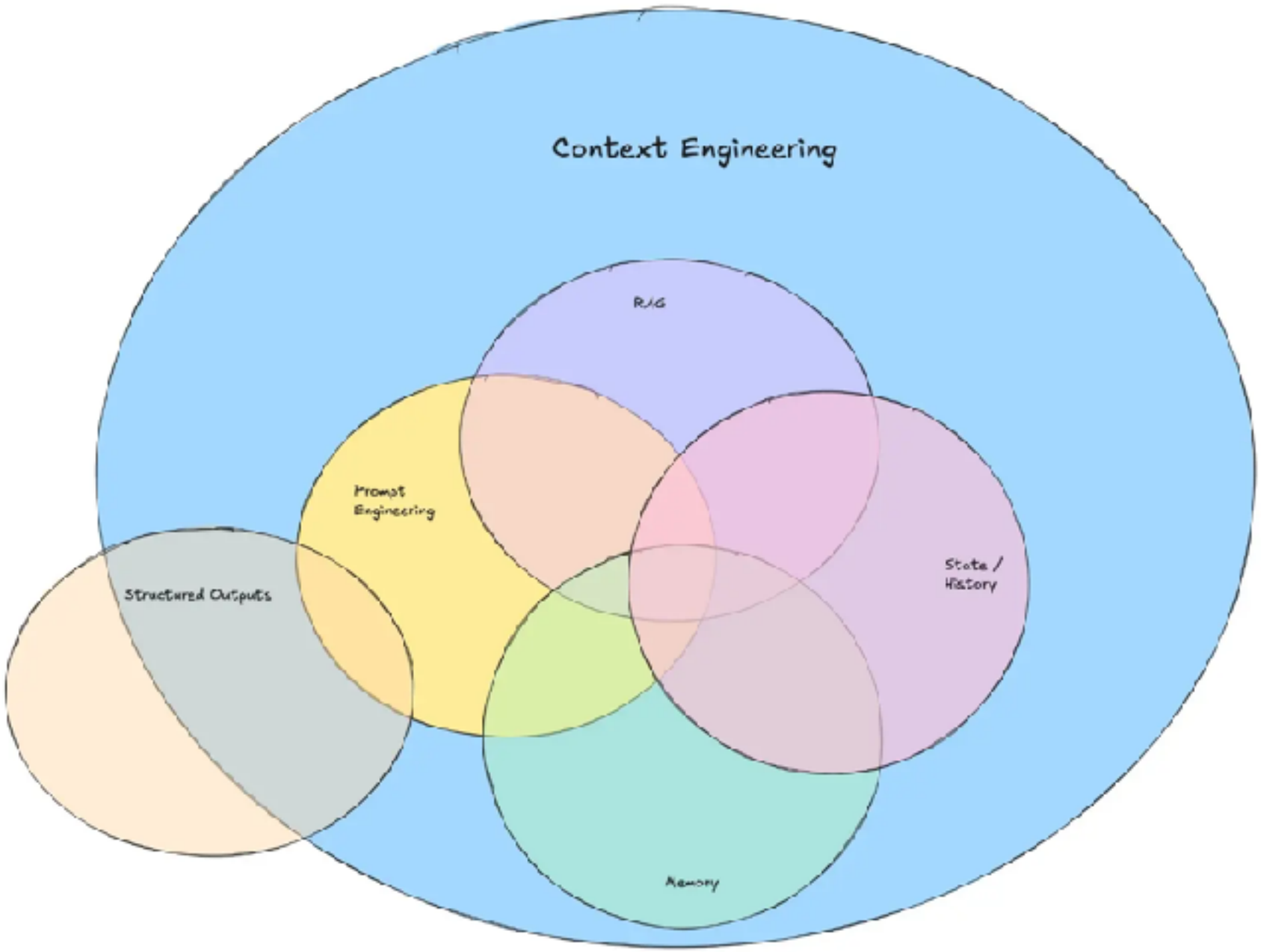
LLM reasoning and additional tools (expert)
Improve better answer




https://www.promptingguide.ai/techniques/react


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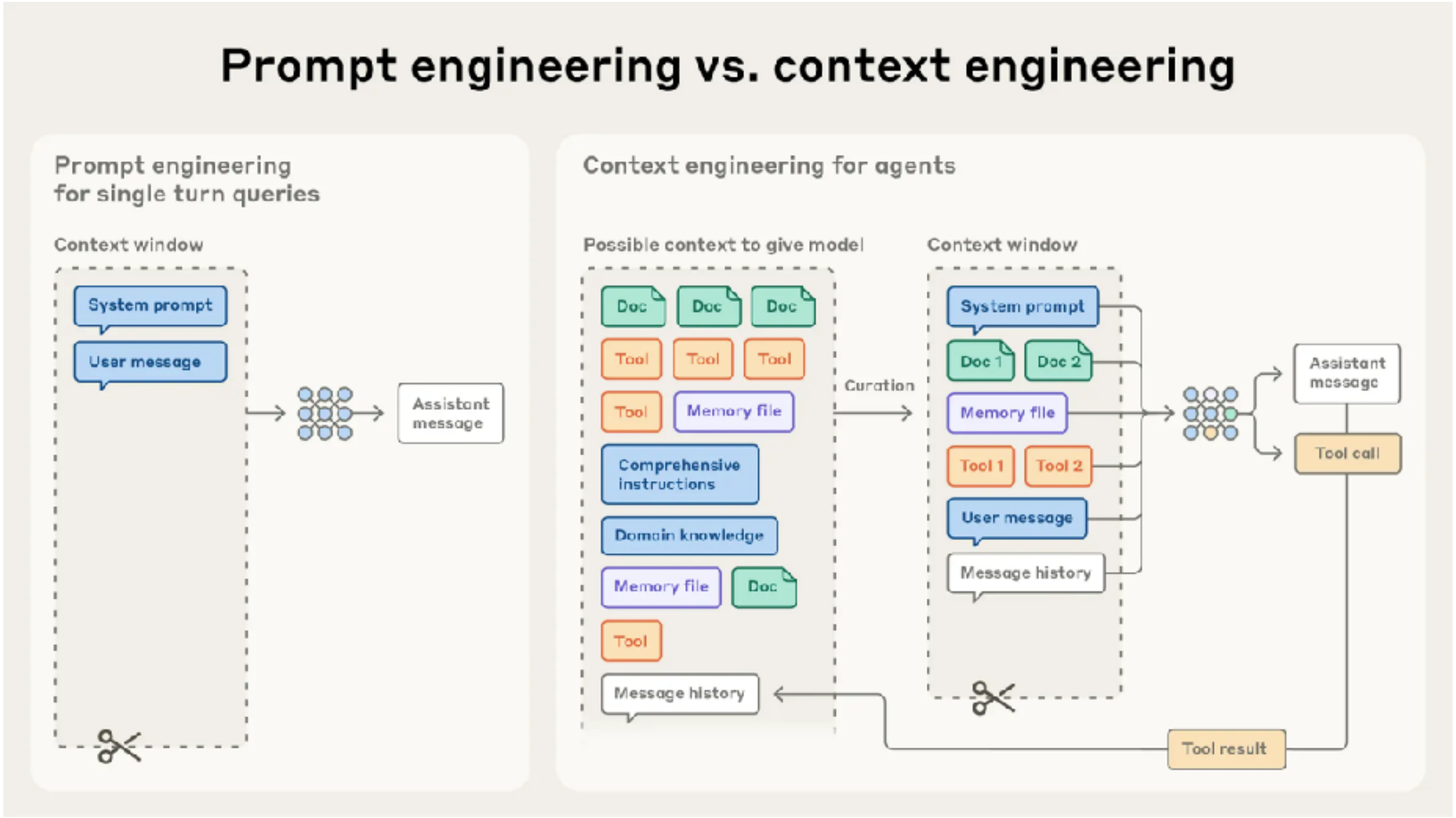
# Context Engineering




https://www.promptingguide.ai/guides/context-engineering-guide

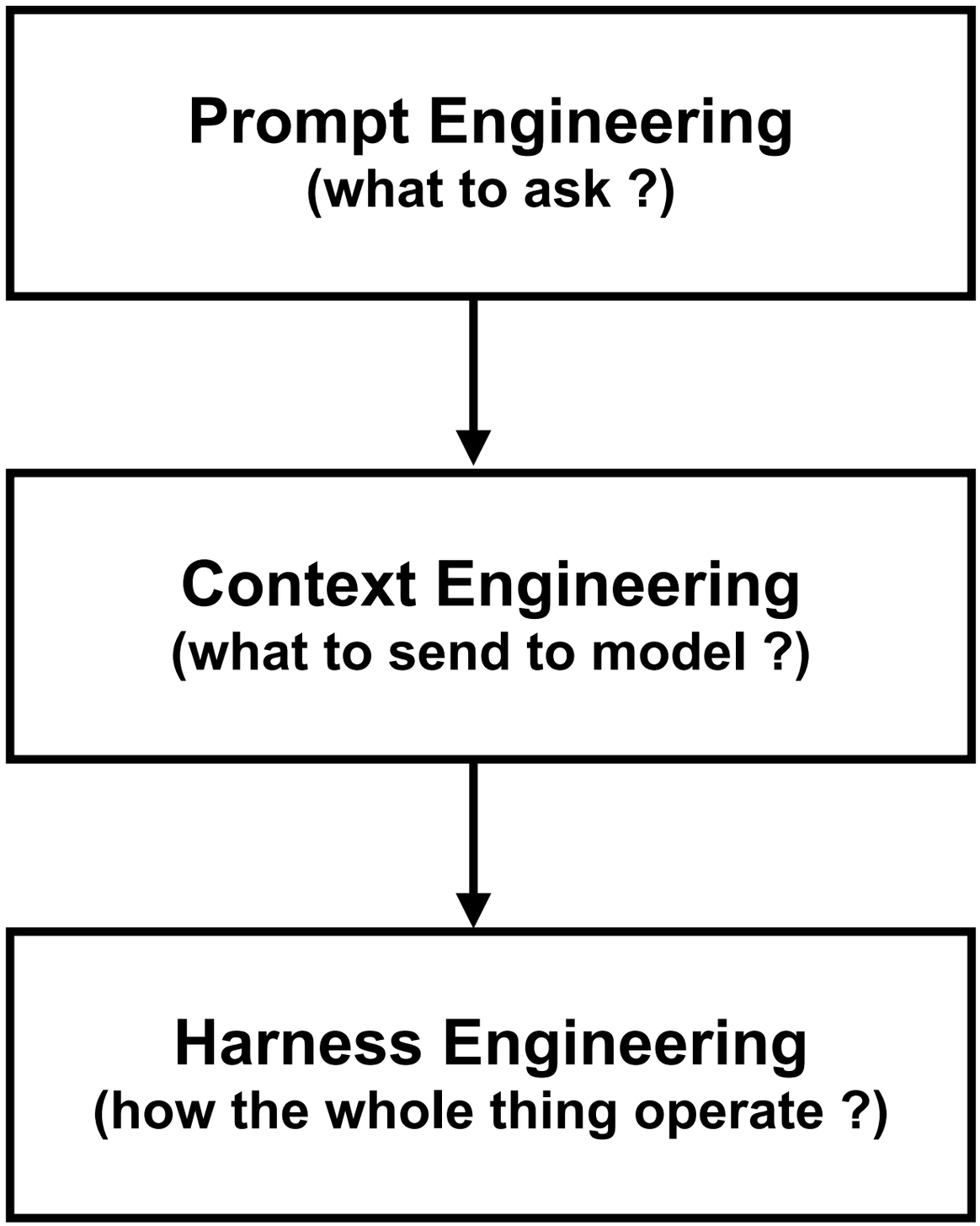

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# Effective Context Engineering



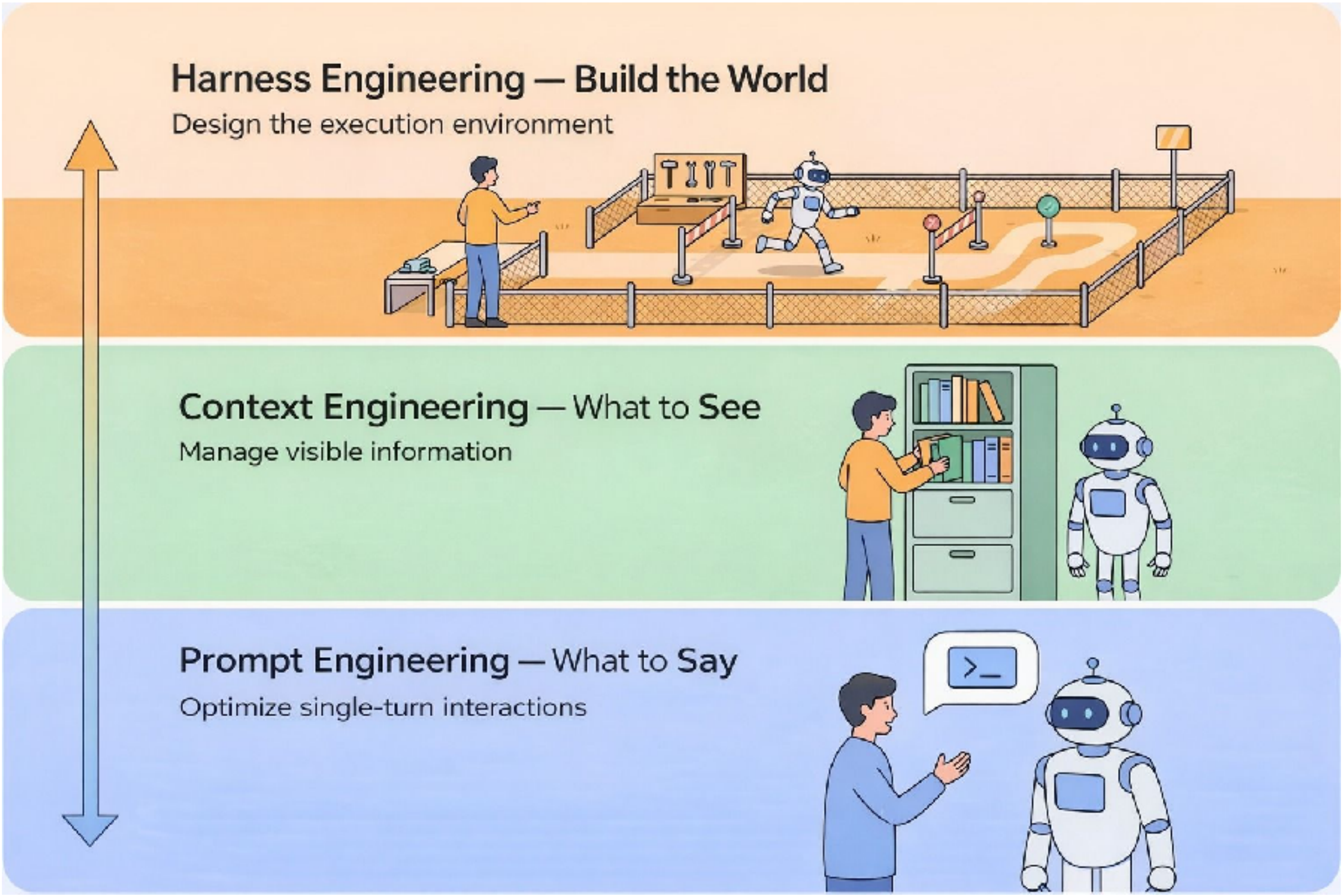

https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents


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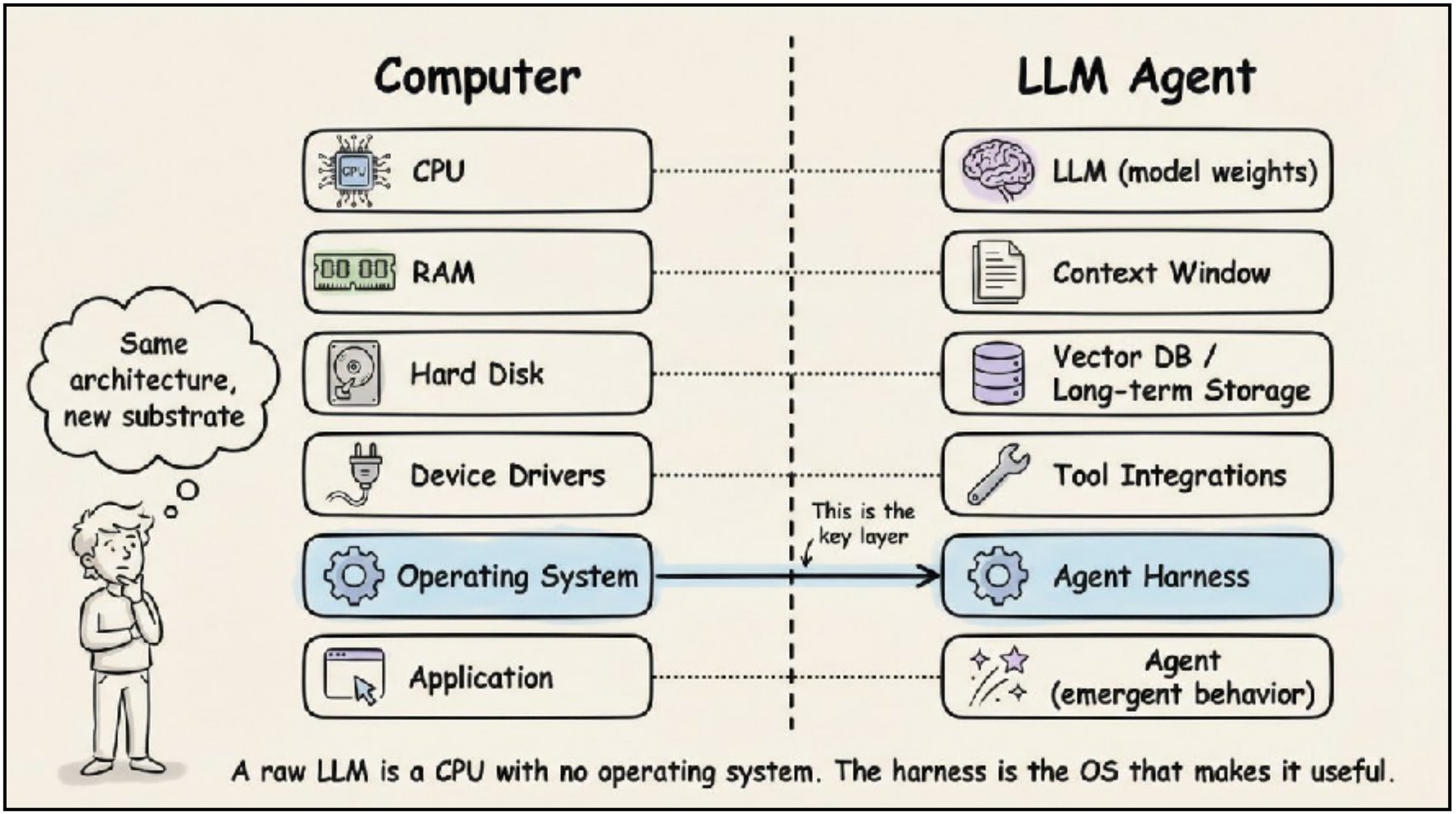
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https://milvus.io/blog/harness-engineering-ai-agents.md


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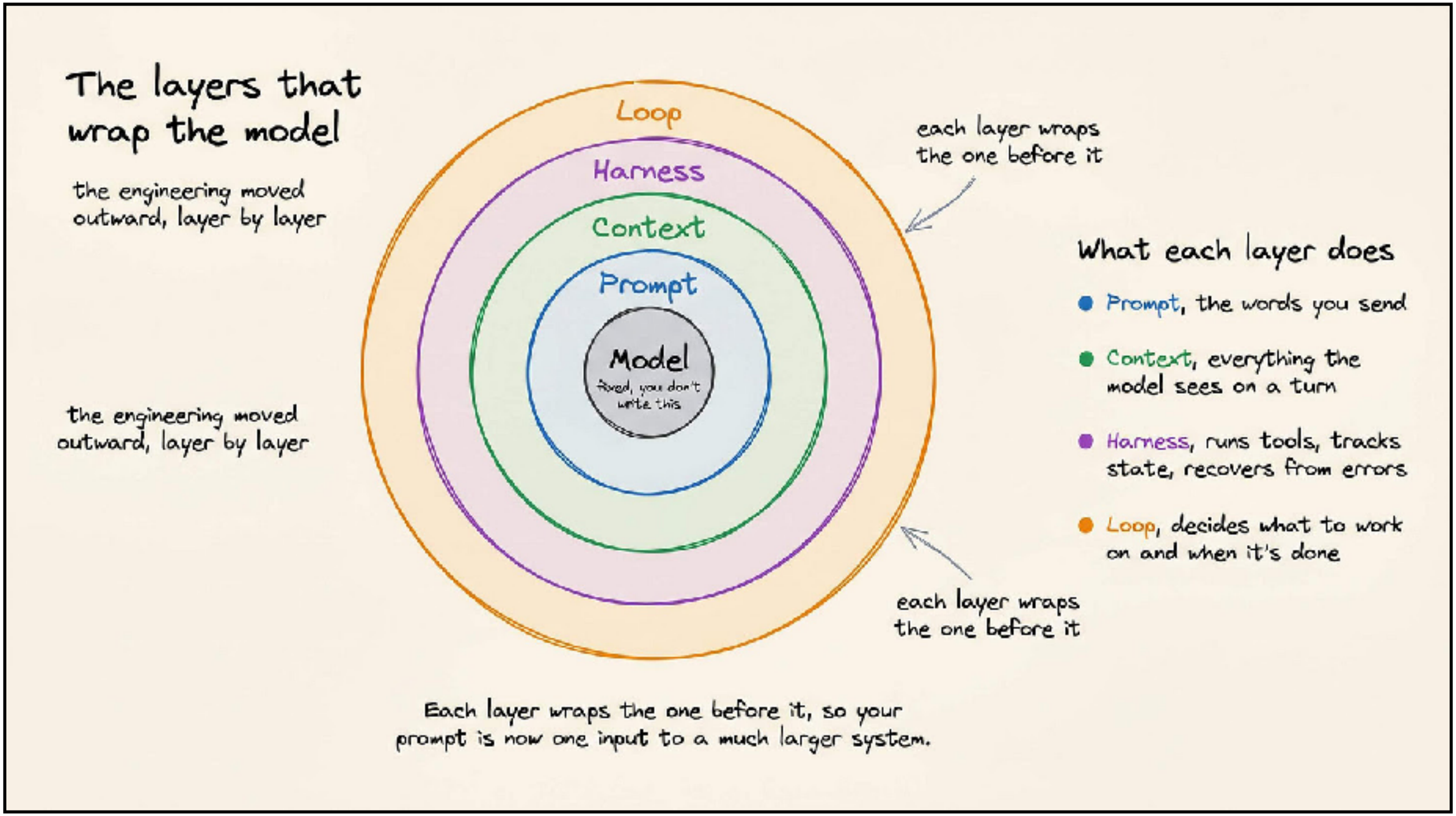
# Harness Engineering




https://x.com/akshay_pachaar/status/2041146899319971922


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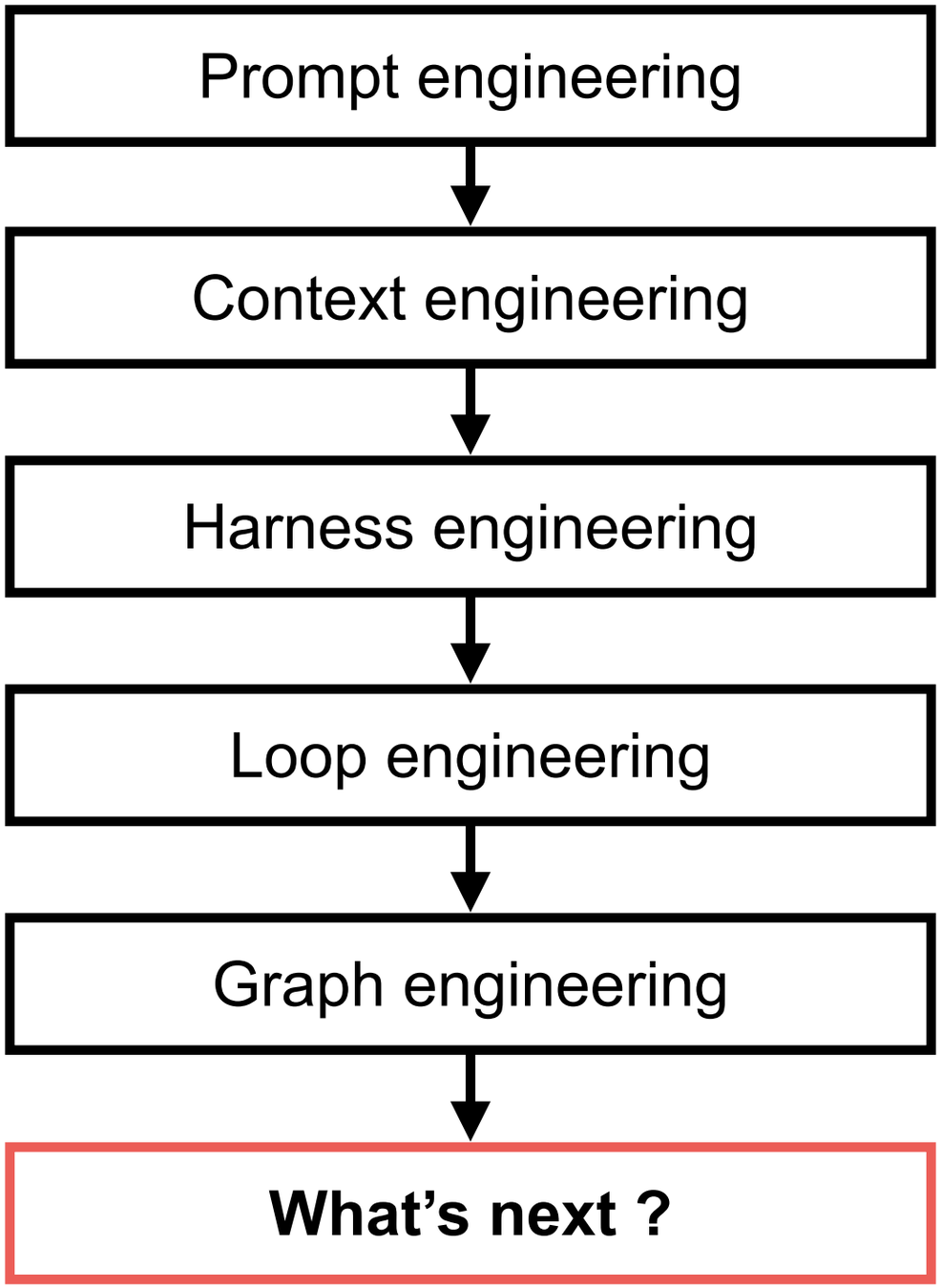
# Loop Engineering





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# Current ?





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# Workshop with Coding

Chat

Text Editor with AI

Pair programming with AI

SDD and Rules

BMAD-METHOD

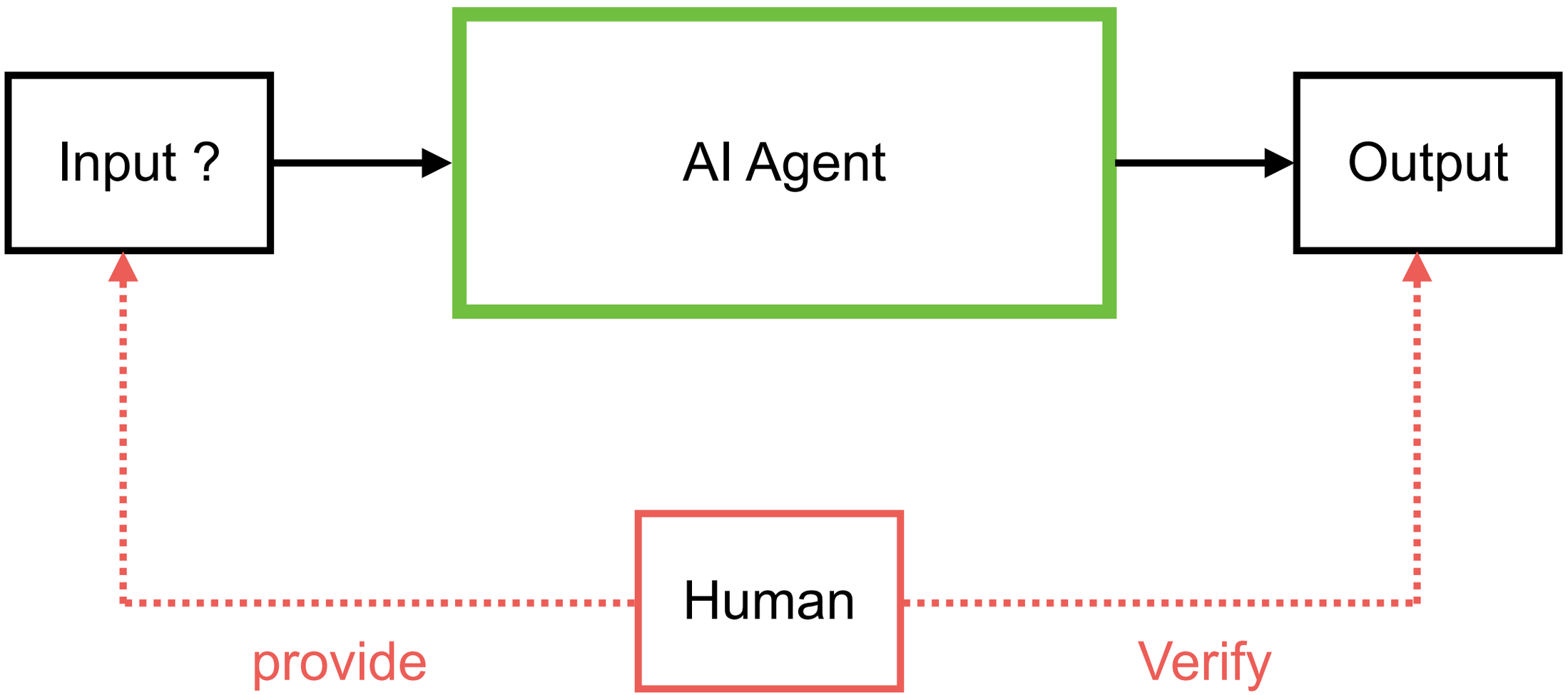

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# AI Agent for Coding

Claude Code
Github Copilot
Cline
Qwen code
Gemini CLI
Pi coding agent


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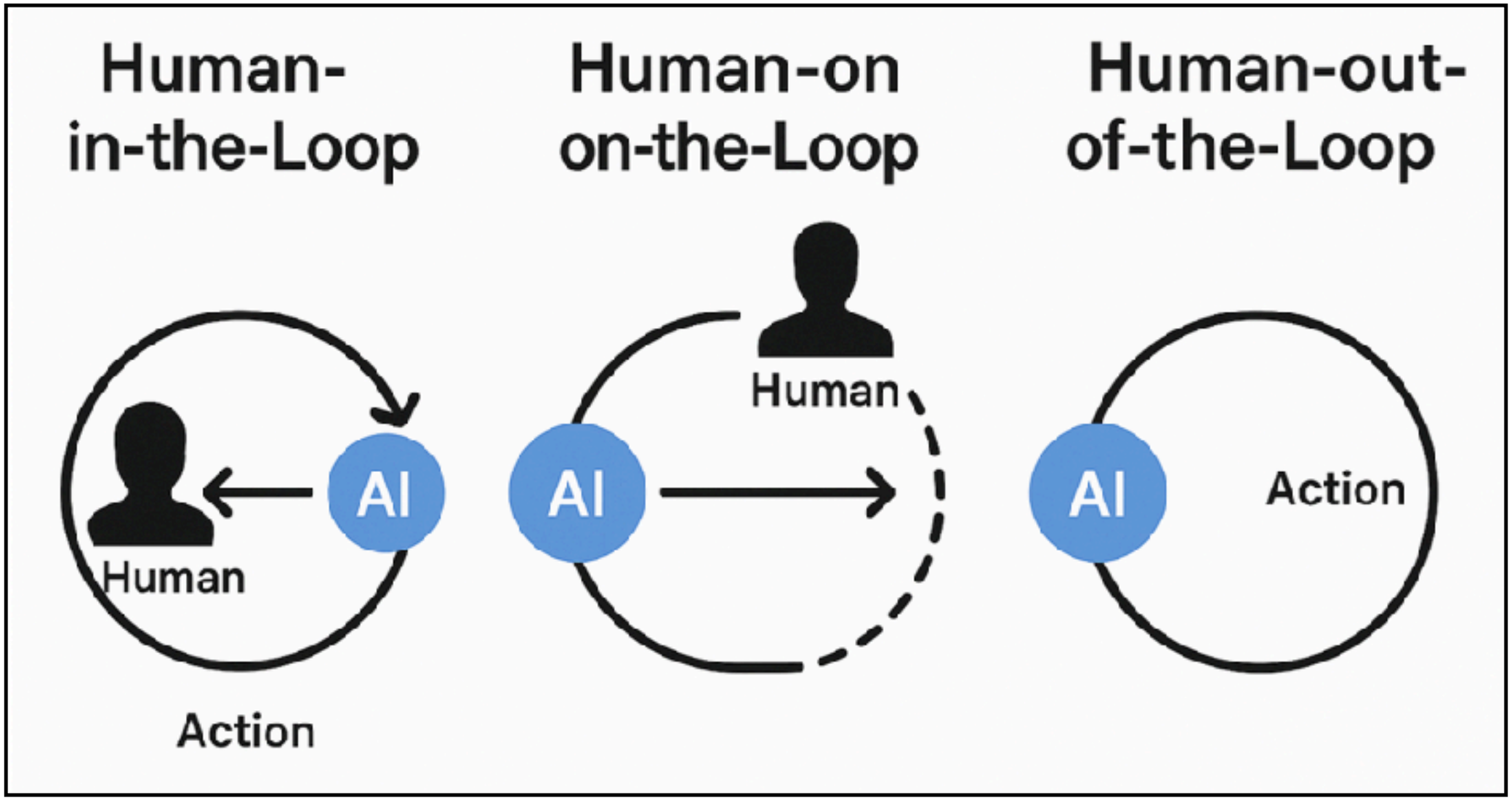
# Coding with AI Agent





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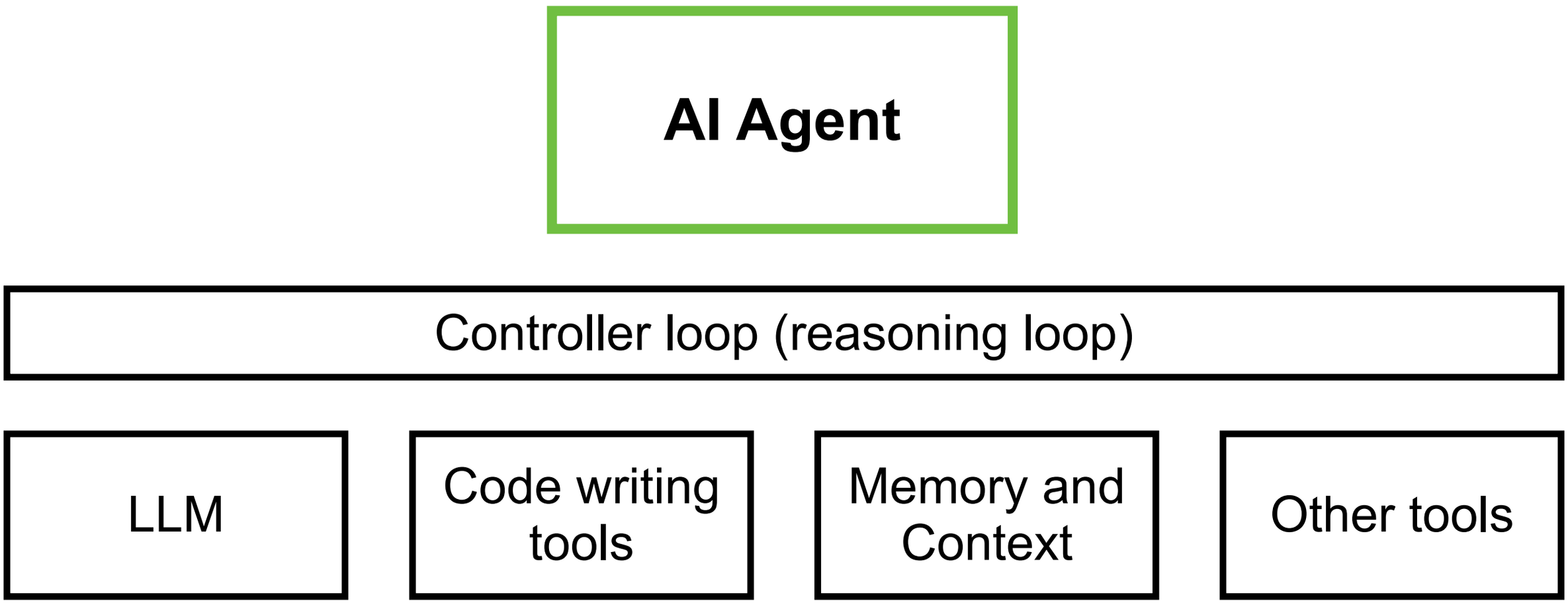
# AI Agent with/without Human





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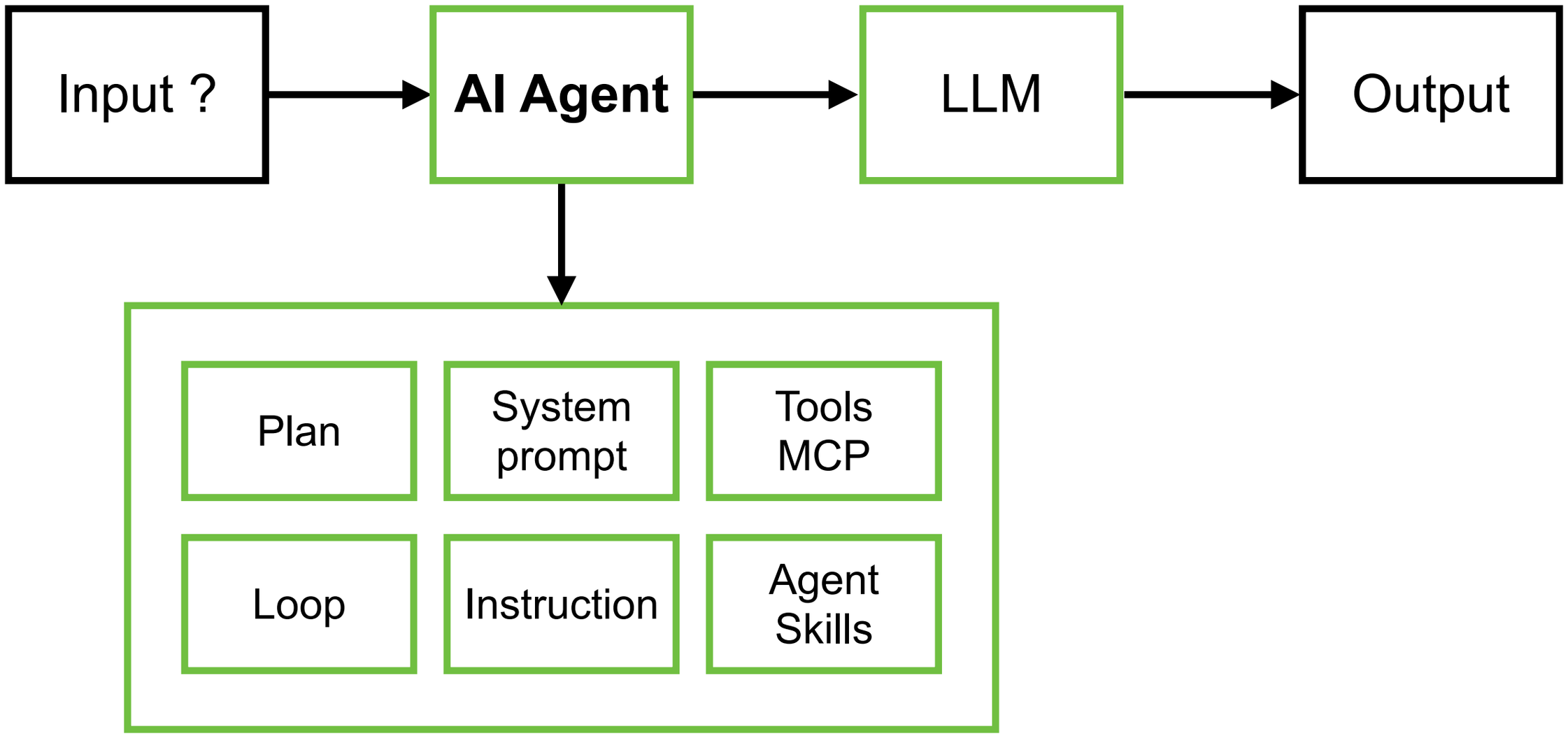
# AI Agent for Coding





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# Working with AI Agent





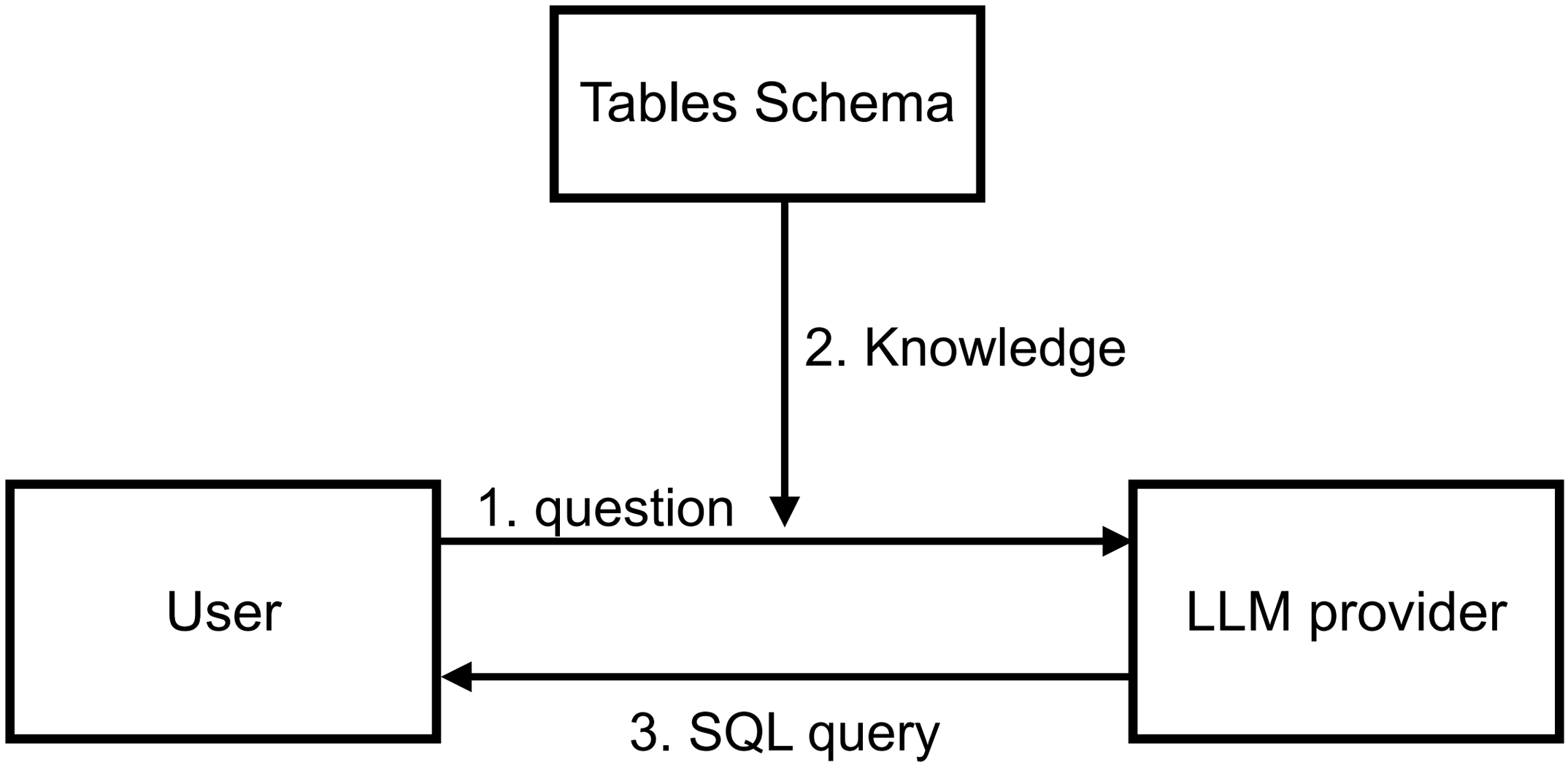
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# Workshop




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# Text-to-SQL




https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-sql




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# Software Development Life Cycle




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# SDLC





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## Planning

- Written planning process
- Arch diagrams

## Testing

- Automated testing
- TDD
- Testing environments
- Testing in prod
- Performance testing
- Load testing
- Generate test data

## Development

- Automated dev env
- CI/CD
- Prototyping
- Code review
- Code generation
- Templates
- Cross-platform dev
- Preview env
- Post-commit code review
- Linting
- Static code analysis
- Project mgmt

## Shipping

- FF & experimentation
- Logging
- Monitoring & alerting
- Staged rollouts

## Maintenance

- Debug production
- Documentation
- Runbook examples
- Mitigation runbook

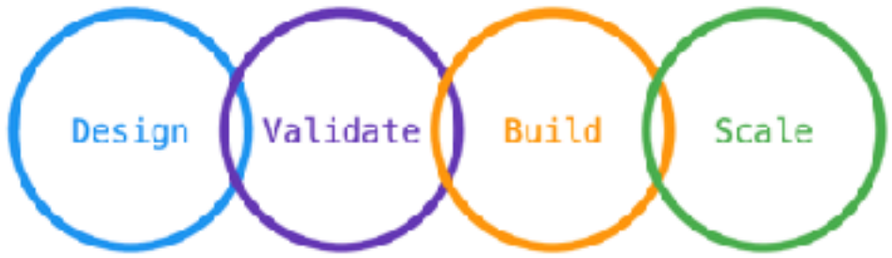
pragmaticengineer.com




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# AI in Software Development

## BEFORE

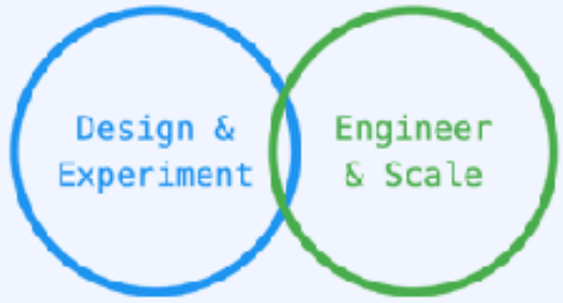



→ Fragmented SDLC Ownership;

→ Scaling Challenges;

→ Siloed Teams Increasing friction.

## NOW




→ Democratized development;

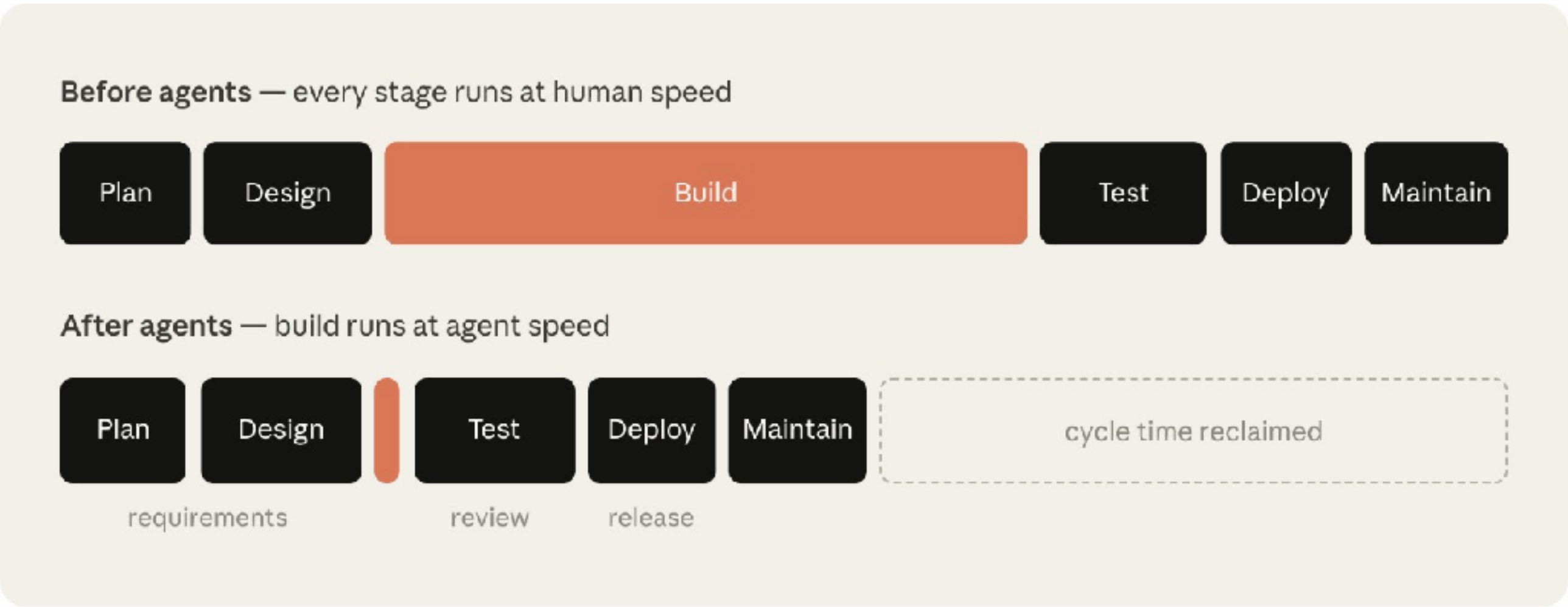
→ Bridge between business and IT;

→ Composability for business agility.

AI-native SDLC drives faster, customer-centric, and higher-quality innovation.


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# AI in SDLC

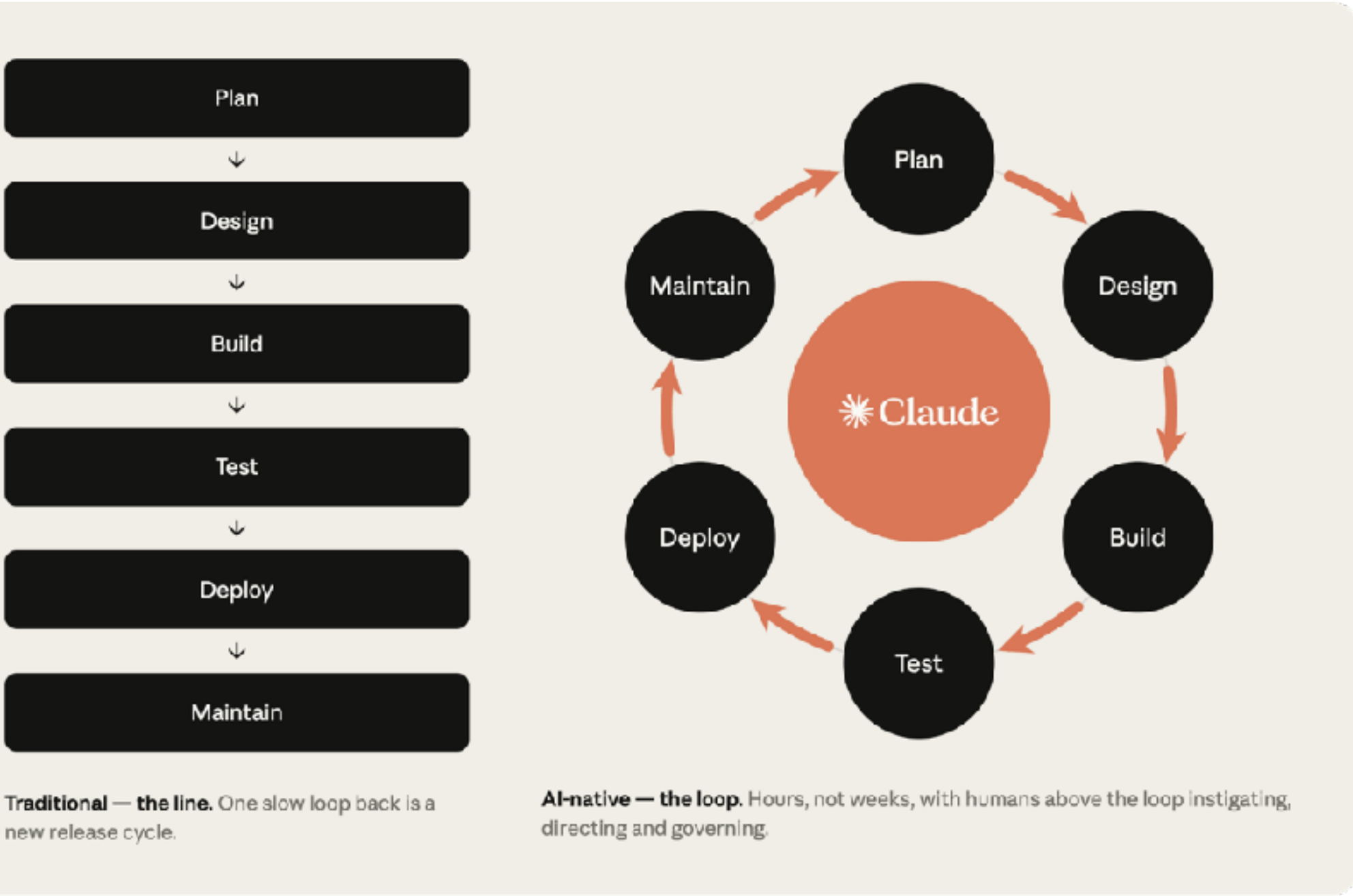



The bottleneck moved !!

https://claude.com/resources/articles/the-ai-native-sdlc-playbook

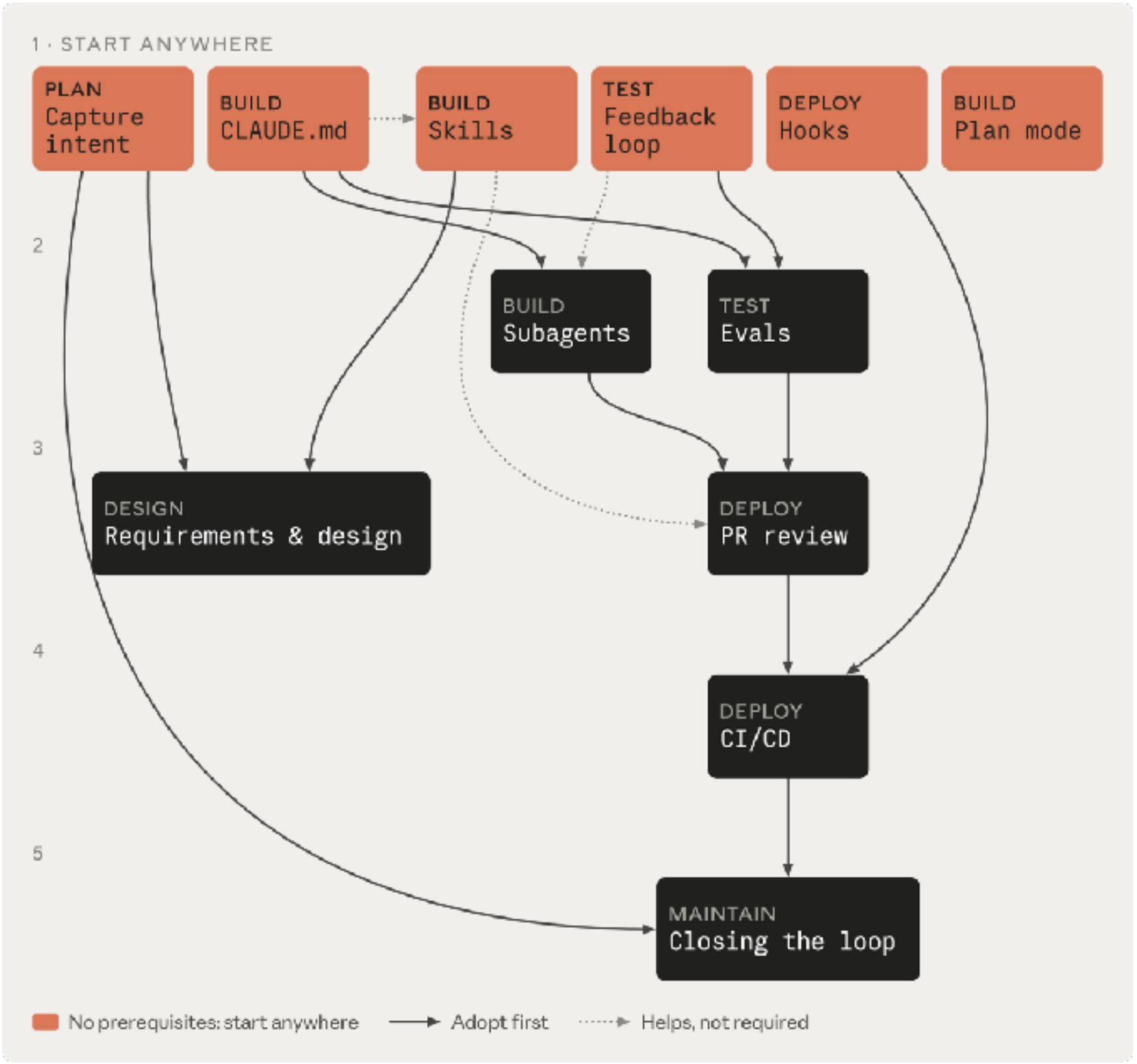

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# AI in SDLC from Anthropic




https://claude.com/resources/articles/the-ai-native-sdlc-playbook

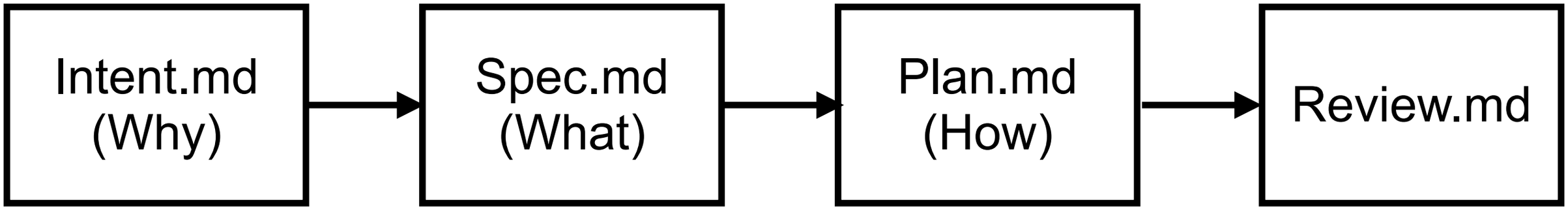

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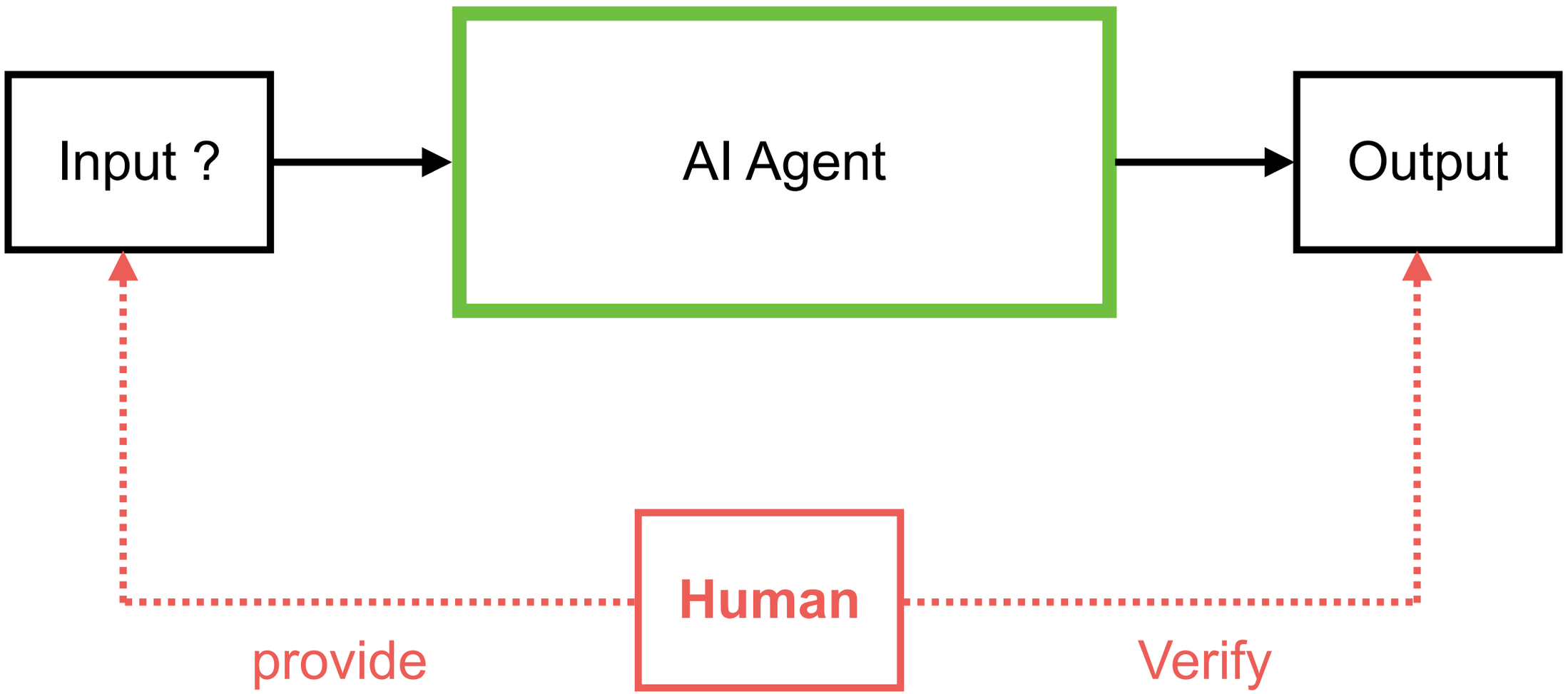
# AI in SDLC from Anthropic




https://claude.com/resources/articles/the-ai-native-sdlc-playbook


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# Coding with AI Agent

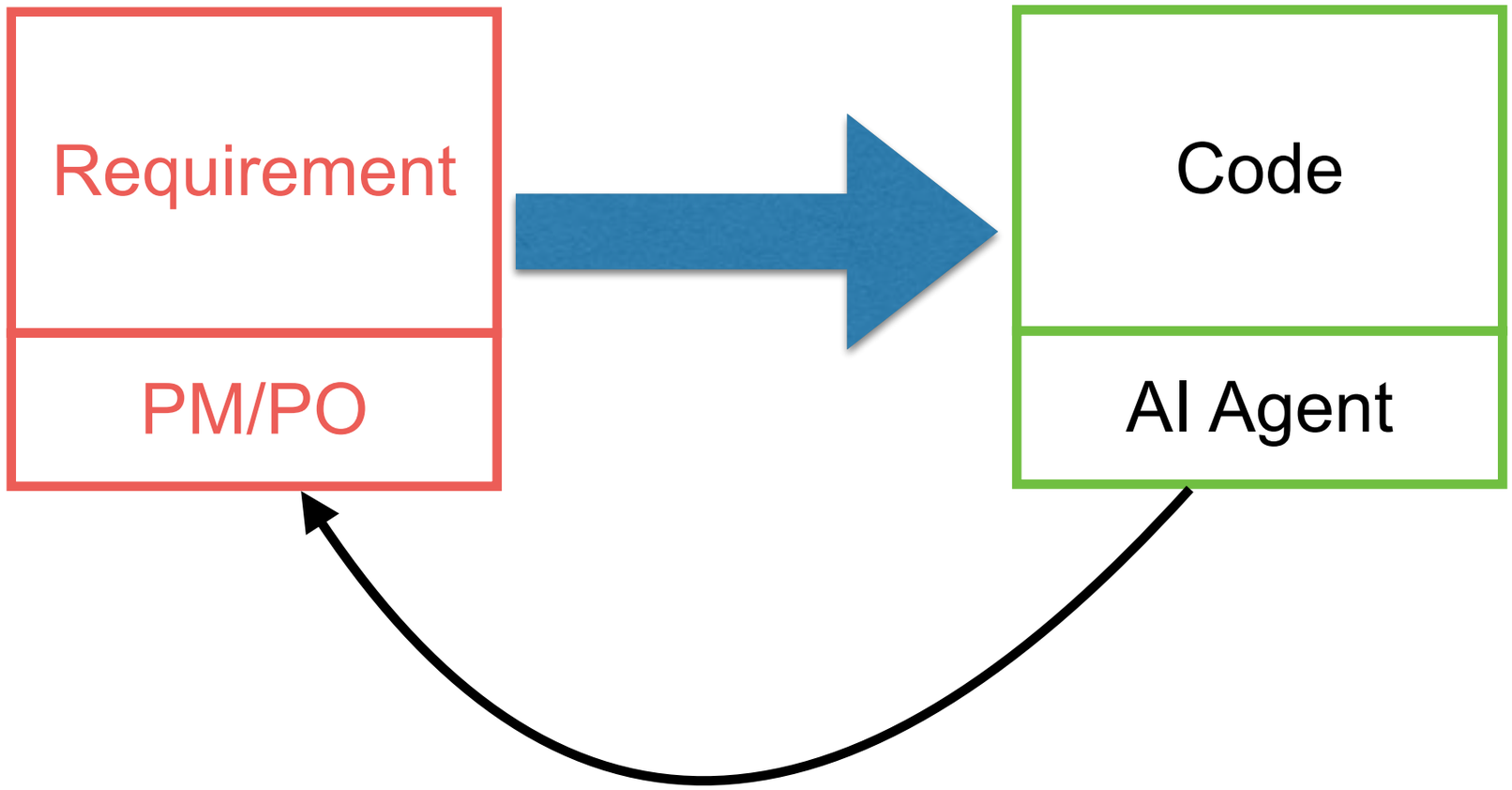




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# New Bottleneck ?

Decide what to **build**, not code




**Get feedback faster**

https://www.theaiopportunities.com/p/is-ai-slowing-down-the-new-bottleneck


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# Impacts with productivity ?

Automated simple tasks

Improve quality and reliability

Improve communication

Faster prototype




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# **Generative AI** isn't just a tool

# it's your **team member**

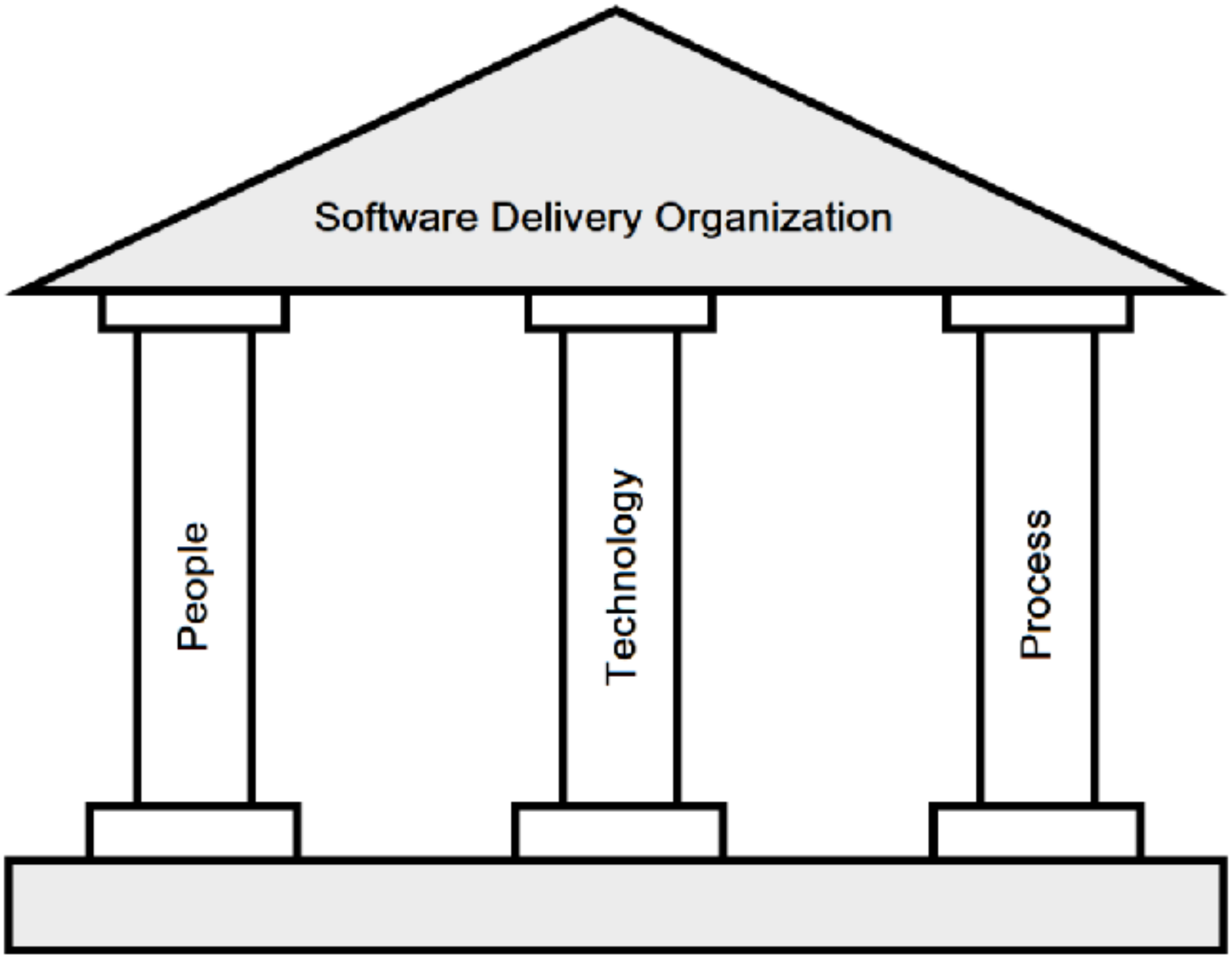

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# Trust, but verify output !!




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# 3 Pillar of Software Development





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# Requirement and Analysis




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# Requirement and Analysis




Clarify requirement
Requirements writing and analysis
User story generation


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# Requirement analysis

Clarify of User requirement ?

https://github.com/up1/workshop-ai-with-technical-team/wiki/Requirement-analysis




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# Design Process




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# Design




Architecture writing assistance
Sequence flow diagram generation
Data modeling
UX/UI design assistance

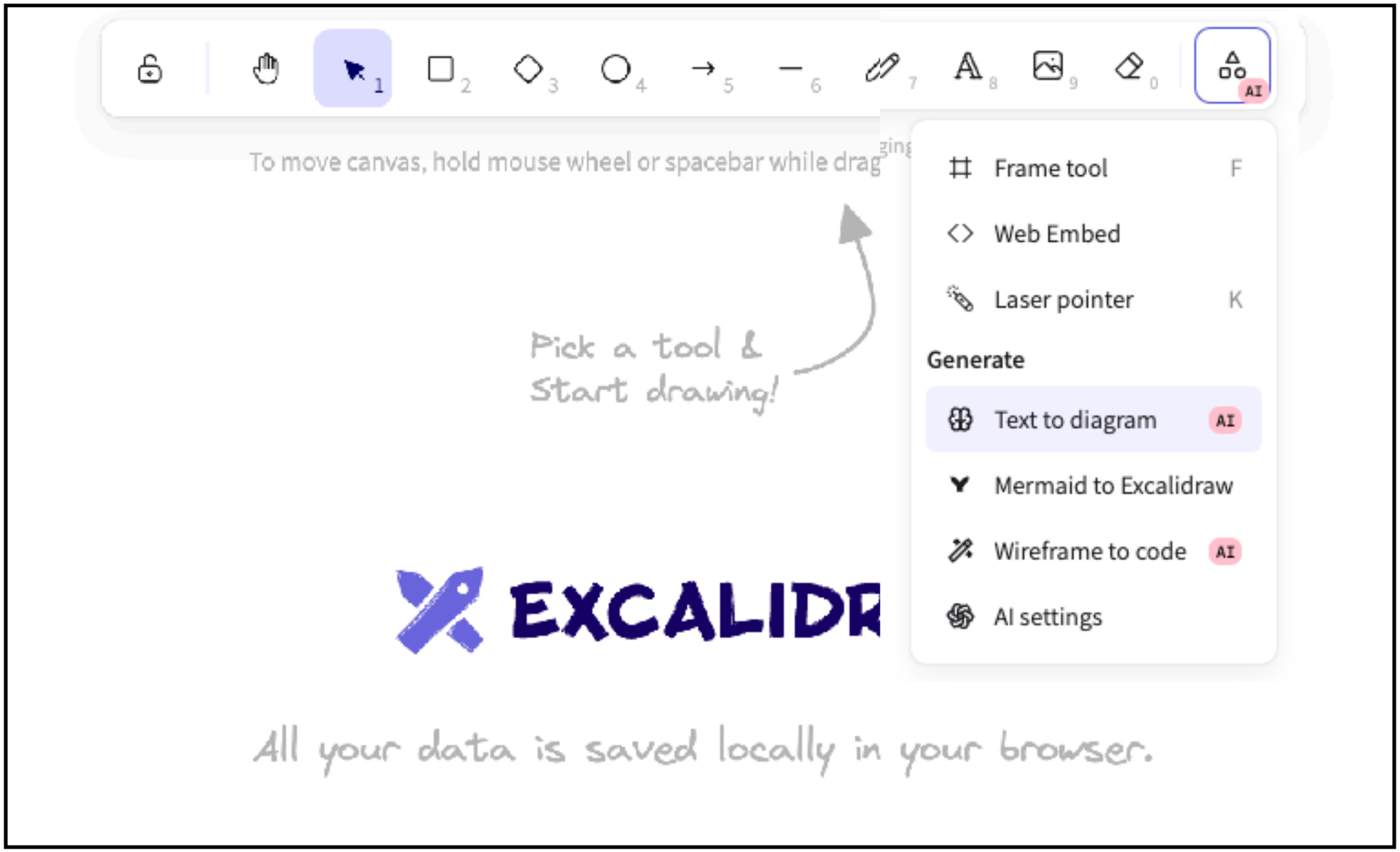

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# Diagram




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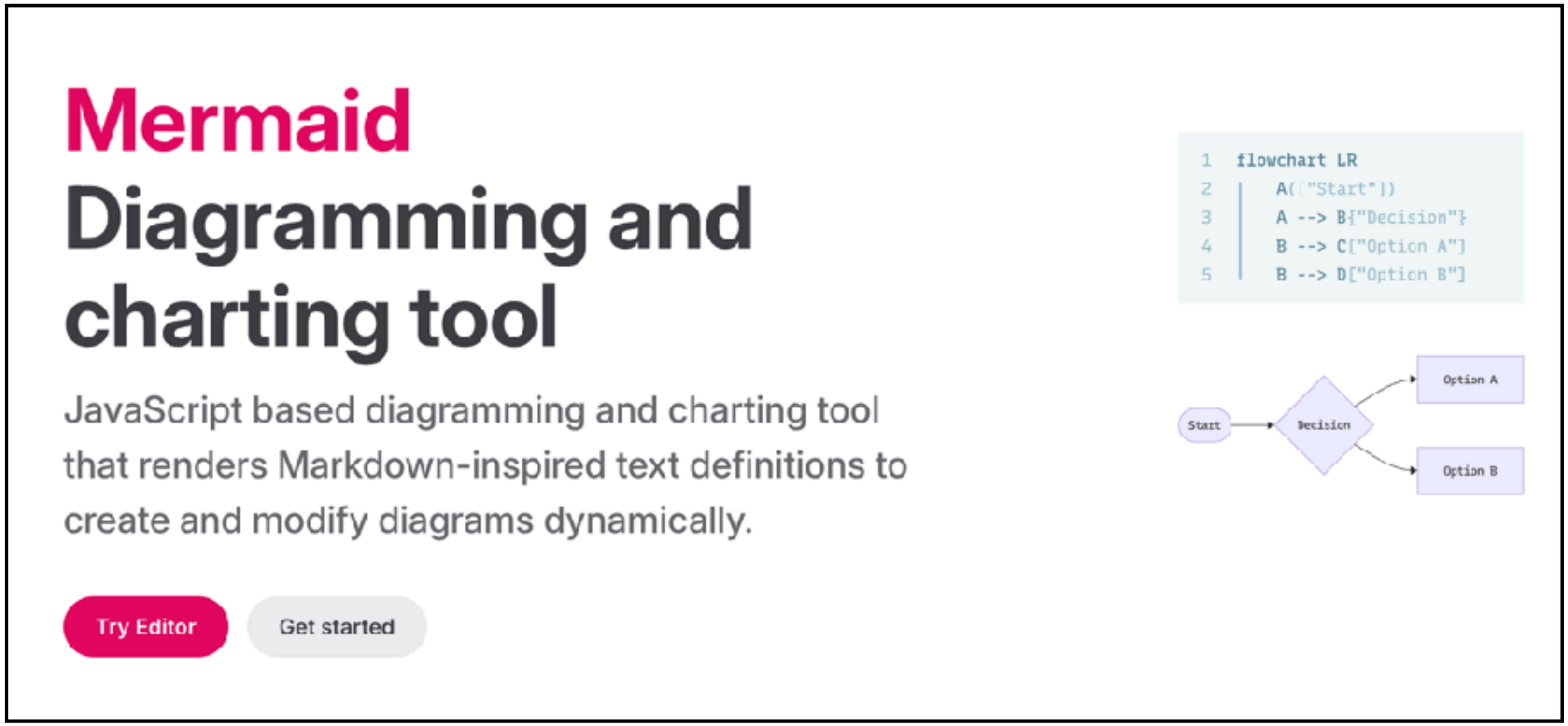
# Excalidraw with AI




https://excalidraw.com/


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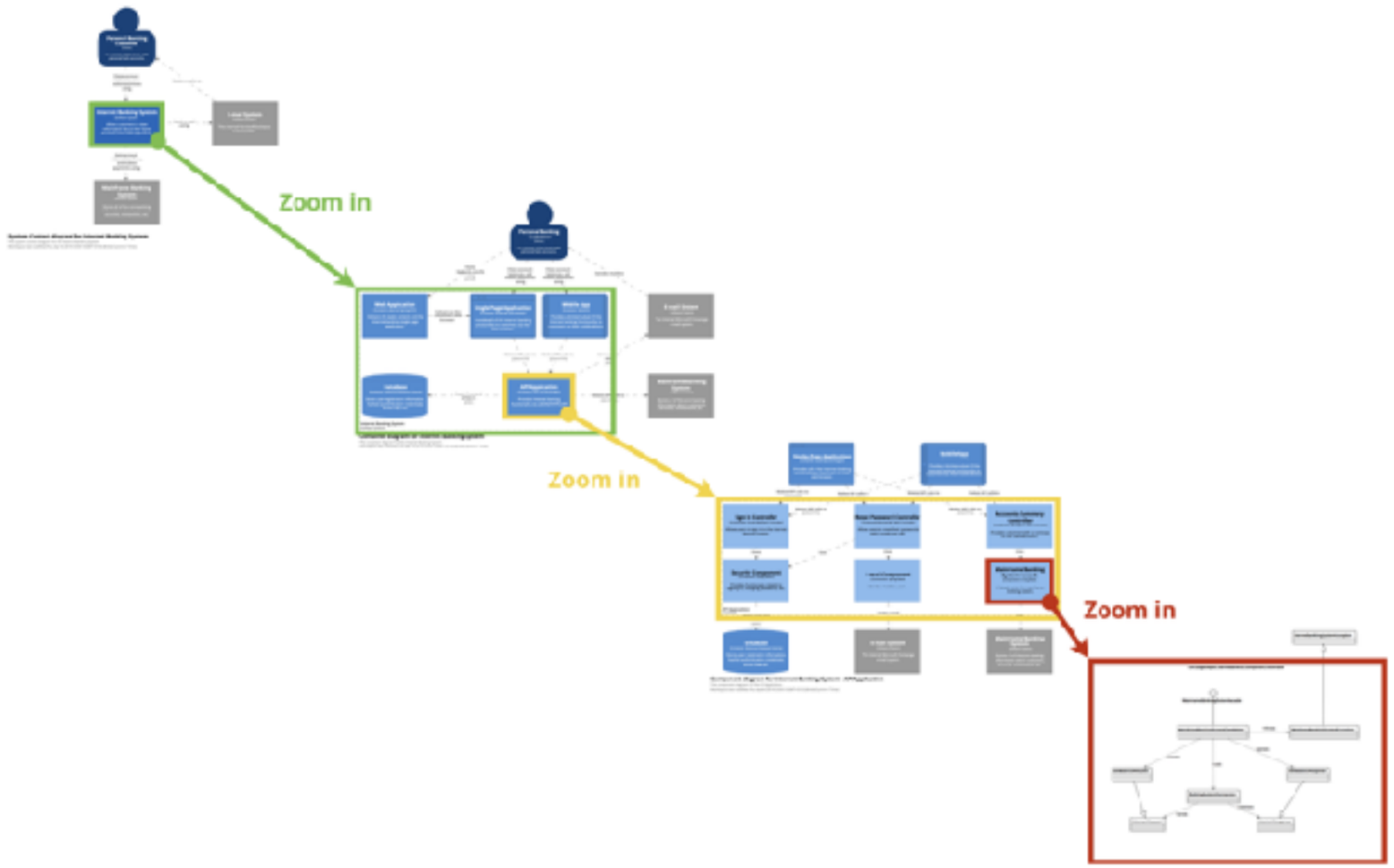
# Mermaid Diagram




https://mermaid.js.org/


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# Software Architecture




Level 1
**Context**

Level 2
**Containers**

Level 3
**Components**

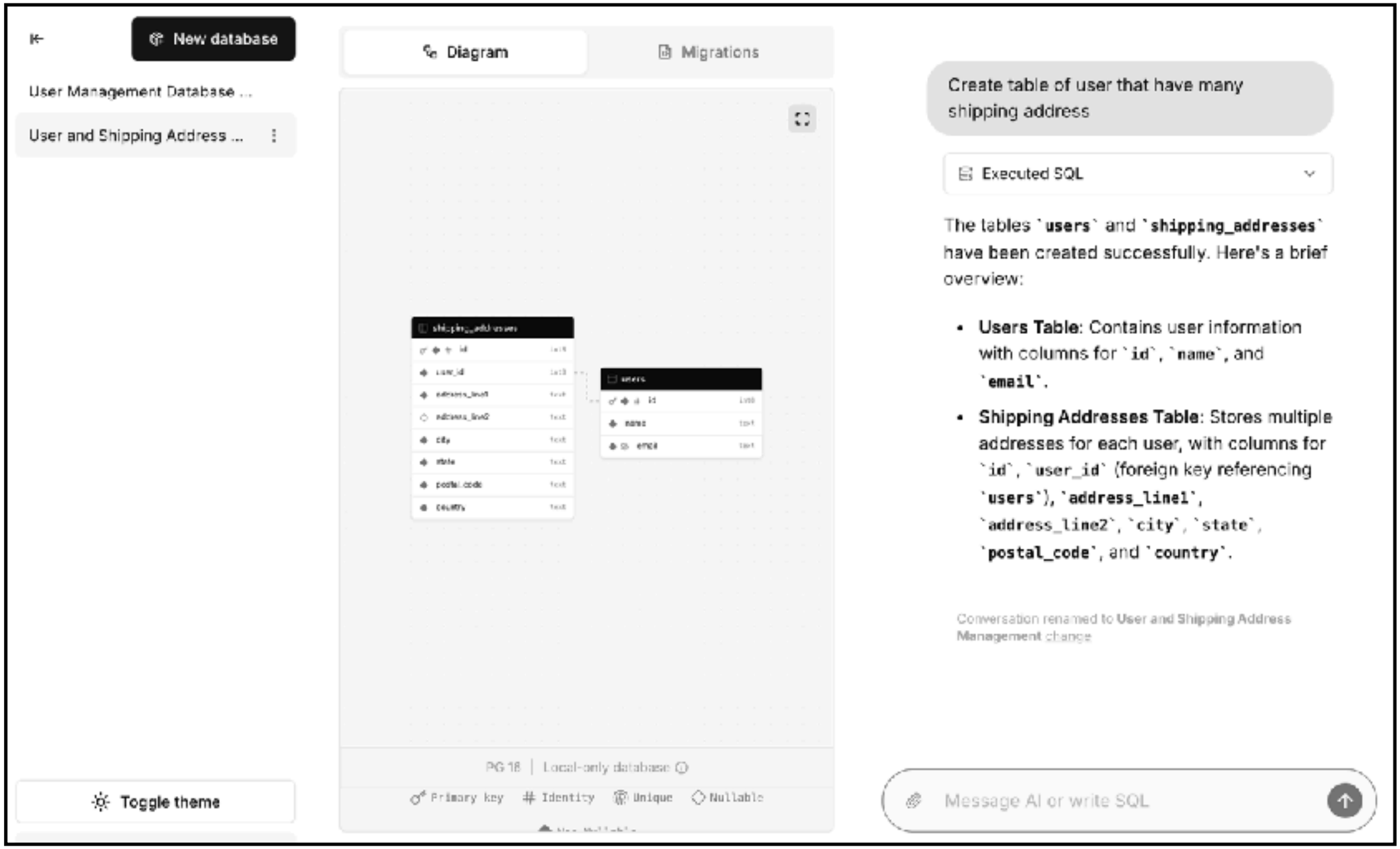
Level 4
**Code**

https://c4model.com/




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# Database



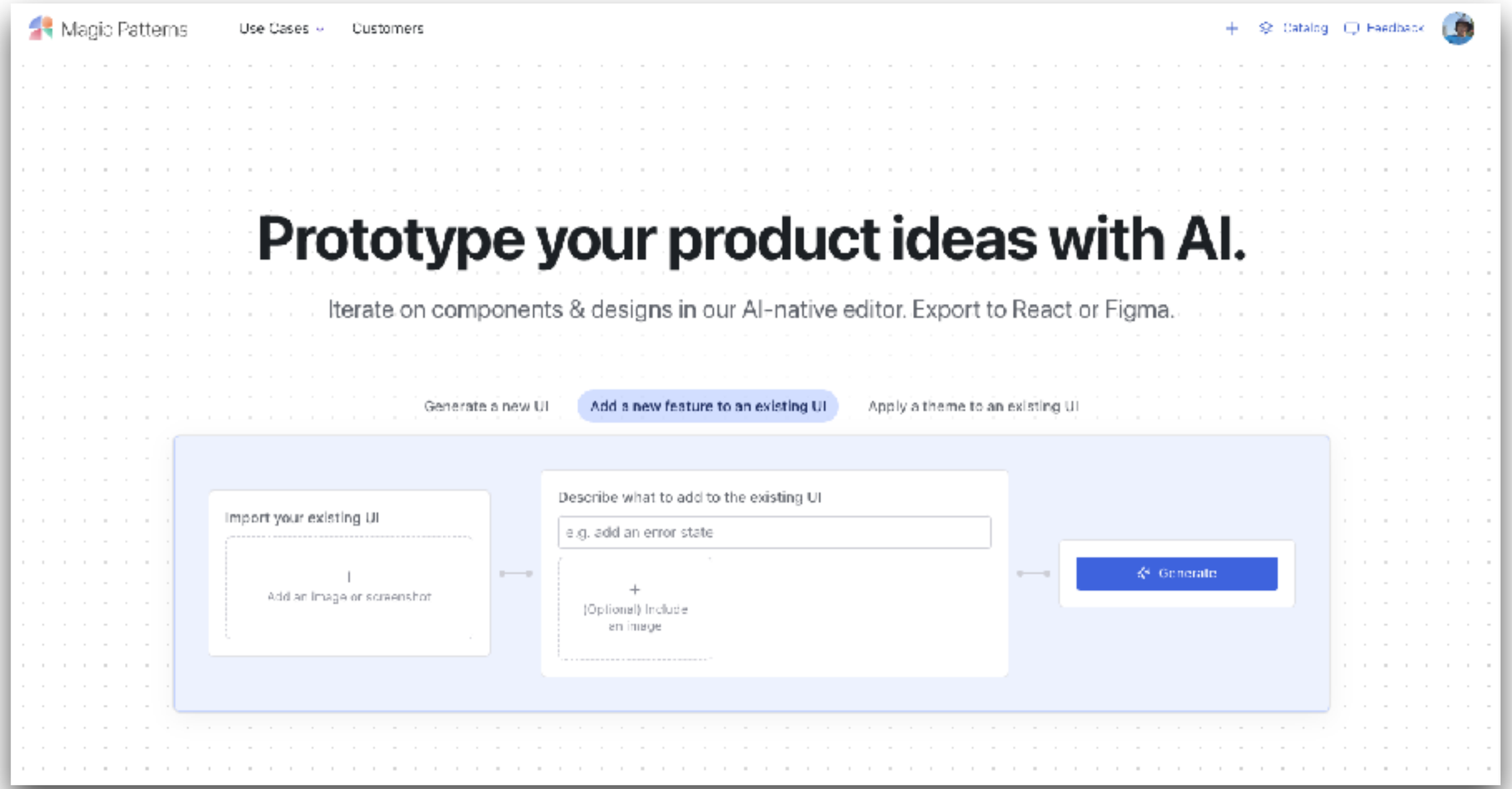

https://database.build/


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# User Interface


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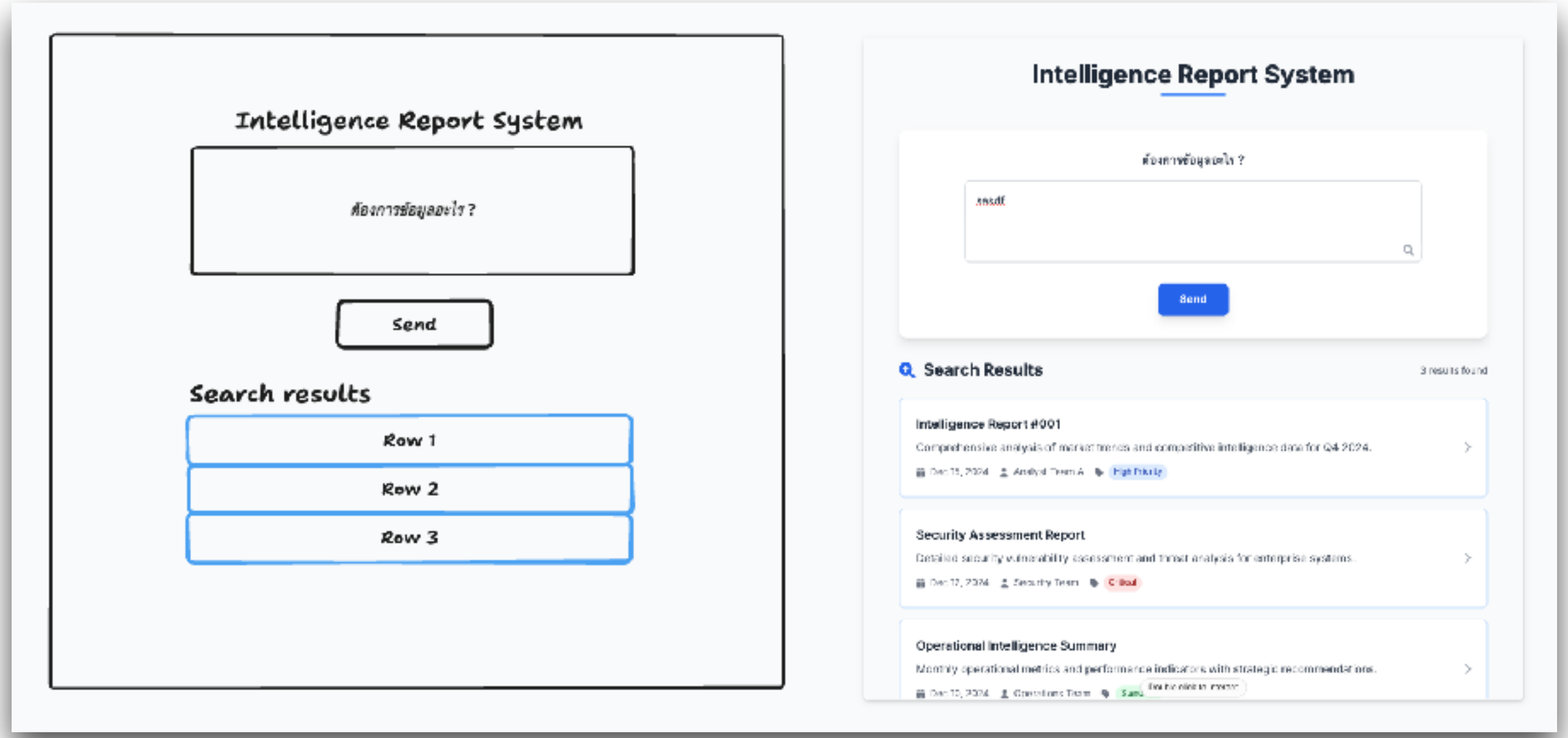
# Magic Pattern




https://www.magicpatterns.com/


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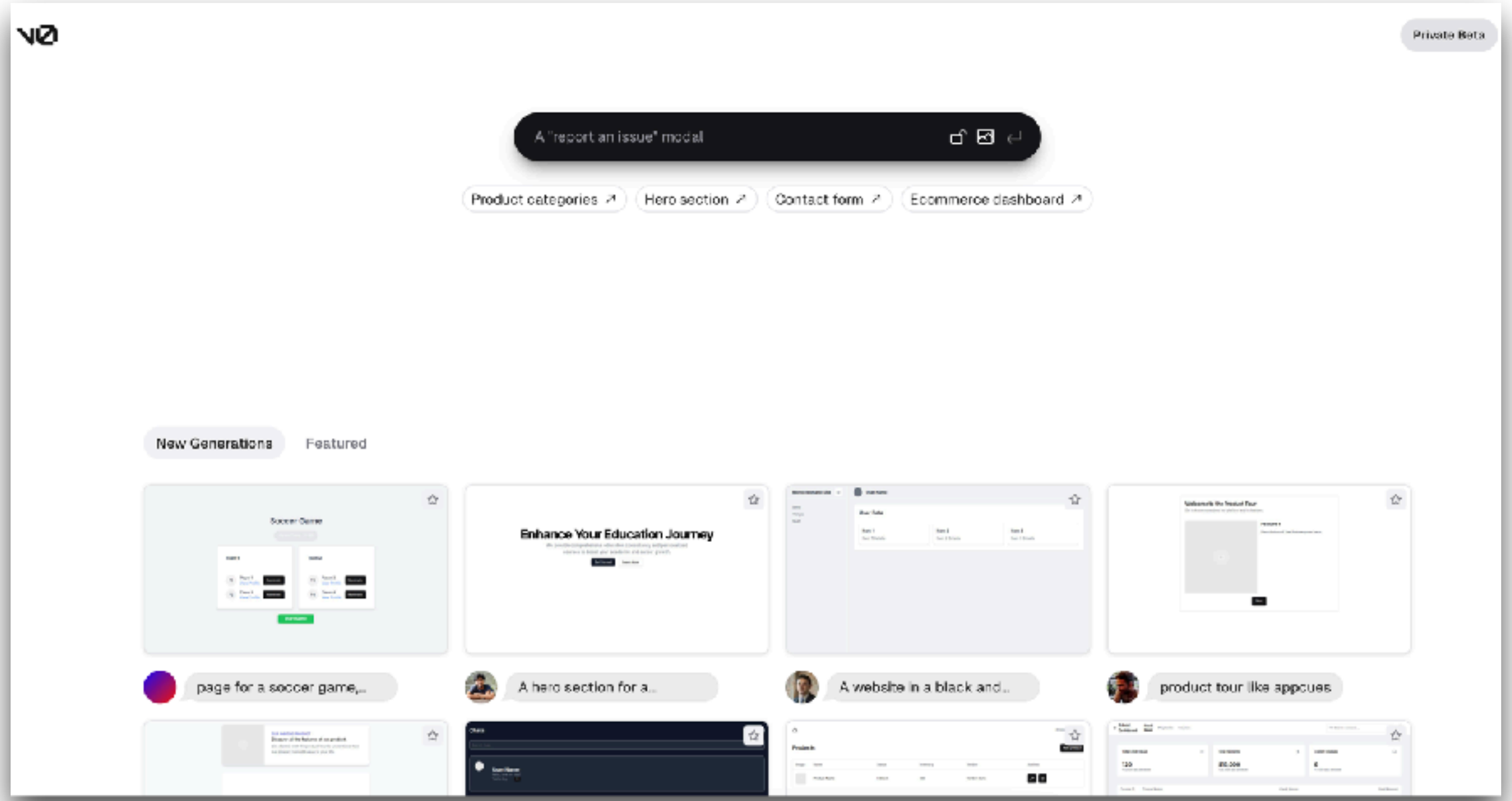
# Make Real




https://github.com/tldraw/make-real


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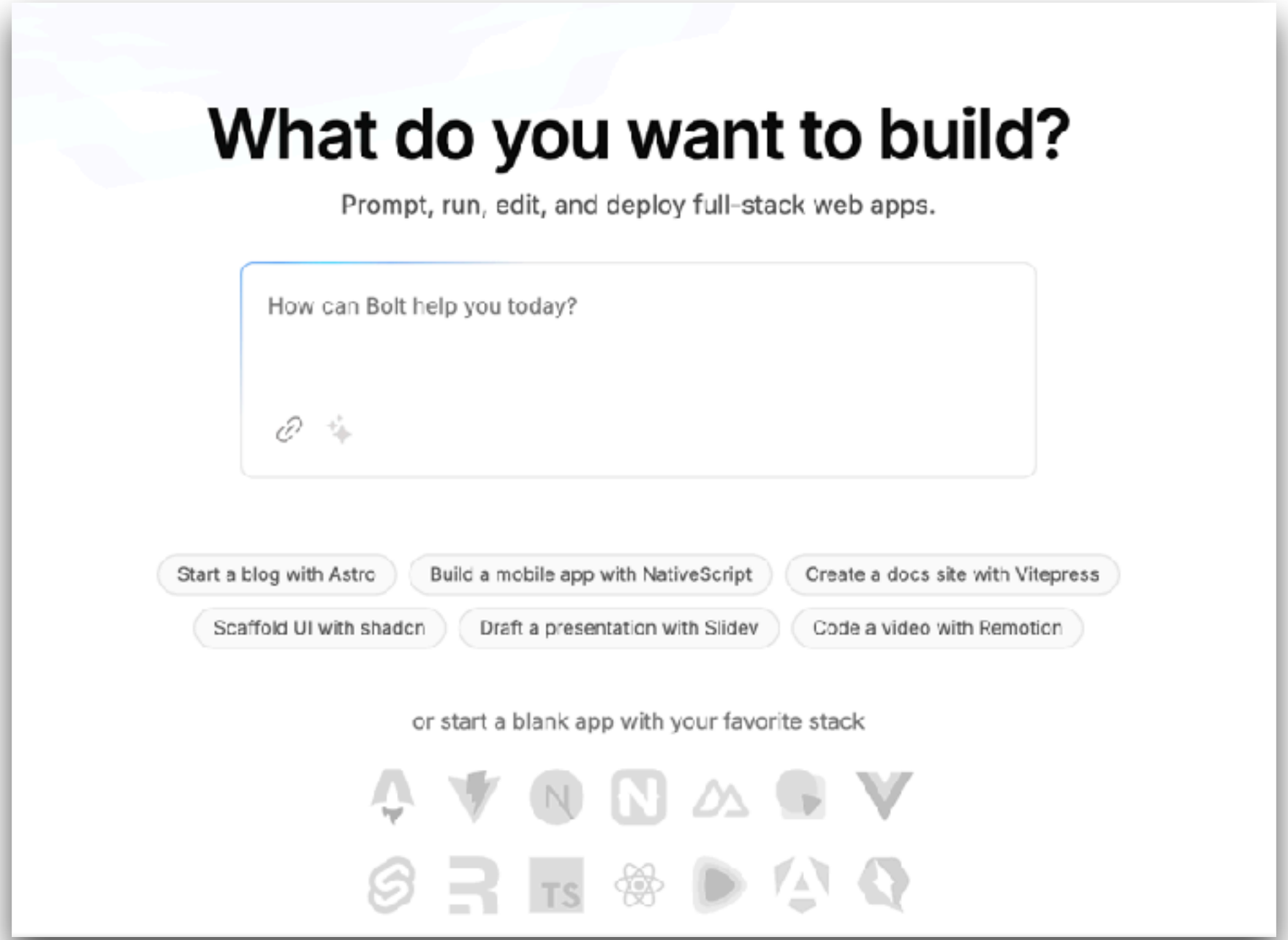
# V0.app




https://v0.dev/


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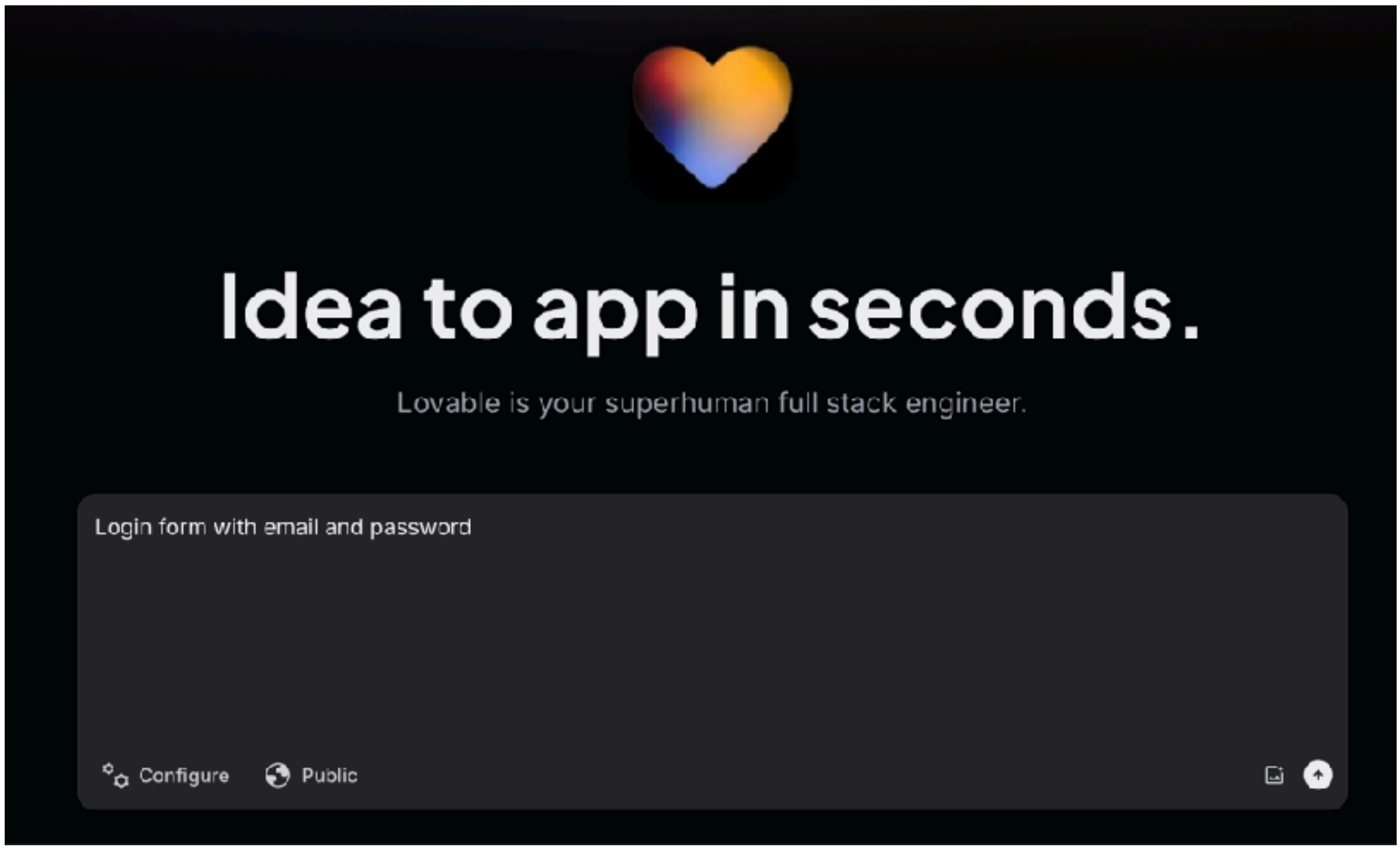
# Bolt.new




https://bolt.new/


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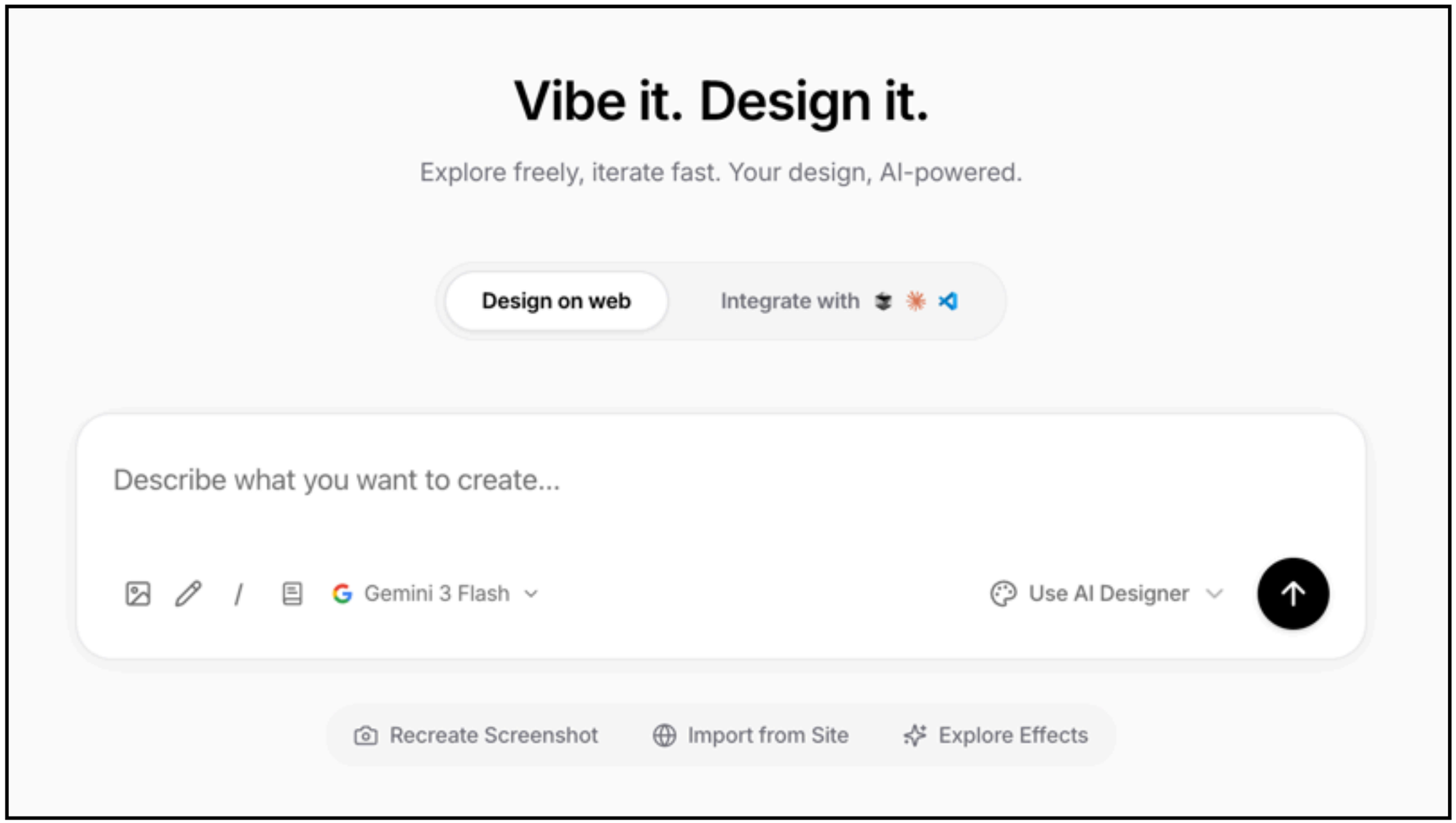
# Lovable




https://lovable.dev/


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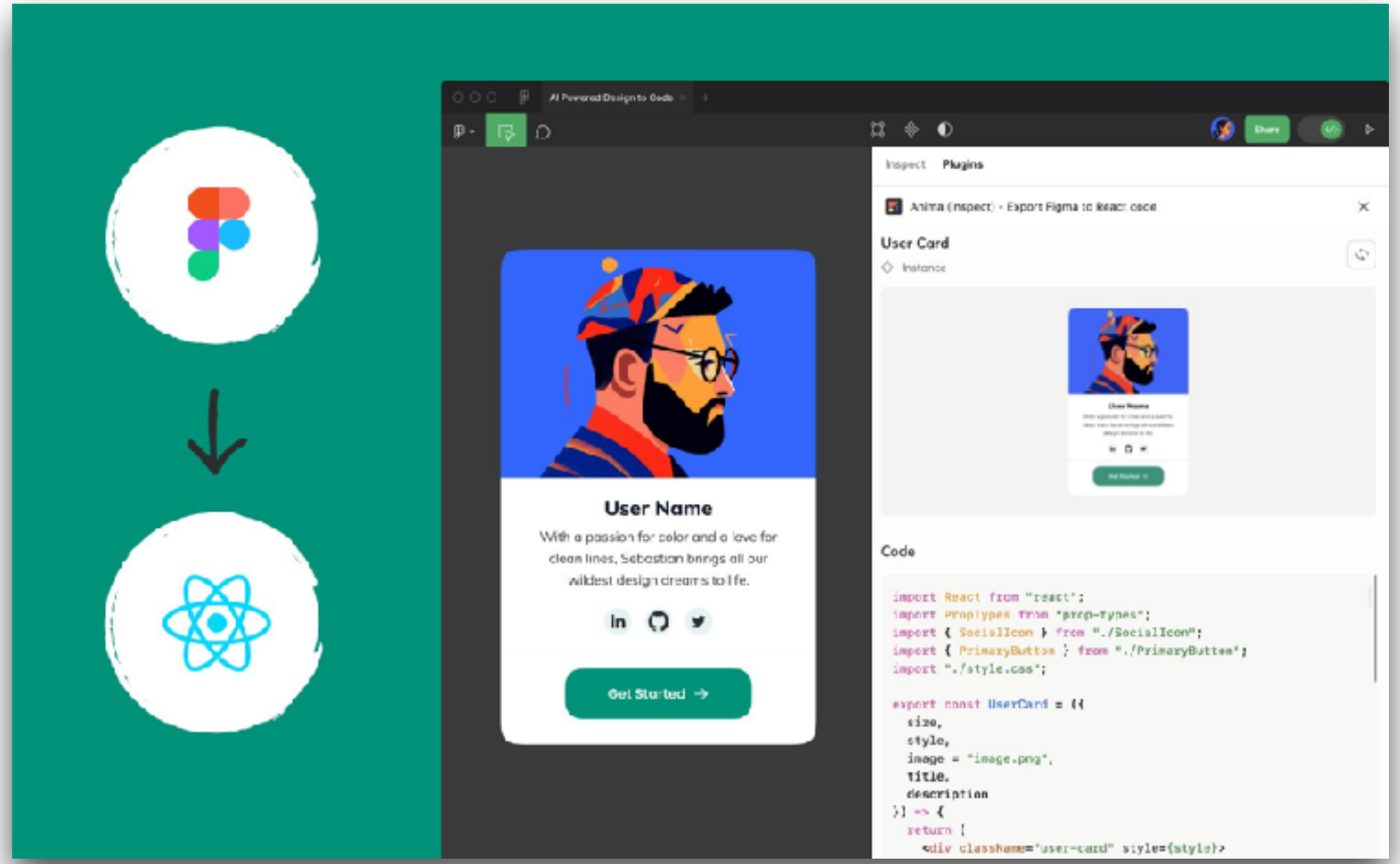
# SuperDesign




https://app.superdesign.dev/


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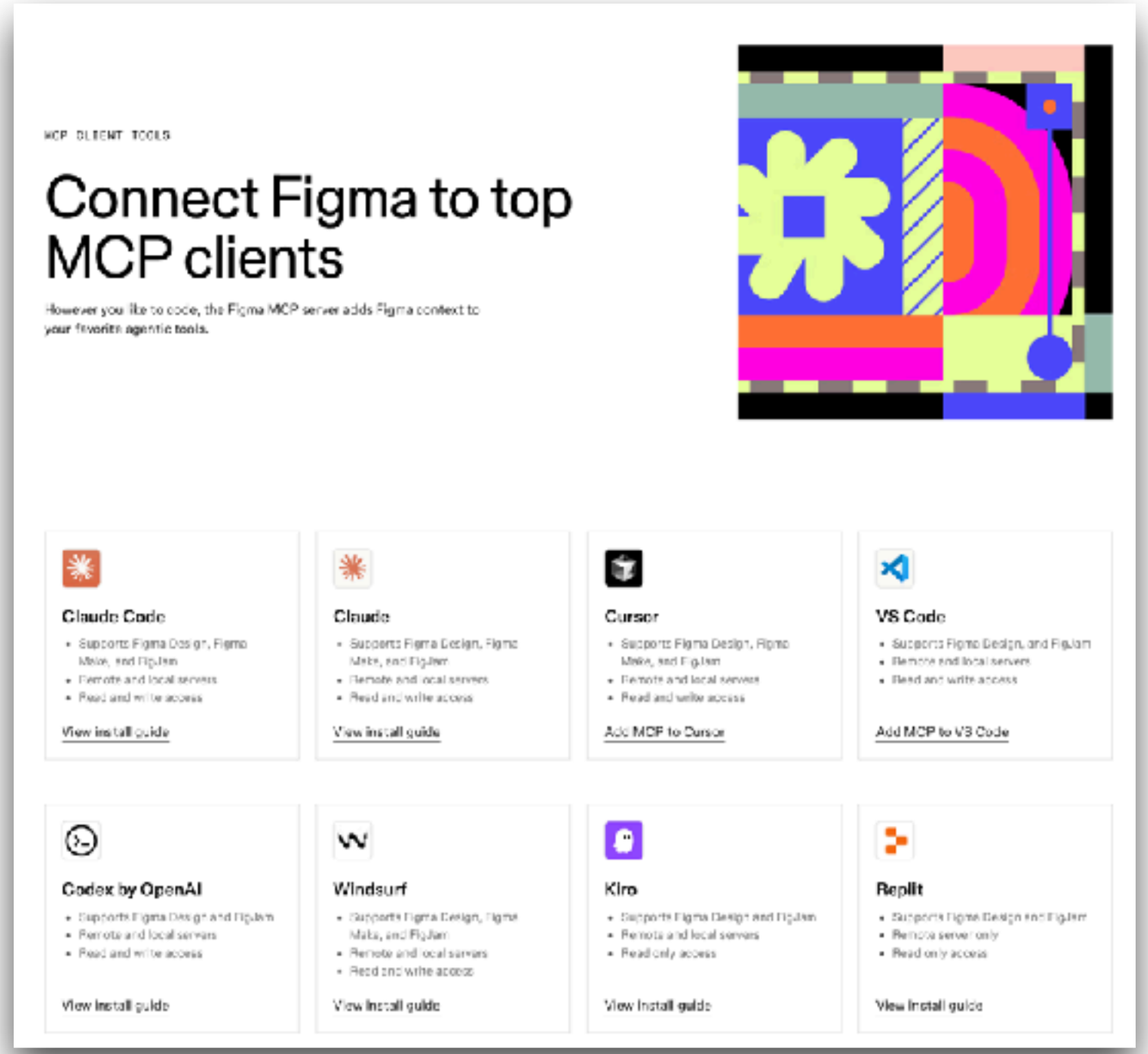
# Figma




https://www.figma.com/mcp-catalog/


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# Figma MCP

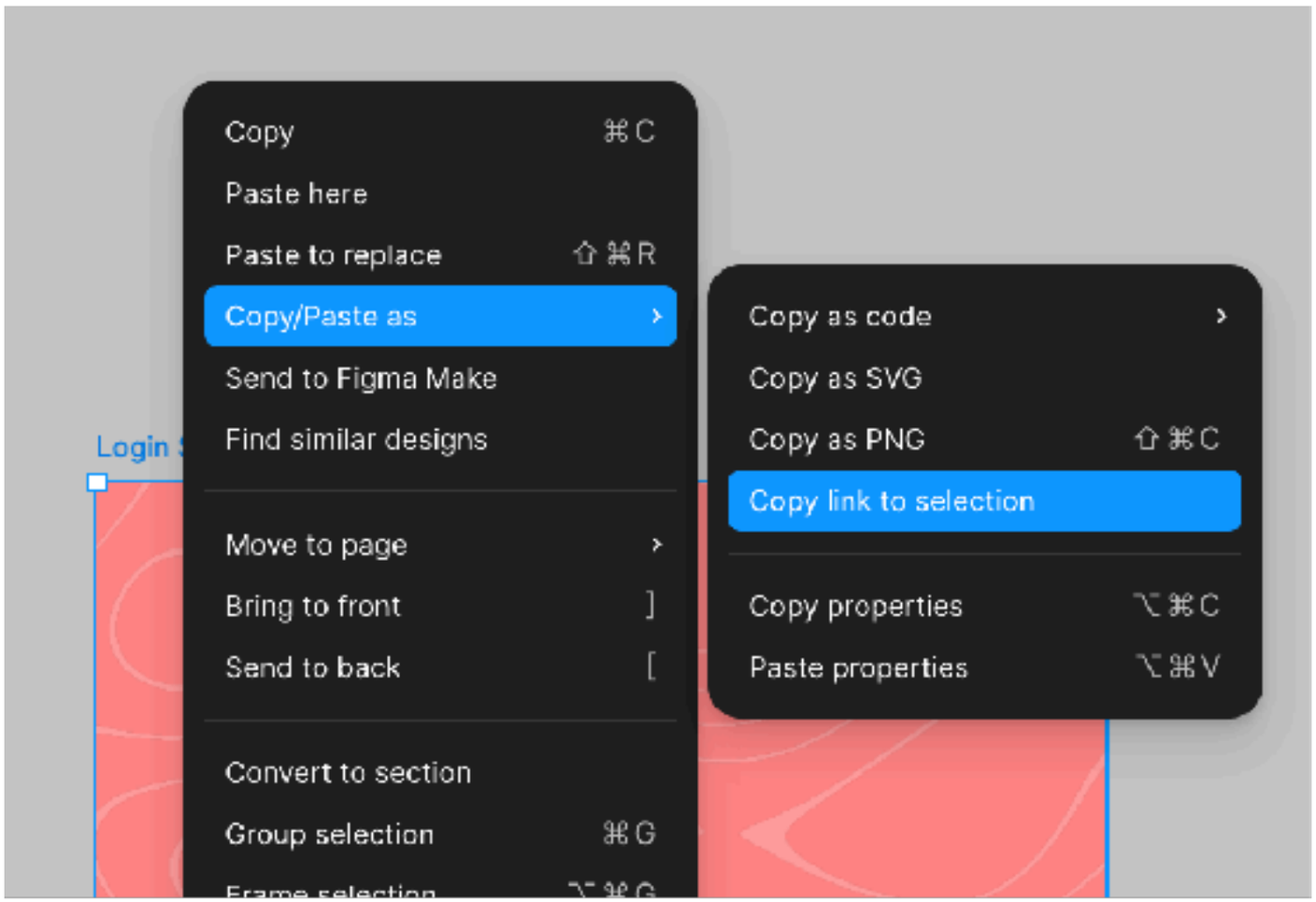



https://www.figma.com/mcp-catalog/


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# Design to Code in Figma

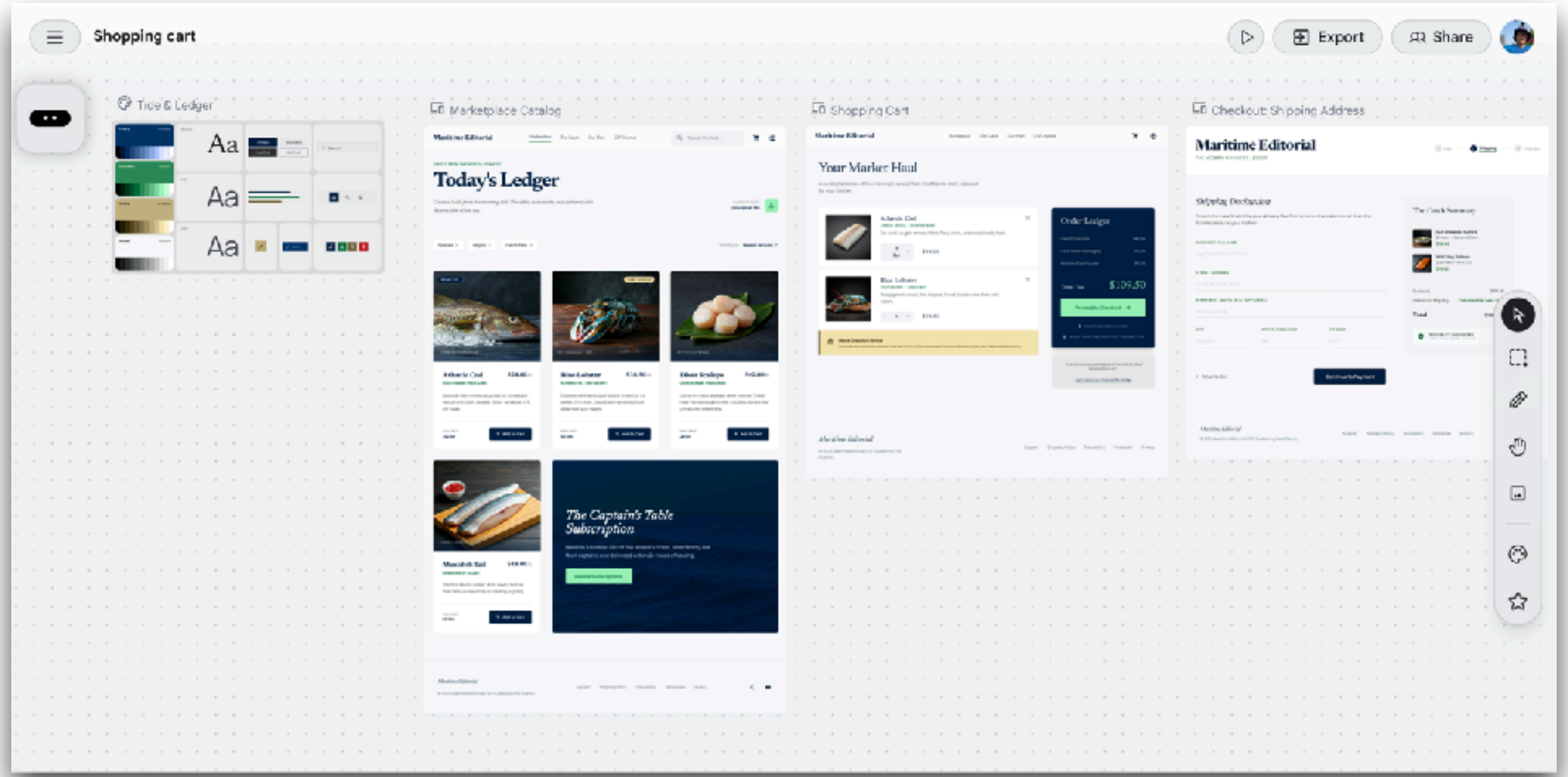
Select frame -> Copy/Paste as -> link to selection




https://www.figma.com/mcp-catalog/


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# Google Stitch

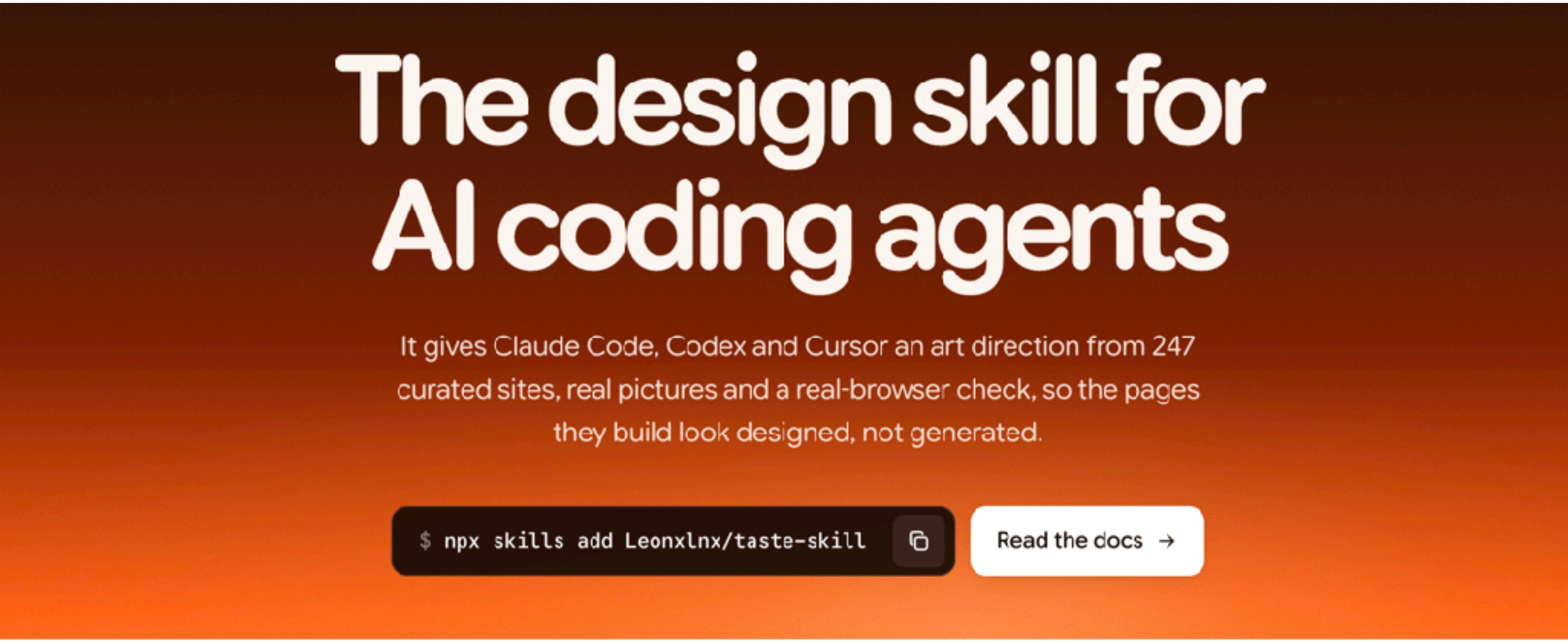



https://stitch.withgoogle.com/


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# Taste Skill

The design skill for AI coding agent




https://www.tasteskill.dev/


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# Development Process

# Plan before Act


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# Develop

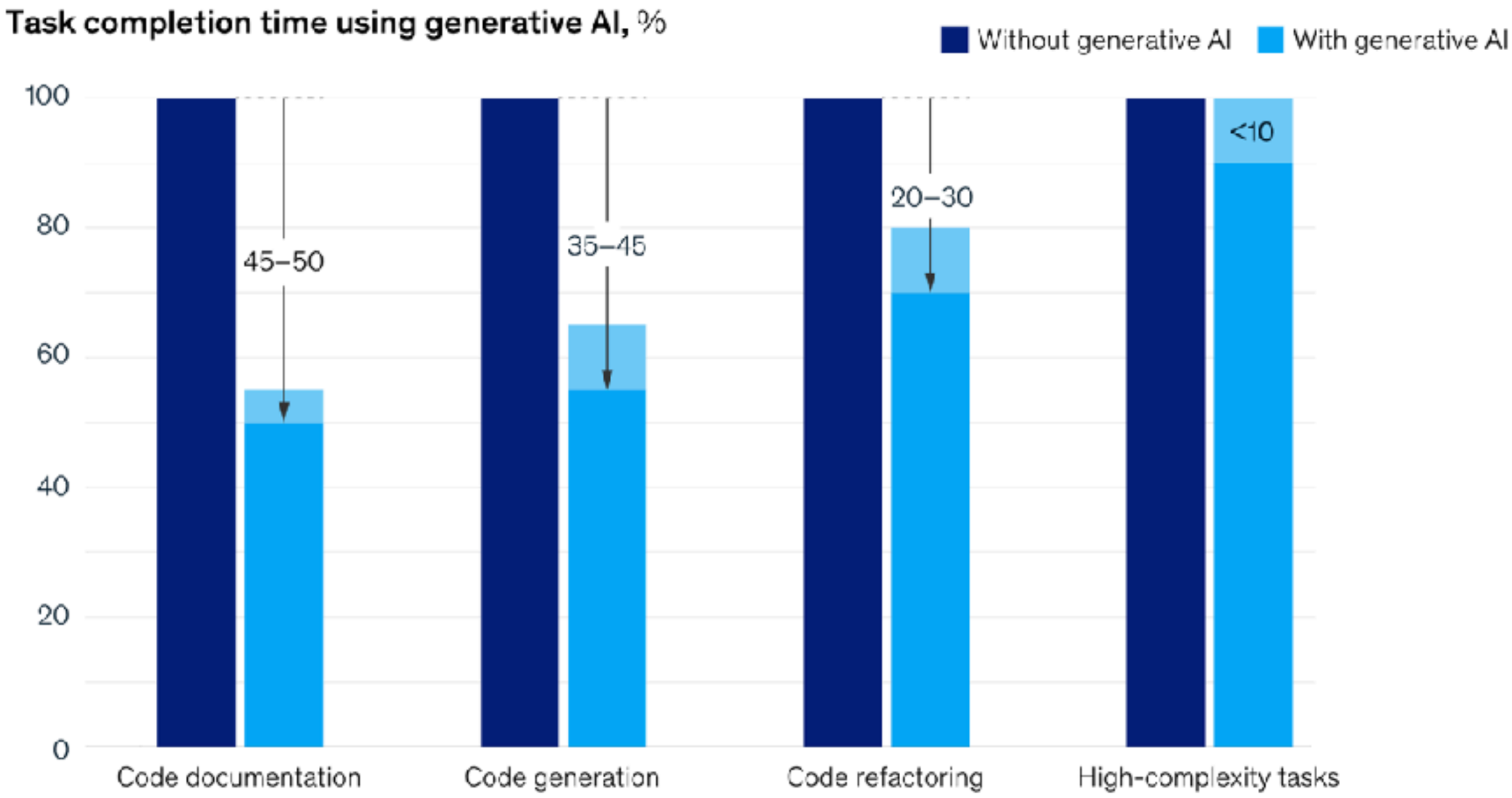



Code generation
Review and explain code
Debugging code
Improve consistency
Code translation


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# Development

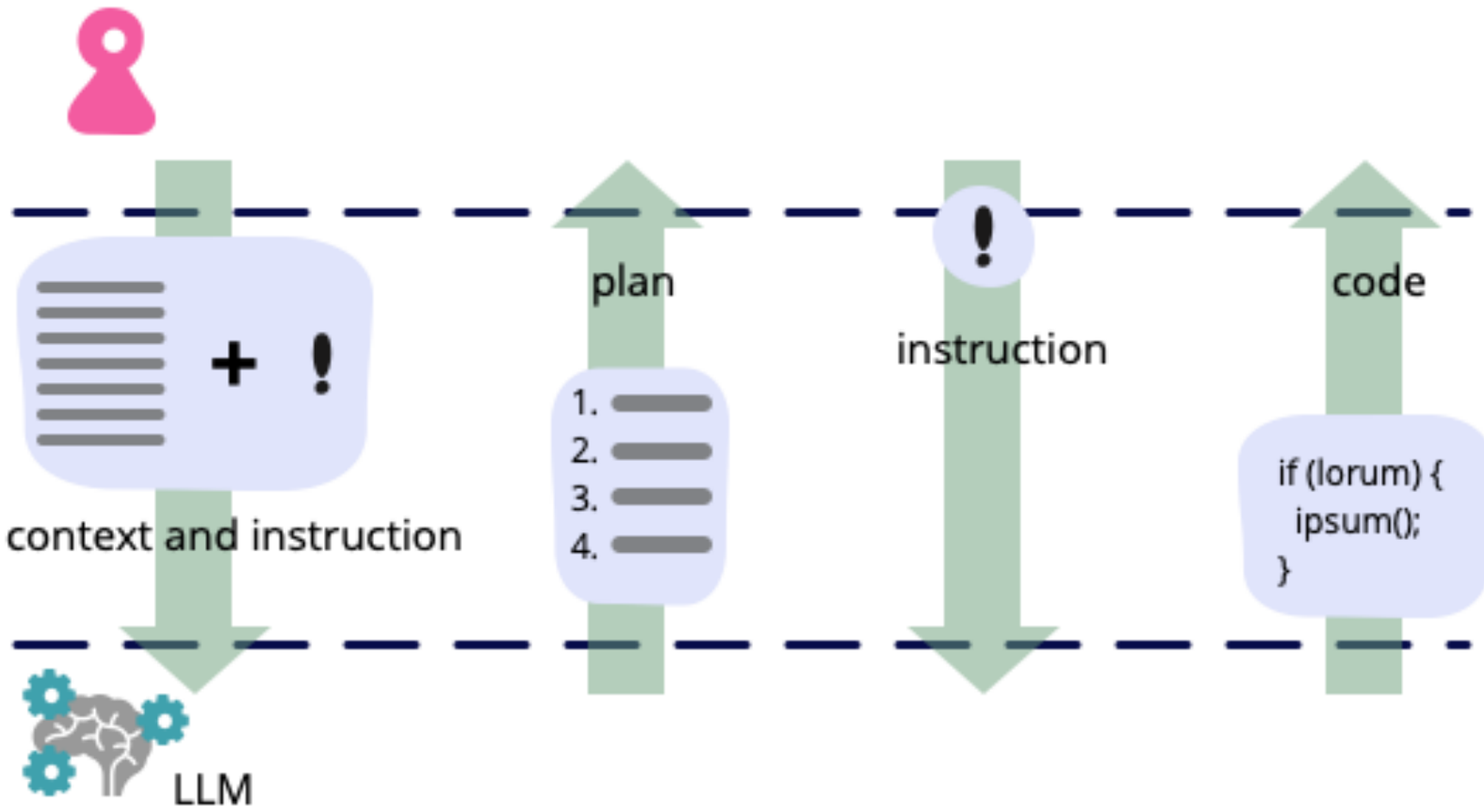
**Generative AI can increase developer speed, but less so for complex tasks.**




https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai


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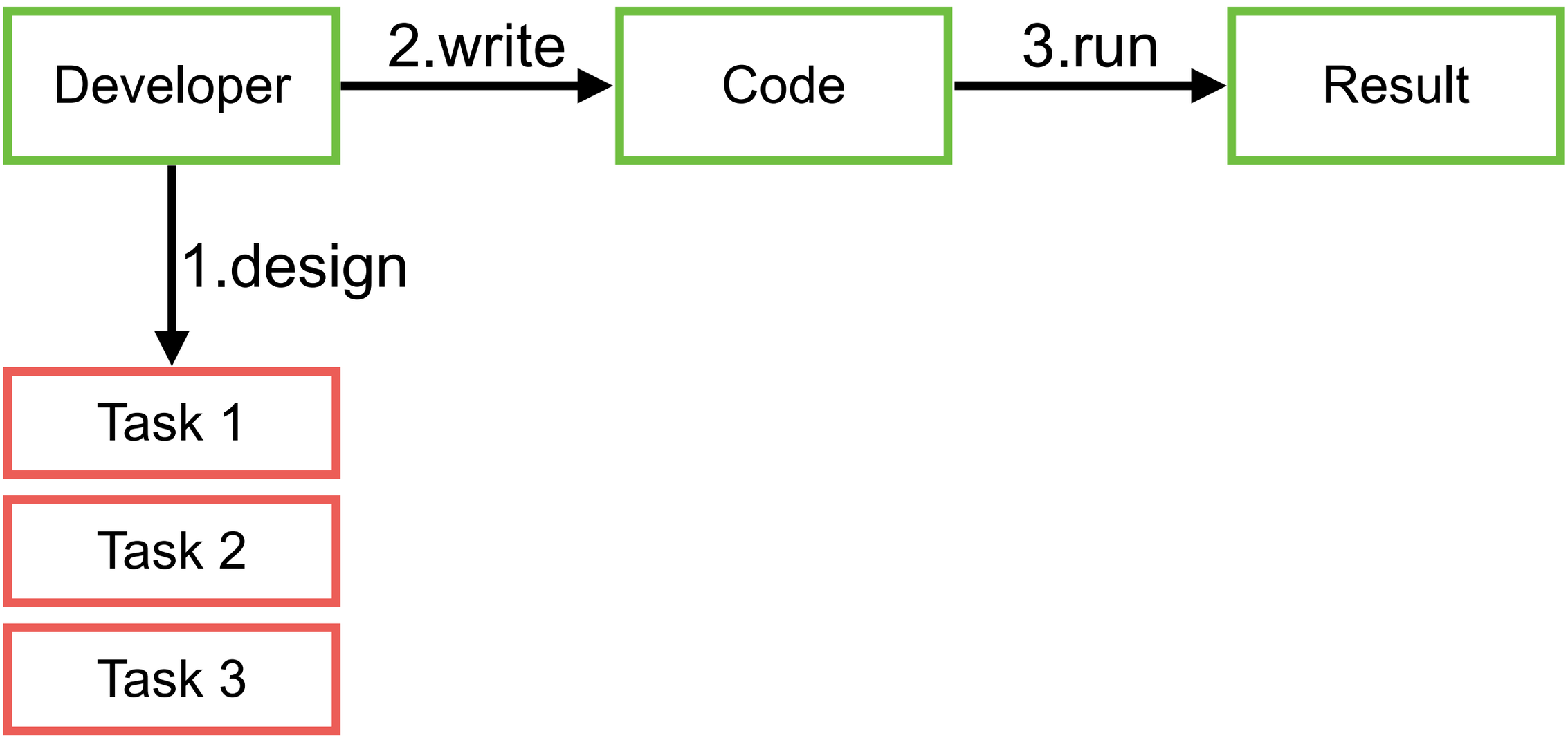
# Development




https://martinfowler.com/articles/2023-chatgpt-xu-hao.html


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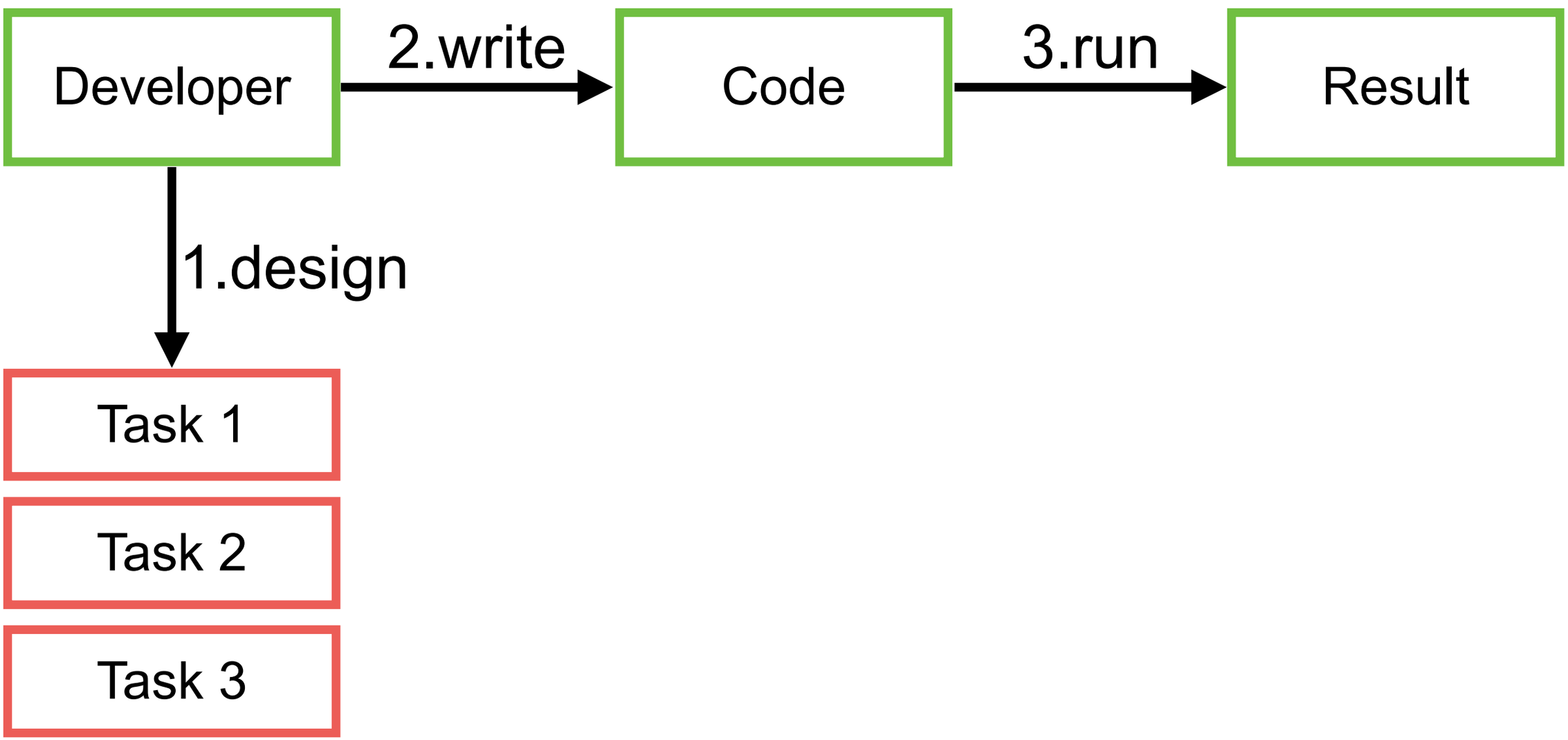
# Development





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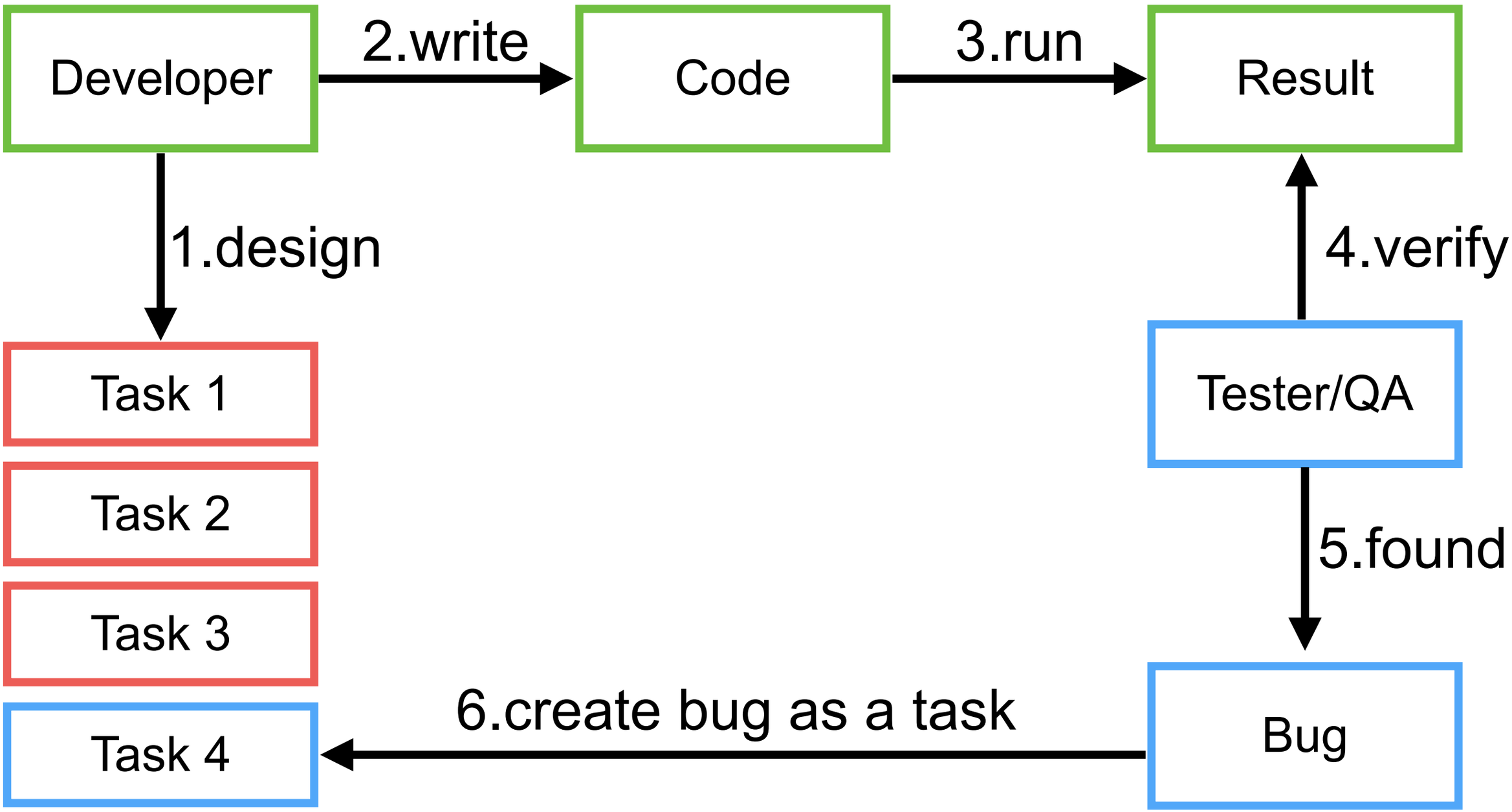
# Development





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# Development + Testing





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# Main Features

Ask question

Generate code

Refactor code

Document code

Find problems in your code


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# Tips and Techniques with AI

Scope of prompt

Error in result

Setup and config project

Latest information

Use technical keywords


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# Tips and Techniques with AI

Understand what you want

Modular approach

Clear and Precise inputs

Make sure you understand the code




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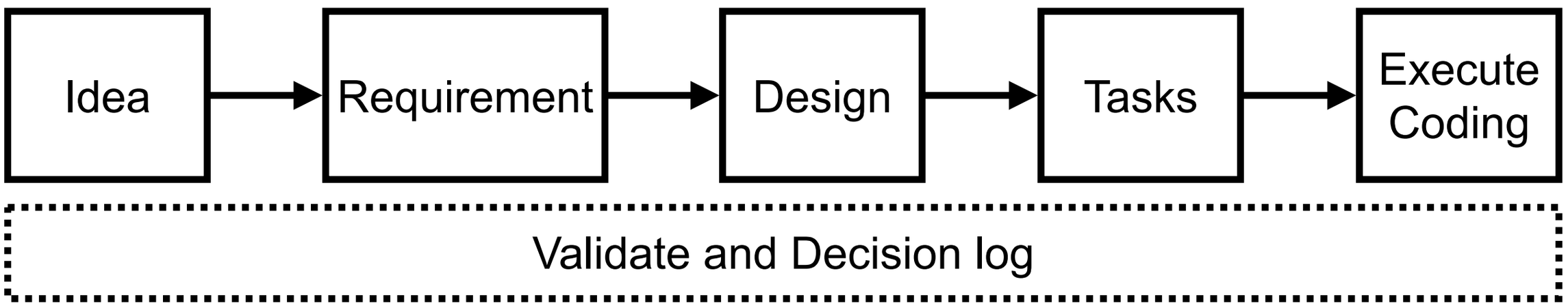
# Specification-Driven Development (SDD)


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# SDD ?

Specification or Documentation-First

Modern software development approach where detailed, structure specifications are created before writing code





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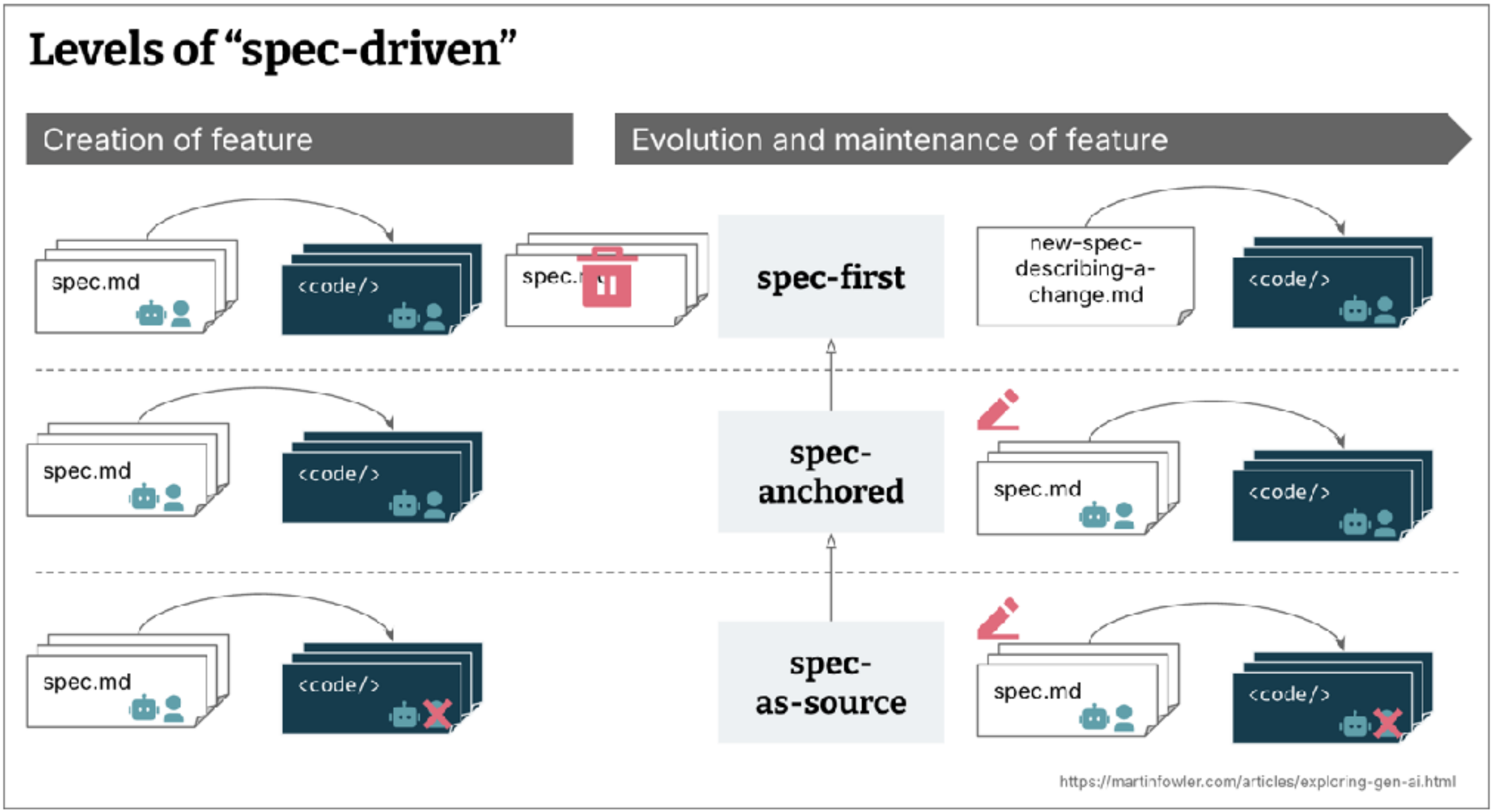
# Types of Specification ?

User requirements
Design documents
Interface contracts (UI, API)
Data schemas
Validation rules
Data/business flows
Security policy and constraints
Resources and performance constraints




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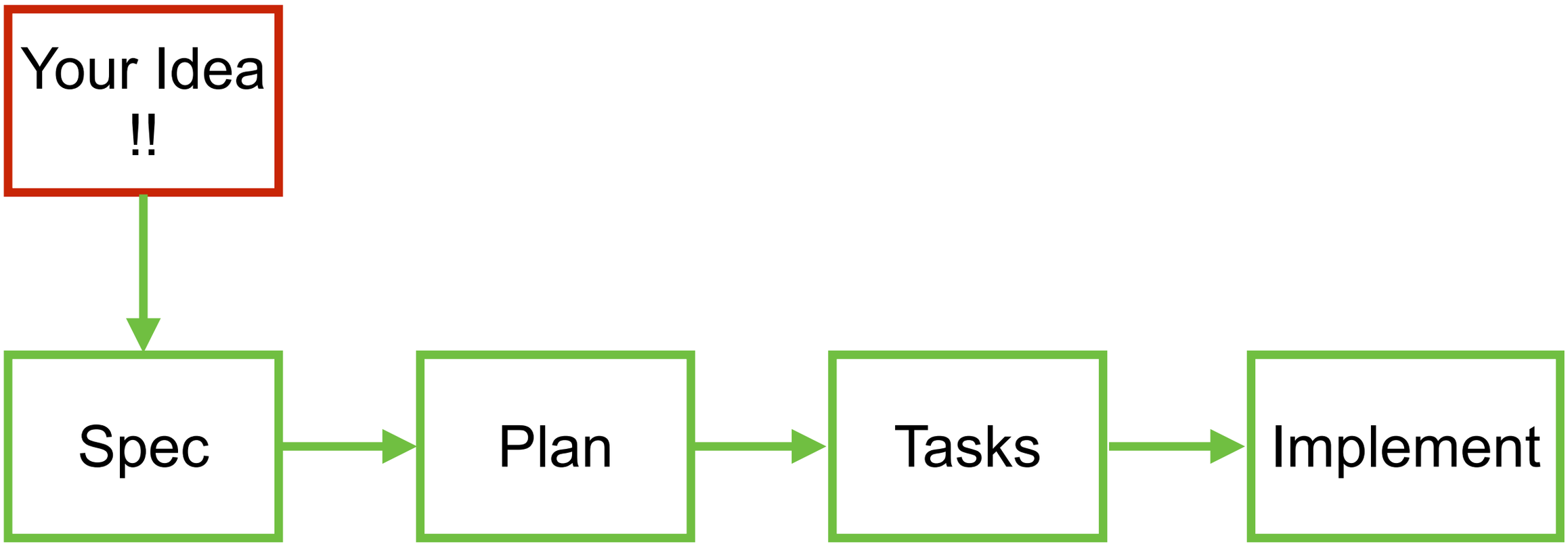
# Workflow SDD




https://martinfowler.com/articles/exploring-gen-ai/sdd-3-tools.html


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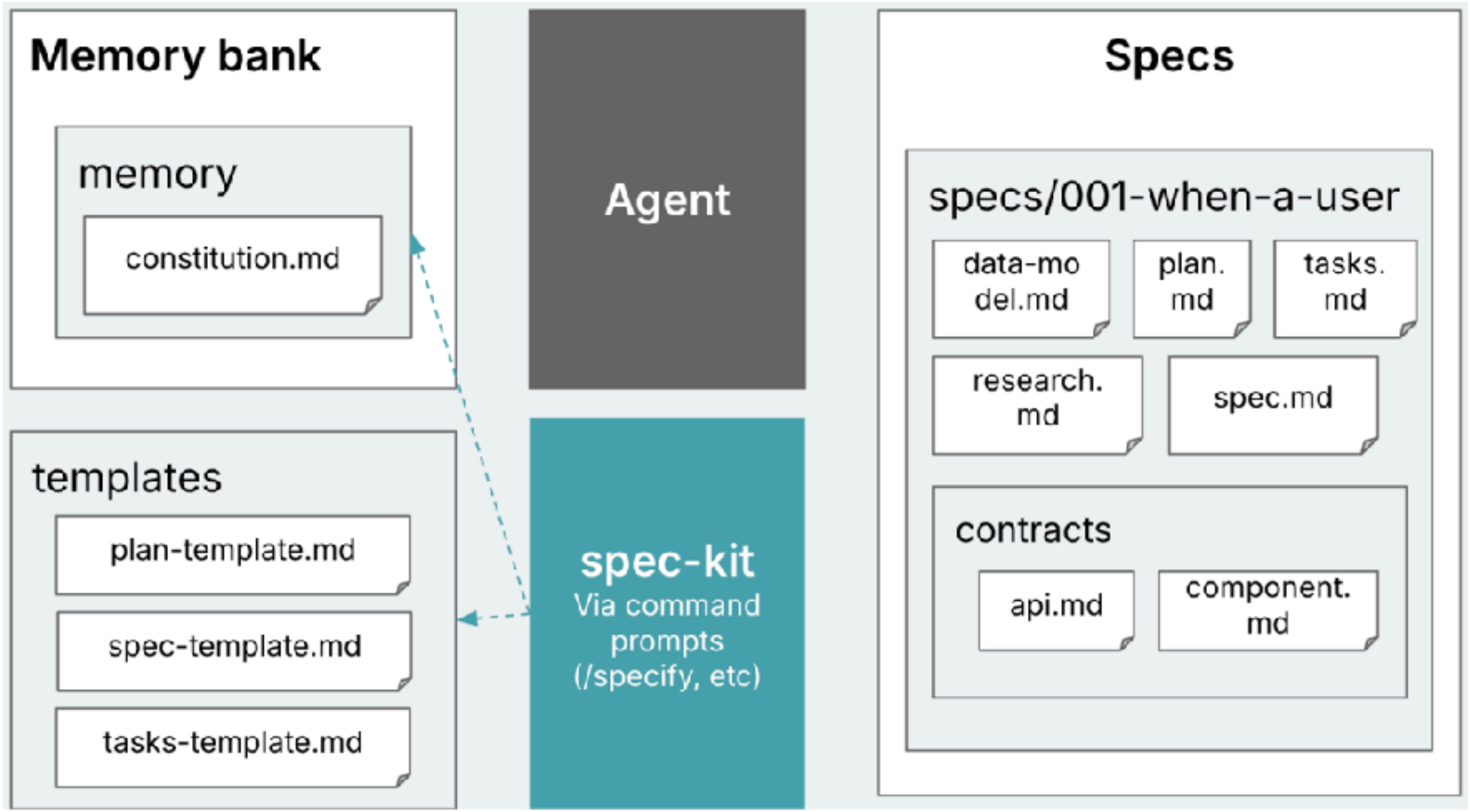
# Workflow SDD





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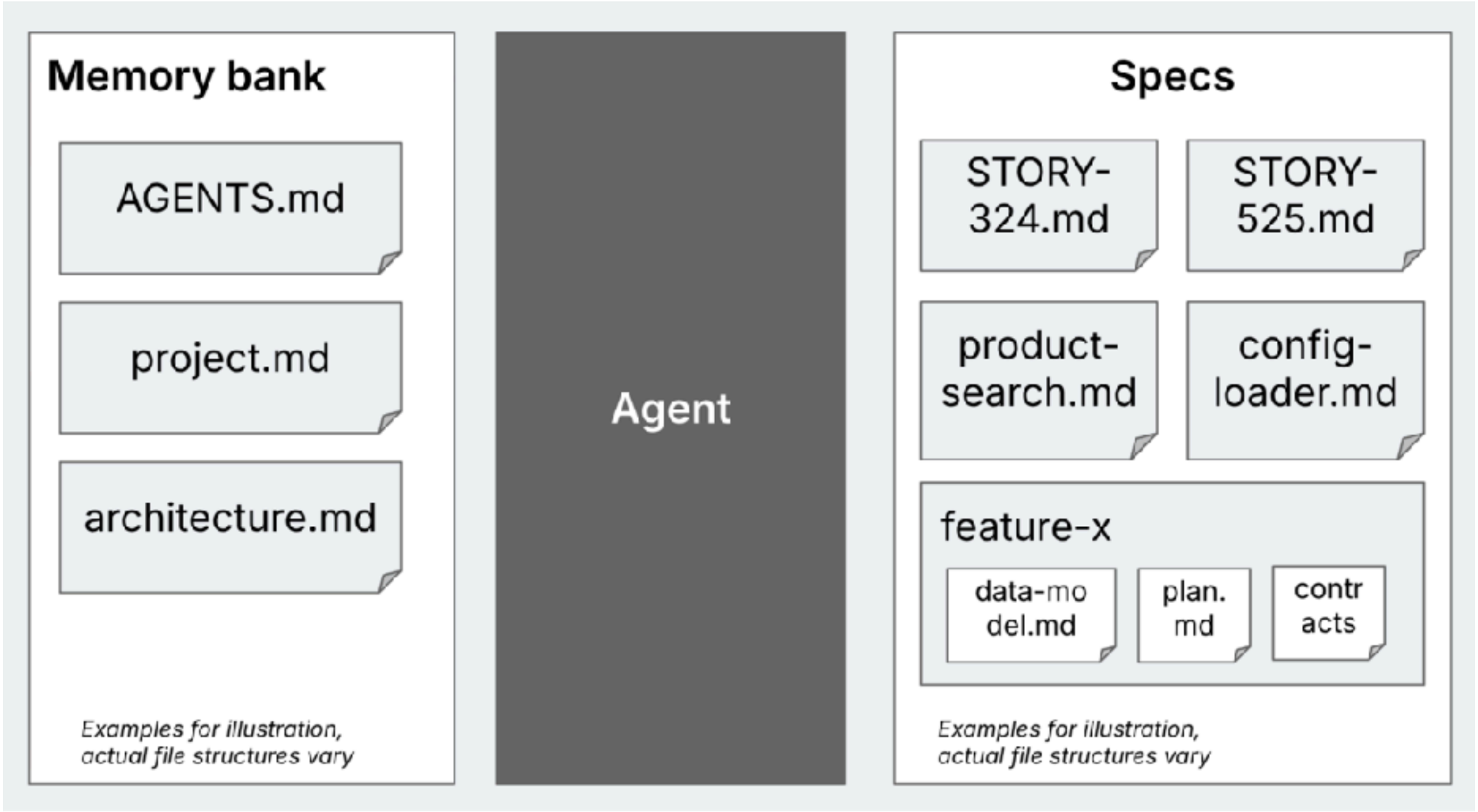
# Spec-Driven Development





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# Memory Bank





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# Testing Process
# with High quality process

Functional

Non-Functional


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# Testing




Test cases writing
Test code generation
Bug detection
Test planning
Data test generation


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# 6 Keys Software Quality

Defect density

Code duplication

Hardcode token/key

Security vulnerabilities

Outdated package

Non-permissive opensource libraries




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# Risks when using Generative AI





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# Risks

Quality of output generated
Explainability of decisions
Security policy !!
Sensitive data !!




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# Start your journey


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